



## **Affirmed Networks Active Intelligent vProbe System Administration Guide**

Release 9.4.0.0

Updated: October 2019

### **Inside ...**

- Affirmed Active Intelligent vProbe Overview
- AI-vProbe Configuration
- Performance Statistics
- EDR Record Format

### **Affirmed Networks Inc.**

35 Nagog Park  
Acton, MA 01720  
USA

Tel: +1-978-268-0800

[www.affirmednetworks.com](http://www.affirmednetworks.com)

Copyright© 2019 Affirmed Networks, Inc. All Rights Reserved.

AFFIRMED & DESIGN<sup>®</sup>, ACUITAS SERVICE MANAGEMENT SYSTEM<sup>®</sup>, AFFIRMED NETWORKS MOBILE CONTENT CLOUD<sup>™</sup>, AFFIRMED OPEN WORKFLOW<sup>™</sup>, AFFIRMED MOBILE CONTENT CLOUD<sup>™</sup> and other trademarks and designs are the registered or unregistered trademarks of Affirmed Networks, Inc. and its subsidiaries in the United States and in foreign countries. All other trademarks are the property of their respective owners. The Affirmed Networks, Inc. trademarks may not be used in connection with any product or service that is not Affirmed Networks' in any manner that is likely to cause confusion among customers or in any manner that disparages or discredits Affirmed Networks, Inc.

This document contains information that is the property of Affirmed Networks, Inc. This document may not be copied, reproduced, reduced to any electronic medium or machine readable form, or otherwise duplicated, and the information herein may not be used, disseminated or otherwise disclosed, except with the prior written consent of Affirmed Networks, Inc.

## **Affirmed NETWORKS, INC. SOFTWARE LICENSE AGREEMENT**

AFFIRMED NETWORKS, INC. ("AFFIRMED") IS WILLING TO LICENSE THE ENCLOSED SOFTWARE AND ACCOMPANYING USER DOCUMENTATION (COLLECTIVELY, THE "PROGRAM") TO YOU ONLY UPON THE CONDITION THAT YOU ACCEPT ALL OF THE TERMS AND CONDITIONS OF THIS LICENSE AGREEMENT. PLEASE READ THE TERMS AND CONDITIONS OF THIS LICENSE AGREEMENT CAREFULLY BEFORE OPENING THE PACKAGE (S) OR USING THE AFFIRMED PRODUCTS CONTAINING THE SOFTWARE, AND BEFORE USING THE ACCOMPANYING USER DOCUMENTATION. OPENING THE PACKAGE (S) OR USING THE AFFIRMED PRODUCTS CONTAINING THE PROGRAM WILL INDICATE YOUR ACCEPTANCE OF THE TERMS OF THIS LICENSE AGREEMENT. IF YOU ARE NOT WILLING TO BE BOUND BY THE TERMS OF THIS LICENSE AGREEMENT, AFFIRMED IS UNWILLING TO LICENSE THE PROGRAM TO YOU, IN WHICH EVENT YOU SHOULD RETURN THE PROGRAM WITHIN TEN (10) DAYS FROM SHIPMENT TO THE PLACE FROM WHICH IT WAS ACQUIRED, AND YOUR LICENSE FEE WILL BE REFUNDED. THIS LICENSE AGREEMENT REPRESENTS THE ENTIRE AGREEMENT CONCERNING THE PROGRAM BETWEEN YOU AND AFFIRMED, AND IT SUPERSEDES ANY PRIOR PROPOSAL, REPRESENTATION OR UNDERSTANDING BETWEEN THE PARTIES.

### **License Grant**

Subject to the provisions of this License, Affirmed grants to Licensee, and Licensee accepts, a non-exclusive, non-transferable license to use the object code form of the software supplied by Affirmed, including all patches, error corrections, updates and revisions thereto in machine-readable, object code form only (the "Software"), and the user documentation, including all updates and revisions thereto (the "User Documentation") for Licensee's internal business purposes (including, without limitation, in conjunction with Licensee's provision of services to its customers) on or in conjunction with the hardware product with which it was originally delivered or on a single designated computer. The Software and User Documentation are sometimes collectively referred to as the "Program". *Licensee acknowledges that certain Software provided by Affirmed may contain other restrictions and Licensee must refer to accompanying license certificates for each such Software licensed.* Licensee agrees that Licensee will not pledge, lease, rent, or share Licensee's rights under this License Agreement, and that Licensee will not, without Affirmed's prior written consent, assign or transfer Licensee's rights hereunder. Licensee agrees that Licensee may not duplicate (except as set forth below), decompile, disassemble, reverse engineer, modify, or otherwise translate the Software or permit a third party to do so and that Licensee shall not make the Program available to any other third party, or create works derivative of the Program, without Affirmed's express written consent. Licensee agrees it shall not publish in any fashion any results of benchmark tests run on the Program.

The Program is copyrighted and is only authorized to reproduce one copy of the Software and the User Documentation solely for backup purposes. Licensee is hereby prohibited from otherwise copying or translating, modifying or adapting the Program or, incorporating in whole or any part in any other product or creating derivative works based on all or any part of the Program. Licensee is not authorized to license others to reproduce any copies of the Program, except as expressly provided in this Agreement. Licensee agrees to ensure that all copyright, trademark and other proprietary notices of Affirmed affixed to or displayed on the Program will not be removed or modified. Licensee shall not decompile, disassemble or reverse engineer, the licensed software or any component thereof, except as may be permitted by applicable law, in which case Licensee must notify Affirmed in writing and Affirmed may provide review and assistance.

U.S. Government Restricted Rights. Notice - Distribution and use of products including computer programs and any related documentation and derivative works thereof, to and by the U.S. Government, are subject to the Restricted Rights provisions of FAR 52.227-19, paragraph (c)(2) as applicable, except for purchases by agencies of the Department of Defense (DOD). If the software is acquired under the terms of a Department of Defense or civilian agency contract, the software is a "commercial item" as that term is defined at 48 C.F.R. 2.101 (Oct. 1995), consisting of "commercial computer software" and "commercial computer software documentation" as such terms are used in 48 C.F.R. 12.212 of the

## Software License Agreement

Federal Acquisition Regulations and its successors and 48 C.F.R. 227.7202-1 through 227.7202-4 (June 1995) of the DoD FAR Supplement and its successors. All U.S. Government end users acquire the software with only those rights set forth in this Agreement. Manufacturer is Affirmed Networks, Inc., 35 Nagog Park, Acton MA 01720. Unpublished - rights reserved under the copyright laws of the United States.

|                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Affirmed's Rights</b>       | Licensee agrees that the Software and the User Documentation are proprietary, confidential products of Affirmed or Affirmed's licensor protected under U.S. copyright law and Licensee will use Licensee's best efforts to maintain their confidentiality. Licensee further acknowledges and agrees that all right, title and interest in and to the Program, including associated intellectual property rights, are and shall remain with Affirmed or Affirmed's licensor. This License Agreement does not convey to Licensee an interest in or to the Program, but only a limited right of use revocable in accordance with the terms of this License Agreement.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>License Fees</b>            | The license fees paid by Licensee are paid in consideration of the license granted under this License Agreement.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Term</b>                    | This License Agreement is effective upon Licensee's execution of this License Agreement and payment of the license fees and shall continue until terminated. Licensee may terminate this License Agreement at any time by returning the Program and all copies or portions thereof to Affirmed. Affirmed may terminate this License Agreement upon the breach by Licensee of any term hereof. Upon such termination by Affirmed, Licensee agrees to return to Affirmed the Program and all copies or portions thereof. Termination of this License Agreement shall not prejudice Affirmed's rights to damages or any other available remedy.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Limited Warranty</b>        | <p>Affirmed warrants that with normal use and service each Software product shall materially conform to Affirmed's then current published specifications for the most current release of such Software product for a period of ninety (90) days from the date of shipment by Affirmed. Program media is warranted for ninety- (90) days from the date of shipment by Affirmed. Software support beyond these periods is available at additional cost under the terms of Affirmed's Annual Maintenance Service Agreement. During the warranty period, as Licensee's sole and exclusive remedy, Affirmed will correct a Software product's failure to conform to the warranty provided that Licensee has notified Affirmed in writing of the nature of the non-conformity. This warranty shall not apply if any Software product has been (i) modified or altered by anyone other than Affirmed, (ii) abused or misapplied, or (iii) used in combination with hardware or software other than the Affirmed manufactured products for which it was designed. Affirmed shall incur no liability under this warranty if Affirmed's tests disclose that the alleged defect is due to causes not within Affirmed's reasonable control, including alteration or abuse of the goods. If a Program is determined not to be defective or to have a defect due to causes not within Affirmed's reasonable control, Licensee agrees to pay for such repair at the repair price as listed in Affirmed's then current applicable price list. In no event does Affirmed warrant that the use of Software products will be error free or uninterrupted. Affirmed's sole obligation under the Software warranty shall be to provide the remedies described above.</p> <p>EXCEPT FOR THE EXPRESS WARRANTIES STATED IN THIS WARRANTY, THE PROGRAM IS LICENSED "AS IS", AND Affirmed DISCLAIMS ANY AND ALL OTHER WARRANTIES, WHETHER EXPRESS, IMPLIED OR STATUTORY WITH RESPECT TO THE SOFTWARE PROVIDED UNDER THIS AGREEMENT, INCLUDING, BUT NOT LIMITED TO, ANY IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE, ANY WARRANTIES OF NONINFRINGEMENT AS WELL AS ANY WARRANTIES ARISING FROM COURSE OF DEALING, USAGE OR TRADE PRACTICE.</p> |
| <b>Limitation of Liability</b> | Affirmed's cumulative liability to Licensee or any other party for any loss or damages resulting from any claims, demands, or actions arising out of or relating to this License Agreement shall not exceed the greater of: (i) ten thousand U.S. dollars (\$10,000) or (ii) the total license fee paid to Affirmed for the use of the Program. In no event shall Affirmed be liable for any indirect, incidental, consequential, special, punitive or exemplary damages or lost profits or the loss of software or data, even if Affirmed has been                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

advised of the possibility of such damages. In the event that the Program contains the intellectual property of a third party pursuant to a license in favor of Affirmed, with a right to sublicense to Licensee, Affirmed hereby disclaims, to the extent permitted by law, such third party's liability of any damages, whether direct, indirect, incidental or consequential, arising out the use of the Program

**Export Control**

Licensee shall not export or re-export, directly or indirectly, any of the Programs (including any part of a Program or any direct product of such Program), Affirmed's Confidential Information or related technical data (a) into, or to a national resident of, any of those countries listed at the time of any shipment of the Products in the applicable U.S. export regulations as "prohibited or restricted" countries, or any other country to which such exports or re-exports may be restricted (collectively, the "Prohibited Countries") or (b) to anyone on the U.S. Treasury Department's list of Specially Designated Nations or the U.S. Commerce Department's Table of Denial Orders. At Licensee's request, Affirmed shall provide a list of the Prohibited Countries, Specially Designated Nations and/or Table of Denial Orders in order to assist Licensee to comply with such requirement.

**Governing Law**

This License Agreement shall be construed and governed in accordance with the laws and under the jurisdiction of the Commonwealth of Massachusetts, U.S.A. Any dispute arising out of this Agreement shall be referred to an arbitration proceeding in Boston, Massachusetts, U.S.A. by the American Arbitration Association.

**Miscellaneous**

If any action is brought by either party to this License Agreement against the other party regarding the subject matter hereof, the prevailing party shall be entitled to recover, in addition to any other relief granted, reasonable attorneys' fees and expenses of arbitration. Should any term of this License Agreement be declared void or unenforceable by any court of competent jurisdiction, such declaration shall have no effect on the remaining terms hereof. The failure of either party to enforce any rights granted hereunder or to take action against the other party in the event of any breach hereunder shall not be deemed a waiver by that party as to subsequent enforcement of rights or subsequent actions in the event of future breaches.

Affirmed has made no commitments or promises orally or in writing with respect to delivery of any future software features or functions. In relation to any future software features or functions, all presentations, RFP responses and/or product outlook documents, information or discussions, either prior to or following the date herein, are for informational purposes only, and Affirmed has no obligation to provide any future releases or upgrades or any features, enhancements or functions, unless specifically agreed to in writing by both parties. Reseller acknowledges that no purchasing decisions are based upon any future software features or functions.

**Third Party  
Beneficiary**

If the Program incorporates or otherwise contains any intellectual Property of any third party pursuant to a license agreement in favor of Affirmed and sublicensed to Licensee, such third party shall, to the extent permitted by law, be a third party beneficiary of the terms and conditions of this Agreement.



# Contents

## Preface

|                                           |        |
|-------------------------------------------|--------|
| Audience .....                            | xv     |
| Release Information .....                 | xv     |
| How to Use This Guide .....               | .xvi   |
| Features in the Release .....             | xvii   |
| Conventions .....                         | .xxxvi |
| How to Obtain Product Documentation ..... | xxxvii |
| Technical Assistance Center .....         | xxxvii |

## Chapter 1 Affirmed Active Intelligent AI-vProbe Overview

|                                                                     |     |
|---------------------------------------------------------------------|-----|
| Event Data Record (EDR) Generation .....                            | 1-1 |
| EDR Record Types .....                                              | 1-3 |
| EDR Server Application Redundancy .....                             | 1-5 |
| EDR Interface Configuration on Geographic Redundancy Clusters ..... | 1-5 |
| Control Plane Traffic Mirroring .....                               | 1-5 |
| Data Plane Traffic Mirroring .....                                  | 1-6 |

## Chapter 2 AI-vProbe Configuration

|                                                                   |      |
|-------------------------------------------------------------------|------|
| Generating SESSION, BEARER, FLOW, and HTTP Transaction EDRs ..... | 2-1  |
| Configuring Subscriber Data Plane Mirroring .....                 | 2-11 |
| Metadata .....                                                    | 2-15 |
| Configuring Control Plane Packet Mirroring .....                  | 2-16 |
| Metadata .....                                                    | 2-19 |
| Sample Protocol Logging .....                                     | 2-21 |
| Configuring UDP Server to Receive Pilot Packets .....             | 2-23 |
| Configuring Real Time Troubleshooting EDRs .....                  | 2-27 |

## Chapter 3 Performance Statistics

|                                        |     |
|----------------------------------------|-----|
| EDR Record Statistics .....            | 3-1 |
| EDR Client and Server Statistics ..... | 3-4 |
| Procedure Statistics .....             | 3-7 |
| Subscriber vTap Statistics .....       | 3-8 |

## Chapter 4 EDR Record Format

|                                              |      |
|----------------------------------------------|------|
| Base Event Data Record . . . . .             | 4-1  |
| STATUS Event Data Record . . . . .           | 4-2  |
| SESSION Event Data Record . . . . .          | 4-3  |
| SessionRecord . . . . .                      | 4-3  |
| SessionRecordKeys . . . . .                  | 4-5  |
| SessionSubscriberInfo . . . . .              | 4-5  |
| ServingNetworkInfo . . . . .                 | 4-6  |
| GatewayInfo . . . . .                        | 4-8  |
| EdrSessionBucketInfo . . . . .               | 4-9  |
| EdrCauseCode . . . . .                       | 4-10 |
| SessionQosInformation . . . . .              | 4-11 |
| SessionUserCSGInfo . . . . .                 | 4-11 |
| EdrSubscriberTimezone . . . . .              | 4-11 |
| EdrUserLocationInfo . . . . .                | 4-12 |
| EdrFullQualTeid . . . . .                    | 4-12 |
| EdrPlmnIdType . . . . .                      | 4-13 |
| EdrCloseInformation . . . . .                | 4-14 |
| EdrCellGlobalIdentity . . . . .              | 4-14 |
| EdrServiceAreaIdentity . . . . .             | 4-15 |
| EdrRoutingAreaIdentity . . . . .             | 4-15 |
| EdrTrackingAreaIdentity . . . . .            | 4-15 |
| EdrEUTRANCellGlobalIdentity . . . . .        | 4-16 |
| EdrLocationAreaIdentity . . . . .            | 4-16 |
| BEARER Event Data Record . . . . .           | 4-17 |
| BearerRecord . . . . .                       | 4-17 |
| BearerRecordKeys . . . . .                   | 4-18 |
| BearerInfo . . . . .                         | 4-18 |
| BearerQosInformation . . . . .               | 4-19 |
| AllocRetentionPriority . . . . .             | 4-19 |
| FLOW Event Data Record . . . . .             | 4-20 |
| FlowRecord . . . . .                         | 4-20 |
| FlowRecordKeys . . . . .                     | 4-23 |
| ipTuple . . . . .                            | 4-23 |
| FlowSubscriberInfo . . . . .                 | 4-23 |
| NetworkPerfInfo . . . . .                    | 4-24 |
| NetworkStatsInfo . . . . .                   | 4-24 |
| NatBindingInfo . . . . .                     | 4-25 |
| HTTP Transaction Event Data Record . . . . . | 4-26 |
| HTTPTransactionRecord . . . . .              | 4-26 |
| timeInfo . . . . .                           | 4-27 |
| subscriberInfo . . . . .                     | 4-28 |
| connectionInfo . . . . .                     | 4-30 |
| requestInfo . . . . .                        | 4-31 |
| responseInfo . . . . .                       | 4-37 |
| compressionInfo . . . . .                    | 4-39 |
| webImageOptInfo . . . . .                    | 4-40 |
| videoInfo . . . . .                          | 4-41 |
| minifyInfo . . . . .                         | 4-46 |
| preemptDnsInfo . . . . .                     | 4-46 |
| cacheInfo . . . . .                          | 4-47 |



|                                                                     |      |
|---------------------------------------------------------------------|------|
| manifestInfo .....                                                  | 4-50 |
| tcpClientConnectionInfo.....                                        | 4-51 |
| filterInfo .....                                                    | 4-52 |
| tcpServerConnectionInfo .....                                       | 4-53 |
| entitlementInfo .....                                               | 4-53 |
| udpProxyInfo .....                                                  | 4-54 |
| DpiInformation .....                                                | 4-55 |
| SHIPInformation.....                                                | 4-56 |
| HttpRecord .....                                                    | 4-57 |
| HttpRecordKeys .....                                                | 4-57 |
| Real Time Troubleshooting (RTT) Event Data Record.....              | 4-58 |
| RTT Common Attribute List .....                                     | 4-59 |
| RTT Attributes for GTP Interface Procedures .....                   | 4-61 |
| RttGtpPCOInfo.....                                                  | 4-63 |
| RttBearerInfo .....                                                 | 4-64 |
| RttGtpBearerFlags .....                                             | 4-64 |
| RTT Attributes for Diameter SWm/Gx/Gy/S6b Interface Procedures..... | 4-65 |
| RTT Attributes for IKE Interface Procedures.....                    | 4-71 |
| EncryptedVideoRateLimiting .....                                    | 4-73 |
| timeInfo.....                                                       | 4-73 |
| subscriberInfo .....                                                | 4-74 |
| connectionInfo .....                                                | 4-75 |
| tcpSplicingInfo .....                                               | 4-76 |



# List of Figures

|            |                                    |            |
|------------|------------------------------------|------------|
| Figure 1-1 | AI-vProbe Network Deployment ..... | <b>1-2</b> |
|------------|------------------------------------|------------|

## *Figures*

# List of Tables

|            |                                            |      |
|------------|--------------------------------------------|------|
| Table 2-1  | Metadata Information (in TLV format).....  | 2-19 |
| Table 4-1  | EventDataRecord Fields.....                | 4-1  |
| Table 4-2  | StatusRecord Fields.....                   | 4-2  |
| Table 4-3  | SessionRecord Fields .....                 | 4-3  |
| Table 4-4  | SessionRecordKeys Subfields.....           | 4-5  |
| Table 4-5  | SessionSubscriberInfo Subfields.....       | 4-5  |
| Table 4-6  | ServingNetworkInfo Subfields.....          | 4-6  |
| Table 4-7  | GatewayInfo Subfields.....                 | 4-8  |
| Table 4-8  | EdrSessionBucketInfo Subfields.....        | 4-9  |
| Table 4-9  | EdrCauseCode Subfields.....                | 4-10 |
| Table 4-10 | SessionQosInformation Subfields .....      | 4-11 |
| Table 4-11 | SessionUserCSGInfo Subfields.....          | 4-11 |
| Table 4-12 | EdrSubscriberTimezone Subfields .....      | 4-11 |
| Table 4-13 | EdrUserLocationInfo Subfields.....         | 4-12 |
| Table 4-14 | EdrFullQualTeid Subfields .....            | 4-12 |
| Table 4-15 | EdrPlmnIdType Subfields.....               | 4-13 |
| Table 4-16 | EdrCloseInformation Subfields.....         | 4-14 |
| Table 4-17 | EdrCellGlobalIdentity Subfields.....       | 4-14 |
| Table 4-18 | EdrServiceAreaIdentity Subfields.....      | 4-15 |
| Table 4-19 | EdrRoutingAreaIdentity Subfields .....     | 4-15 |
| Table 4-20 | EdrTrackingAreaIdentity Subfields .....    | 4-15 |
| Table 4-21 | EdrEUTRANCellGlobalIdentity Subfields..... | 4-16 |
| Table 4-22 | EdrLocationAreaIdentity Subfields.....     | 4-16 |
| Table 4-23 | BearerRecord Subfields.....                | 4-17 |
| Table 4-24 | BearerRecordKeys Subfields .....           | 4-18 |
| Table 4-25 | BearerInfo Subfields .....                 | 4-18 |
| Table 4-26 | BearerQosInformation Subfields.....        | 4-19 |
| Table 4-27 | AllocRetentionPriority Subfields .....     | 4-19 |
| Table 4-28 | FlowRecord Subfields .....                 | 4-20 |
| Table 4-29 | FlowRecordKeys Subfields.....              | 4-23 |
| Table 4-30 | ipTuple Subfields .....                    | 4-23 |
| Table 4-31 | FlowSubscriberInfo Subfields.....          | 4-23 |
| Table 4-32 | NetworkPerfInfo Subfields .....            | 4-24 |
| Table 4-33 | NetworkStatsInfo Subfields .....           | 4-24 |
| Table 4-34 | NatBindingInfo Subfields .....             | 4-25 |
| Table 4-35 | HTTPTransactionRecord Fields.....          | 4-26 |
| Table 4-36 | timeInfo Subfields.....                    | 4-27 |
| Table 4-37 | subscriberInfo Subfields.....              | 4-28 |
| Table 4-38 | connectionInfo Subfields.....              | 4-30 |
| Table 4-39 | requestInfo Subfields.....                 | 4-31 |
| Table 4-40 | responseInfo Subfields .....               | 4-37 |
| Table 4-41 | compressionInfo Subfields .....            | 4-39 |
| Table 4-42 | webImageOptInfo Subfields .....            | 4-40 |
| Table 4-43 | videoInfo Subfields .....                  | 4-41 |
| Table 4-44 | MinyInfo Subfields .....                   | 4-46 |
| Table 4-45 | PreemptDnsInfo Subfields .....             | 4-46 |
| Table 4-46 | cacheInfo Subfields.....                   | 4-47 |
| Table 4-47 | ManifestInfo Fields .....                  | 4-50 |

## Tables

|            |                                                      |      |
|------------|------------------------------------------------------|------|
| Table 4-48 | tcpClientConnectionInfo Fields .....                 | 4-51 |
| Table 4-49 | filterInfo Fields.....                               | 4-52 |
| Table 4-50 | tcpServerConnectionInfo Fields.....                  | 4-53 |
| Table 4-51 | entitlementInfo Fields .....                         | 4-53 |
| Table 4-52 | udpProxyInfo Fields.....                             | 4-54 |
| Table 4-53 | DpiInformation Fields.....                           | 4-55 |
| Table 4-54 | SHIPInformation Fields .....                         | 4-56 |
| Table 4-55 | HttpRecord Fields .....                              | 4-57 |
| Table 4-56 | HttpRecordKeys Fields .....                          | 4-57 |
| Table 4-57 | Procedures that Result in RTT EDR Generation .....   | 4-58 |
| Table 4-58 | RTT Common Attribute List.....                       | 4-59 |
| Table 4-59 | GTP Interface Attribute List .....                   | 4-61 |
| Table 4-60 | RttGtpPCOInfo Subfields .....                        | 4-63 |
| Table 4-61 | RttBearerInfo Subfields.....                         | 4-64 |
| Table 4-62 | RttGtpBearerFlags Subfields.....                     | 4-64 |
| Table 4-63 | Diameter SWm/Gx/Gy/S6b Interface Attribute List..... | 4-65 |
| Table 4-64 | Diameter IKE Interface Attribute List .....          | 4-71 |
| Table 4-65 | EncryptedVideoRateLimiting Fields.....               | 4-73 |
| Table 4-66 | timeInfo Subfields .....                             | 4-73 |
| Table 4-67 | subscriberInfo Subfields.....                        | 4-74 |
| Table 4-68 | connectionInfo Subfields.....                        | 4-75 |
| Table 4-69 | tcpSplicingInfo Subfields.....                       | 4-76 |

# Preface

This guide provides an overview of the Affirmed Networks Active Intelligent vProbe (AI-vProbe) and describes its features, configuration, and EDR record formats.

---

**Note:** *The Affirmed Networks Mobile Content Cloud™ (MCC) is the mobile services software solution that operates on virtual platforms such as blade systems.*

---

## Audience

This guide is intended for network operators (sometimes referred to as Operators in this guide) and other qualified service personnel responsible for provisioning and monitoring the network/system. The audience should be familiar with Wi-Fi access networks, Long Term Evolution (LTE), 3G, or 4G networks.

## Release Information

See the Release Notes for important information, requirements, limitations, and restrictions that apply to this release.

## How to Use This Guide

Read the following chapters and appendices in the order in which they appear:

| Chapter/Appendix                                                          | Describes                                                                   |
|---------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| <a href="#">Chapter 1, Affirmed Active Intelligent AI-vProbe Overview</a> | An overview of the Affirmed Active Intelligent vProbe solution.             |
| <a href="#">Chapter 2, AI-vProbe Configuration</a>                        | The CLI commands to configure AI-vProbe.                                    |
| <a href="#">Chapter 3, Performance Statistics</a>                         | The AI-vProbe performance statistics that you can view through the MCC CLI. |
| <a href="#">Chapter 4, EDR Record Format</a>                              | The EDR data record fields and data types.                                  |



## Features in the Release

This release includes the following new features:

---

**Note:** For a complete list of the new features supported in this release, see the release notes. In addition to the information in this guide, see the additional Affirmed Networks documentation listed in the Affirmed Networks Documentation Library. For a comprehensive list of terms and acronyms used in Affirmed Networks documentation, see Affirmed Networks Terms and Acronyms.

---

| Feature                                                                                                                                                   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Release 9.4.0.0 Features</b>                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| In the FLOW records, added the GQUIC user agent value to the tetheringAlgorithm field<br>See <a href="#">FLOW Event Data Record</a> .                     | In the FLOW EDR, added the GQUIC user agent value to the tetheringAlgorithm field:<br><ul style="list-style-type: none"> <li>tetheringAlgorithm &lt;string&gt; {ttl, userAgent, tcpipfingerprinting, gquic-user-agent}</li> </ul>                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Release 9.3.2.0 Features</b>                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| In the HTTP Transaction data records (HTDRs), added the SHIPInformation field to the HTTPTransactionRecord<br>See <a href="#">HTTPTransactionRecord</a> . | In the HTTP Transaction data records (HTDRs), added the following field to the HTTPTransactionRecord:<br><b>SHIPInformation Data Record:</b><br>string Content Provider /<br>uint64 AppID / not used<br>ConsentStatus Consent Status /<br>ConsentStatus contains the following enumerations:<br>0 = Consent Not Available<br>1 = Consent Not Matched<br>2 = Consent Matched<br>ShipStatus Ship Status /<br>ShipStatus contains the following enumerations:<br>0 = SHIP success<br>1 = Content Provider Not Found<br>2 = Info Not Available<br>3 = Consent Validation Failed<br>4 = SHIP Internal Server Error |

| Feature                                                                                                                   | Description                                                                                                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Release 9.3.1.0 Features</b>                                                                                           |                                                                                                                                                                                                                    |
| Added TTL value to the dpiStringInfo subfield of the FlowRecord field of the Flow EDR<br>See <a href="#">FlowRecord</a> . | Added the TTL value to the dpiStringInfo subfield of the FlowRecord field of the Flow EDR.                                                                                                                         |
| <b>Release 9.3.0.0 Features</b>                                                                                           |                                                                                                                                                                                                                    |
| Support for the realApn field in the SESSION Event Data Record (EDR)<br>See <a href="#">SESSION Event Data Record</a> .   | In the SESSION EDR, added the following field to the SessionSubscriberInfo data record:<br><br><code>optional string realApn // logical name of the mapped access point to the external packet data network</code> |

| Feature                                                                                                         | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Additional Event Data Record (EDR) fields</p> <p>See <a href="#">HTTP Transaction Event Data Record</a>.</p> | <p>In the HTTP Transaction data records (HTDRs), added the following fields to the HTTPTransactionRecord:</p> <p><b>timeInfo Data Record:</b></p> <p>optional uint32 transactionId// Proxy Transaction ID</p> <p>optional string recordEntity// Proxy Process Name</p> <p><b>subscriberInfo Data Record:</b></p> <p>optional string virtualGwName// Proxy Service Configuration Name</p> <p>optional string sessOperatorSpecific1// MIND data</p> <p>optional bytes userLocationInfo</p> <p><b>requestInfo Data Record:</b></p> <p>optional HttpSvcErrCode failureReason// HTTP Failure Reason</p> <p>HttpSvcErrCode contains enumerations.</p> <p><b>tcpClientConnectionInfo Data Record:</b></p> <p>optional string congestionAlgorithm// TCP congestion algorithm</p> <p><b>tcpServerConnectionInfo Data Record:</b></p> <p>optional string congestionAlgorithm// TCP congestion algorithm</p> <p><b>udpProxyInfo Data Record:</b></p> <p>optional uint64 bytesDroppedDueToBufferLimiting// Number bytes dropped by buffer limiting</p> <p>optional UdpProxyFailureReason failureReason// UDP failure reason</p> <p>UdpProxyFailureReason contains the following enumerations:</p> <p>0 = UDPPROXY_FAILURE_NO_REASON</p> <p>1 = UDPPROXY_FAILURE_REASON_SERVER_CONNECTION_INIT_FAILURE</p> <p>2 = UDPPROXY_FAILURE_REASON_OUT_OF_SOCKET</p> <p>3 = UDPPROXY_FAILURE_REASON_NO_CONNECTION_FOUND</p> <p>4 = UDPPROXY_FAILURE_REASON_NO_PACKETS</p> <p>5 = UDPPROXY_FAILURE_REASON_OTHER_REASONS</p> |

| Feature                                                                                                                                                    | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Release 9.2.0.0 Features</b>                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Added support to include IMSI in the bearerInfo data record of the Start/Stop/Interim/Attempt BEARER EDR<br>See <a href="#">BEARER Event Data Record</a> . | Added support to include IMSI in the bearerInfo data record of the Start/Stop/Interim/Attempt BEARER Event Data Record.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Release 9.1.0.0 Features</b>                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Added vProbe support for PFCP protocol logging<br>See <a href="#">EDR Record Format</a> .                                                                  | <ul style="list-style-type: none"> <li>■ Added vProbe support for PFCP protocol logging:</li> <li>■ Control Plane packet mirroring support for PFCP protocol</li> </ul> <p>Support to populate control plane function or user plane function session IDs in SESSION, BEARER, FLOW, and HTTP Transaction EDRs. Added the following fields to the RecordKeys data records:</p> <p><b>SessionRecordKeys</b></p> <ul style="list-style-type: none"> <li>– 2 = optional uint64 uPlaneSessionId</li> </ul> <p><b>BearerRecordKeys</b></p> <ul style="list-style-type: none"> <li>– 3 = optional uint64 uPlaneSessionId</li> </ul> <p><b>FlowRecordKeys</b></p> <ul style="list-style-type: none"> <li>– 4 = optional uint64 cPlaneSessionId</li> </ul> <p><b>HTTPRecordKeys</b></p> <ul style="list-style-type: none"> <li>– 5 = optional uint64 cPlaneSessionId</li> </ul> |
| <b>Release 8.4.0.0 Features</b>                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Enhanced tethering detection logic using TCP-IP fingerprinting<br>See <a href="#">FLOW Event Data Record</a> .                                             | <p>In the FLOW EDR, added the tcpipfingerprinting value to the tetheringAlgorithm field:</p> <ul style="list-style-type: none"> <li>■ tetheringAlgorithm &lt;string&gt; {ttl, userAgent, tcpipfingerprinting}</li> </ul> <p>Added the following FLOW EDR statistic:</p> <ul style="list-style-type: none"> <li>■ tetheringQualityIndex &lt;float&gt;</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

| Feature                                                                                            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Release 8.3.0.0 Features</b>                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Support for Content-Categorization Filtering for HTTPS Traffic<br>See <a href="#">filterInfo</a> . | <p>Added support for HTTPS traffic to:</p> <ul style="list-style-type: none"> <li>■ Apply content-filtering</li> <li>■ Apply content online-categorization</li> <li>■ Allow or terminate/block HTTPS connections</li> </ul> <p>With this feature, the existing content filter statistics are incremented:<br/> <b>show services content-filter statistics cluster-level summary</b></p> <p>With this feature, the following existing content filter statistics have been changed:<br/> <b>show services content-filter statistics cluster-level content-categorization</b></p> <ul style="list-style-type: none"> <li>■ Content Categorization – URL-based Lookups =&gt; Content Categorization – HTTP Request Lookups</li> <li>■ Content Categorization – Response-based Lookups =&gt; Content Categorization – HTTP Response Lookups</li> </ul> <p>With this feature, the following content filter statistics have been added:<br/> <b>show services content-filter statistics cluster-level content-categorization</b></p> <ul style="list-style-type: none"> <li>■ Content Categorization – HTTPS Request Lookups</li> <li>■ Content Categorization – Lookup Failures</li> </ul> <p>With this feature, the HTTP Transaction Event Data Record filter-information data record fields are populated:</p> <ul style="list-style-type: none"> <li>■ filterName</li> <li>■ filterType</li> <li>■ actionApplied</li> <li>■ filterInstanceName</li> <li>■ onlineCategory</li> </ul> |

| Feature                                                                                      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Additional Event Data Record (EDR) field</p> <p>See <a href="#">NetworkStatsInfo</a>.</p> | <p>Added vProbe support for TCP/UDP closure reason reporting in the Flow EDR NetworkStatsInfo data record.</p> <p>Flow EDR NetworkStatsInfo data record field:</p> <p>closureReason</p> <p>Indicates the reason for TCP/UDP closure. The following values are defined:</p> <ul style="list-style-type: none"> <li>■ TCP_FIN: (TCP flows only) MCC detects FIN bit set in TCP flags of TCP flow.</li> <li>■ TCP_RESET: (TCP flows only) MCC detects RESET bit set in TCP flags of TCP flow. A TCP flow can be reset either by UE or Server or by MCC (for example, due to redirection)</li> <li>■ IDLE_TIMEOUT: (TCP and UDP/ICMP flows) MCC detects the TCP/UDP flow is idle for configured time interval (no traffic seen on flow for idle timeout interval).</li> <li>■ SESSION_CLOSURE: (TCP and UDP/ICMP flows) MCC detects the UE session is terminated due to any reason. The EDR for all the active and idle TCP/UDP/ICMP flows of that session are populated with this closure reason.</li> <li>■ SESSION_MIGRATION: (TCP and UDP/ICMP flows) After recovery when on-going connection is migrated to Active SSM and existing connection on standby SSM times out. The connection on the standby SSM is populated with this closure reason.</li> <li>■ NO_RESOURCE: (TCP and UDP/ICMP flows) Connection is reclassified and resource allocation fails.</li> <li>■ BEARER_CLOSURE: (TCP and UDP/ICMP flows) Connection closed as a result of Bearer deletion.</li> </ul> <p>Use the following command to specify to send the statistics to the EDR server:</p> <pre>an3000-mcm-slot7cpu1(config)# service-construct edr-interface-profile &lt;name&gt; edr-service-profile &lt;name&gt; [session flow]-edr-profile record-info network-stats-info enabled</pre> |

| Feature                                                                                                              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Optimization and support for QUIC/UDP videos (similar to HTTP/HTTPS)</p> <p>See <a href="#">udpProxyInfo</a>.</p> | <p>This feature enables Operators to selectively apply rate limiting to Quick UDP Internet Connection (QUIC) protocol flows. This feature complies with Google's Quic software. Using this feature, UDP traffic with a configured destination network port is sent to the udp-proxy.</p> <p>This feature supports the following:</p> <ul style="list-style-type: none"> <li>■ Quic network domain detection</li> <li>■ Quic service rules</li> <li>■ Transparent and non-transparent modes</li> <li>■ Quic video rate limiting</li> <li>■ Quic statistics and KPIs</li> <li>■ Quic flow data records</li> </ul> <p>This feature includes the following configuration changes and new objects:</p> <ul style="list-style-type: none"> <li>■ Renamed <b>tcp-splicing-profile</b> to <b>traffic-control-profile</b>. <ul style="list-style-type: none"> <li>– The <b>action</b> field in the services http-proxy traffic-control profile object is hidden.</li> <li>– Added the <b>segment-threshold</b> field to the services http-proxy traffic-control-profile object. Configure the segment threshold in bytes for bypassing rate-limiting.</li> <li>– Added the <b>steering-group</b> field to the services http-proxy traffic-control-profile object. The steering-group allows the UDP flow to be steered to the destination configured within itself, when the profile is selected. In the packet processing, after UDP policies are evaluated, the associated traffic-control-profile is selected. The steering group within the profile can have multiple steering-destinations. A round-robin approach is used between destinations. In case of a single steering-destination, the current UDP flow is steered to that destination.</li> <li>– If the destination is a host-name, DNS lookup is invoked and the flow is diverted to the resolved IP address. This action causes MCC to issue the Profile - Steered to Host statistic or Profile - Steered to IP statistic.</li> </ul> </li> <li>■ Enhanced the domain-rule-group to support <b>url-list</b>. Each rule configured in the domain-rule-group can be associated with the url-list (although not mandatory). For example: <pre>an3000-mcm-slot8cpu1(config)# service-construct domain-rule-group DRG rule url domain-name url-list</pre> </li> <li>■ Enhanced the service-rule to match a domain-rule-group, along with others, while making a decision about service-rule matching. It supports matching packet-filter, time-of-day-rule-group, and session-rule-group, if configured. If configured in the udp-policy, the service-rule can not have http-rule-group and/or tls-rule-group configured.</li> </ul> |

| Feature                                                                                                                                                   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Optimization and support for QUIC/UDP videos (similar to HTTP/HTTPS)<br>(continued)                                                                       | <ul style="list-style-type: none"> <li>■ Added the following additional objects under services http-proxy: <ul style="list-style-type: none"> <li>– <b>udp idle-timeout</b>, <b>udp quic-reset-timeout</b>, and <b>udp quic-rate-limiting</b> objects</li> <li>– <b>udp-policy</b> object</li> <li>– <b>udp-policy</b> object under http-ip-flow</li> <li>– <b>enum</b> for udp to the protocol field</li> <li>– <b>http-ip-flow</b> under instance.</li> <li>– <b>tcp-traffic-control-profile</b> and <b>udp-traffic-control-profile</b> under services http-proxy instance</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Release 8.2.0.0 Features</b>                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Support for additional fields in SESSION Event Detail Record (EDR)<br>See <a href="#">EDR Record Format</a> and <a href="#">AI-vProbe Configuration</a> . | <p>In the SESSION EDR, added the following enumerations to the dataStatistics field:</p> <ul style="list-style-type: none"> <li>■ 9 = uplinkSessionActivityDuration</li> <li>■ 10 = downlinkSessionActivityDuration</li> <li>■ 11 = uplinkSessionPeakThroughput</li> <li>■ 12 = downlinkSessionPeakThroughput</li> <li>■ 13 = edrSessionBucket</li> </ul> <p>edrSessionBucket is a repeated message of data type EdrSessionBucketInfo. It contains the following subfields:</p> <ul style="list-style-type: none"> <li>– bucketName: (string) Name of the uplink/downlink bucket</li> <li>– min: (uint32) Bucket minimum value (in bytes) [default = 0]</li> <li>– max: (uint32) Bucket maximum value (in bytes) [default = 4294967295]</li> <li>– direction: (string) Bucket direction (uplink, downlink)</li> <li>– time: (uint32) Accumulated time in bucket</li> </ul> <p>Use the following command to enable EDR session volume measurement reporting. This profile contains configuration required for session-level data statistics record:</p> <pre>an3000-mcm-slot7cpu1(config)# service-construct edr-session-data-stats-profile &lt;name&gt;</pre> <p>Possible completions:</p> <pre>measurement-layer          TCP/IP layer to be used for bucket volume calculation  include-icmp                Include/exclude ICMP traffic in peak throughput and activity duration calculation  activity-duration-time-interval  Time interval to be used for session activity duration calculation in uplink/downlink direction  peak-throughput-time-interval    Time interval to be used for session peak throughput calculation in uplink/downlink direction</pre> |



| Feature                                                                           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Support for additional fields in SESSION Event Detail Record (EDR)<br>(Continued) | <p><code>bucket-volume-time-interval</code> Time interval to be used for session volume throughput calculation in uplink/downlink direction</p> <p>Use the following command to configure the EDR flow network stats profile. This profile contains configuration required for flow-level network-stats-info record:</p> <pre>an3000-mcm-slot7cpu1(config)# service-construct edr-flow-network-stats-profile &lt;name&gt;</pre> <p>Possible completions:</p> <p><code>measurement-layer</code> TCP/IP layer to be used for bucket volume calculation</p> <p><code>include-icmp</code> Include/exclude ICMP traffic in peak throughput and activity duration calculation</p> <p><code>activity-duration-time-interval</code> Time interval to be used for session activity duration calculation in uplink/downlink direction</p> <p><code>peak-throughput-time-interval</code> Time interval to be used for session peak throughput calculation in uplink/downlink direction</p> <p>Use the following commands to map the configured profiles to an EDR Interface Profile:</p> <pre>an3000-mcm-slot7cpu1(config)# service-construct edr-interface-profile &lt;name&gt; edr-session-data-stats-profile &lt;name&gt; edr-flow-network-stats-profile &lt;name&gt;</pre> <p><b>Notes:</b></p> <p>If <b>edr-session-data-stats-profile</b> is configured and mapped to <b>edr-interface-profile</b>, only session-related attributes are reported in EDR.</p> <p>If <b>edr-flow-network-stats-profile</b> is omitted from the <b>edr-interface-profile</b>, default values for the parameters are used to calculate the peak throughput and activity duration.</p> |

| Feature                                                                                                                                                                                                                                                   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Enhanced SESSION Attempt EDRs to provide the local/external SESSION setup failure reason and SESSION Stop EDRs to provide local/external normal/failure session close reason for generating new KPIs</p> <p>See <a href="#">EDR Record Format</a>.</p> | <p>Modified the EdrCloseInformation protobuf to add the localExternalCloseReason. This attribute is set in the SESSION Attempt EDRs for SESSION setup failures and in the SESSION Stop EDRs for session de-activations. It is used by the Mediation Service Function to generate new KPIs. The closeInitiator attribute is also filled to provide information on the originator for the SESSION close or SESSION setup failure:</p> <ul style="list-style-type: none"> <li>■ New localExternalCloseReason subfield added to the EdrCloseInformation data record: <ul style="list-style-type: none"> <li>– optional uint32 localExternalCloseReason = 3; // Session close or session setup failure due to local or external normal or failure reason</li> </ul> </li> <li>■ The values for the localExternalCloseReason can be: <ul style="list-style-type: none"> <li>CLOSE_REASON_INVALID = 0,</li> <li>LOCAL_CLOSE_NORMAL = 1,</li> <li>LOCAL_CLOSE_FAILURE = 2,</li> <li>EXTERNAL_CLOSE_NORMAL = 3,</li> <li>EXTERNAL_CLOSE_FAILURE = 4</li> </ul> </li> </ul> <p>Added MCC support for the following new APN and service-level performance statistics for validating the statistics derived on the Mediation Service Function:</p> <ul style="list-style-type: none"> <li>■ Number of session setup failures due to local failures</li> <li>■ Number of session setup failures due to external failures</li> <li>■ Number of session deactivations due to local failures</li> <li>■ Number of session deactivations due to external failures</li> </ul> <p>In the SESSION Event data records, added the following:</p> <ul style="list-style-type: none"> <li>■ CloseInitiator enum updates: <ul style="list-style-type: none"> <li>– 8 = CLOSE_INITIATOR_S6B</li> <li>– 9 = CLOSE_INITIATOR_DHCP</li> <li>– 255 = CLOSE_INITIATOR_INVALID</li> </ul> </li> </ul> <p>In the STATUS Event data records, added the following:</p> <ul style="list-style-type: none"> <li>■ Updated the minorVersion field of the StatusRecord field: <ul style="list-style-type: none"> <li>– For release 6.5 = 16</li> </ul> </li> </ul> |

| Feature                                                                                                  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Release 8.1.0.0 Features</b>                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Additional Event Data Record (EDR) fields                                                                | <p>Added support for the following extensions of the iEDR records:</p> <p>Flow EDR NetworkStatsInfo data record fields:</p> <ul style="list-style-type: none"> <li>■ Activity Duration DL</li> <li>■ Activity Duration UL</li> <li>■ Peak Throughput DL</li> <li>■ Peak Throughput UL</li> <li>■ Initial Uplink Round Trip Time</li> <li>■ Initial Downlink Round Trip Time</li> </ul> <p>See <a href="#">NetworkStatsInfo (page 4-24)</a>.</p> <p>Use the following command to specify to send the statistics to the EDR server:</p> <pre>an3000-mcm-slot7cpu1(config)# service-construct edr-interface-profile &lt;name&gt; edr-service-profile &lt;name&gt; flow-edr-profile record-info network-stats-info enabled</pre> |
| <b>Release 8.0.2.0 Features</b>                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <p>Added tethering fields to the FLOW Event Data Record (EDR)</p> <p>See <a href="#">FlowRecord</a>.</p> | <p>Added the following subfields to the dpiInfo field of the FLOW EDR:</p> <ul style="list-style-type: none"> <li>■ tetheredFlow: Indicates whether the flow is tethered.</li> <li>■ tetheringEntitled: Indicates whether the subscriber is entitled to perform tethering.</li> <li>■ tetheringAlgorithm: Specifies the tethering detection algorithm used to detect the flow as tethered. The following values are defined: <ul style="list-style-type: none"> <li>– 1 = ttl</li> <li>– 2 = userAgent</li> <li>– 3 = NA</li> </ul> </li> </ul>                                                                                                                                                                              |

| Feature                                                                                             | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Added field in HTDR (HTTP EDR) for entitlement information<br>See <a href="#">entitlementInfo</a> . | <p>This feature enables Operators to match DPI/flow record information to verify if non-entitled users are tethering. This feature supports both HTDRs and EDRs. All transactions, including TCP connections and HTTP requests, are recorded.</p> <p>In the HTTP Transaction Event Data Record, added entitlement-info field to the HTTPTransactionRecord field. The entitlement-info field contains the Entitlements field listing entitlement values.</p> <p>Operators enable/disable (default) this field as follows:</p> <pre>an3000-mcm-slot7cpu1 config# service-construct edr-interface-profile ... http-edr-profile record-info entitlement-info enabled</pre> <p>Operators enable/disable (default) this field in the HTDR template as follows:</p> <pre>an3000-mcm-slot7cpu1 config# service-construct data-record-template http-proxy http-transaction entitlement-information service-construct data-record-template http-proxy http-transaction http-tx entitlement-information present no entitlement-information position 16 entitlement-information format 0 entitlement-information length 0 entitlement-information alignment left entitlement-information padding " "</pre> |

| Feature                                                                                                                             | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                          |                            |                         |                              |                   |                            |                              |                                   |                            |                                                  |                        |                     |                       |                                          |                         |                                |
|-------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|----------------------------|-------------------------|------------------------------|-------------------|----------------------------|------------------------------|-----------------------------------|----------------------------|--------------------------------------------------|------------------------|---------------------|-----------------------|------------------------------------------|-------------------------|--------------------------------|
| <b>Release 7.2.0.0 Features</b>                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                          |                            |                         |                              |                   |                            |                              |                                   |                            |                                                  |                        |                     |                       |                                          |                         |                                |
| <p>Added PGW/GGSN support for sending Pilot Packets</p> <p>See <a href="#">Configuring UDP Server to Receive Pilot Packets</a>.</p> | <p>Support for sending Pilot Packets to multiple destinations on Session setup/tear-down, RAT-type change, and NAT address/port portion allocation/de-allocation. The Pilot Packet provides key information about a subscriber session, including Subscriber Identity (MSISDN, IMSI, NAI), Subscriber IP address, and Subscriber NAT IP address and port range. This information can be used to enable new services, such as Subscriber Analytics.</p> <p>A Pilot Packet contains a number of attributes in a string of type-length-values that are sent by UDP transport to multiple destinations.</p> <p>Use the following command to configure the UDP server to receive Pilot Packets in the type-length-value encoding format:</p> <pre><b>service-construct edr-server-profile &lt;name&gt;</b></pre> <p>Possible completions:</p> <table> <tr> <td><code>admin-state</code></td><td>Admin state of the profile</td></tr> <tr> <td><code>ip-address</code></td><td>IP address of the EDR server</td></tr> <tr> <td><code>port</code></td><td>TCP port of the EDR server</td></tr> <tr> <td><code>network-context</code></td><td>Network context of the EDR server</td></tr> <tr> <td><code>edr-source-ip</code></td><td>Source IP address to use to reach the EDR server</td></tr> <tr> <td><code>transport</code></td><td>Transport type: udp</td></tr> <tr> <td><code>encoding</code></td><td>Encoding format: tlv (type-length-value)</td></tr> <tr> <td><code>keep-alive</code></td><td>EDR server keep alive interval</td></tr> </table> | <code>admin-state</code> | Admin state of the profile | <code>ip-address</code> | IP address of the EDR server | <code>port</code> | TCP port of the EDR server | <code>network-context</code> | Network context of the EDR server | <code>edr-source-ip</code> | Source IP address to use to reach the EDR server | <code>transport</code> | Transport type: udp | <code>encoding</code> | Encoding format: tlv (type-length-value) | <code>keep-alive</code> | EDR server keep alive interval |
| <code>admin-state</code>                                                                                                            | Admin state of the profile                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                          |                            |                         |                              |                   |                            |                              |                                   |                            |                                                  |                        |                     |                       |                                          |                         |                                |
| <code>ip-address</code>                                                                                                             | IP address of the EDR server                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                          |                            |                         |                              |                   |                            |                              |                                   |                            |                                                  |                        |                     |                       |                                          |                         |                                |
| <code>port</code>                                                                                                                   | TCP port of the EDR server                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                          |                            |                         |                              |                   |                            |                              |                                   |                            |                                                  |                        |                     |                       |                                          |                         |                                |
| <code>network-context</code>                                                                                                        | Network context of the EDR server                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                          |                            |                         |                              |                   |                            |                              |                                   |                            |                                                  |                        |                     |                       |                                          |                         |                                |
| <code>edr-source-ip</code>                                                                                                          | Source IP address to use to reach the EDR server                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                          |                            |                         |                              |                   |                            |                              |                                   |                            |                                                  |                        |                     |                       |                                          |                         |                                |
| <code>transport</code>                                                                                                              | Transport type: udp                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                          |                            |                         |                              |                   |                            |                              |                                   |                            |                                                  |                        |                     |                       |                                          |                         |                                |
| <code>encoding</code>                                                                                                               | Encoding format: tlv (type-length-value)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                          |                            |                         |                              |                   |                            |                              |                                   |                            |                                                  |                        |                     |                       |                                          |                         |                                |
| <code>keep-alive</code>                                                                                                             | EDR server keep alive interval                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                          |                            |                         |                              |                   |                            |                              |                                   |                            |                                                  |                        |                     |                       |                                          |                         |                                |

| Feature                                                               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Added PGW/GGSN support for sending Pilot Packets ( <i>Continued</i> ) | <p>Use the following command to configure the edr-interface-profile to enable/disable the Pilot Packet subtypes. For the primary Pilot Packet server, transport must be set to UDP and encoding must be set to TLV.</p> <pre><b>service-construct edr-interface-profile &lt;name&gt; edr-service-profile &lt;name&gt;</b></pre> <p>Possible completions:</p> <pre>admin-state      Admin state of the EDR service profile primary-server   Primary pilot packet server</pre> <p><b>Note:</b> Because the Pilot Packets are sent by connectionless UDP, the secondary-server is not used, even if configured in the edr-service-profile.</p> <p>Use the following command to configure the pilot-edr-interface-profile and specify the Pilot EDR record type (START, STOP, NAT Binding, RAT Type Change) to send to the UDP server.</p> <pre><b>service-construct edr-interface-profile &lt;name&gt; edr-service-profile &lt;name&gt; pilot-edr-profile</b></pre> <pre>admin-state      Admin state of the pilot EDR profile record-type start-records Enable/Disable Start records record-type stop-records Enable/Disable Stop records record-type nat-binding-records Enable/Disable NAT Binding records record-type rat-change-records Enable/Disable RAT Change records</pre> <p>If desired, enable/disable the optional record information to send to the UDP server:</p> <pre>record-info serving-network-info Enable/Disable serving network info record-info subscriber-info Enable/Disable subscriber info</pre> <p>Use the following command to associate the EDR Interface Profile with a Workflow Control Profile:</p> <pre><b>workflow control-profile &lt;name&gt; edr-interface &lt;name&gt;</b></pre> <p>APN-level statistics are available that count the number of session-start, session-stop, and rat-type-change Pilot Packet records generated on a per-APN level:</p> <pre><b>show zone default gateway statistics cluster-level call-performance apn</b></pre> <pre>■ num-session-start-pilot-packets-sent ■ num-session-stop-pilot-packets-sent ■ num-rat-change-pilot-packets-sent</pre> <p>Added statistics of the total number of Pilot Packet records generated for the cluster:</p> <pre><b>show infrastructure edr edr-pilot-stats</b></pre> <p>Added server statistics and statistics of the number of start/stop/rat-type-change/nat-binding records generated for each client and EDR Record type:</p> <pre><b>show cluster node edr-server-statistics show cluster node edr-pilot-client-statistics</b></pre> |

| Feature                                                                                                                   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>ABR Rate Limiting based on Toll Free Data Sponsored Website Manifest</p> <p>See <a href="#">EDR Record Format</a>.</p> | <p>Added support to:</p> <ul style="list-style-type: none"> <li>■ Apply rate limiting based on Max-Rate field in the Content Provider (CP) Manifest, when present</li> <li>■ Bypass rate limiting when no Max-Rate field/ invalid max-rate is present in the CP Manifest</li> </ul> <p>Max-Rate field takes the highest preference for Rate Limiting over existing Opt-In/ Opt-Out logic.</p> <p><b>Prerequisite:</b> Toll-Free Data functionality (toll-free-profile) must be configured for CP Manifest rate limiting to be operational.</p> <p>Added the following statistics to the ABRRATELIMITINGVIDEOSEGMENTSTATS statistics table:</p> <ul style="list-style-type: none"> <li>■ Video Segment – CP Manifest with Max-Rate</li> <li>■ Video Segment Bypass – CP Manifest with no Max-Rate</li> <li>■ Video Segment Bypass – CP Manifest with Invalid Max-Rate</li> </ul> <p>Added the following statistics to the TCPSPLICINGRATELIMITINGSTATS statistics table:</p> <ul style="list-style-type: none"> <li>■ Rate Limiting Summary – CP Manifest with Max-Rate</li> <li>■ Rate Limiting Bypasses – CP Manifest with no Max-Rate</li> <li>■ Rate Limiting Bypasses – CP Manifest with Invalid Max-Rate</li> </ul> <p>In the HTTP Transaction data records (HTDRs), added the following:</p> <ul style="list-style-type: none"> <li>■ Manifest Information field. It contains the following subfields: <ul style="list-style-type: none"> <li>– manifestMatched – Indicates whether the Content Provider (CP) Manifest is matched</li> <li>– manifestMaxRateMax – Rate received from Manifest (kbps)</li> </ul> </li> <li>■ Max Rate Based Rate Limiting Applied Reason Video Information subfield to record the number of requests going through Rate Limiting using Max-Rate in the CP Manifest. It contains the following enumerations: <ul style="list-style-type: none"> <li>– 0 = CP_MANIFEST_MAXRATE_HTTP_PROXY</li> <li>– 1 = CP_MANIFEST_MAXRATE_TCP_PROXY</li> </ul> </li> <li>■ Enumerations to the ABR Rate Limiting Bypass Reason subfield (within Video Information field): <ul style="list-style-type: none"> <li>– 5 = ABR_CANCEL_TOOSHORT</li> <li>– 6 = ABR_CP_MANIFEST_NO_MAX_RATE</li> <li>– 7 = ABR_CP_MANIFEST_INVALID_MAX_RATE</li> </ul> </li> <li>■ Enumerations to the Rate Limiting Bypass Cancel Reason (within TCP Splicing Information field): <ul style="list-style-type: none"> <li>– 4 = Bypass_No_MaxRate_in_CP_Manifest</li> <li>– 5 = Bypass_Invalid_MaxRate_in_CP_Manifest</li> </ul> </li> </ul> |

| Feature                                                                                                                                       | Description                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Added vProbe support for refererUrl Request Information subfield in the HTDR Transaction iEDRs<br><br>See <a href="#">EDR Record Format</a> . | In the HTTP Transaction data records (HTDRs), added the refererUrl subfield (within Request Information field), which is a percent-encoded URL string that refers to the original domain that originated a particular request.                                                                                                                                                 |
| Mediation Server support for conversion of Real Time Troubleshooting (RTT) EDRs to CSV files                                                  | Mediation Server supports conversion of Real Time Troubleshooting (RTT) EDRs to CSV files. The Operator configures which EDR fields to map to the CSV file and which fields to ignore.<br><br>The Mediation Server can maintain a minimum of 4 hours of CSV data, in case of connectivity failure from the MCC to the Mediation Server. CSV files are closed every 60 seconds. |



| Feature                                                                                                                                                 | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Added support for managing when a subscriber vTap is started and stopped</p> <p>See <a href="#">Configuring Subscriber Data Plane Mirroring</a>.</p> | <p>Added support for managing when a subscriber vTap is started and stopped. The commands require the <b>requires-external-activation</b> of the subscriber vTap to be enabled:</p> <pre>an3000-mcm-slot7cpu1(config)# vtap subscriber-tap &lt;name&gt; admin-state [enabled disabled] requires-external-activation [yes no] subscriber-filter-set &lt;name&gt; reference-point [access core intermediate] sink &lt;name&gt;</pre> <p>The <b>admin-state</b> of a subscriber vTap must be enabled for the <b>requires-external-activation</b> setting to take effect.</p> <p>Use the following command to configure the vTap to be started at a future time. Either specify a start time and end time or specify a start time and a duration, in seconds:</p> <pre>an3000-mcm-slot7cpu1(config)# vtap start &lt;name&gt; start-time &lt;start time&gt; end-time &lt;end time&gt; duration &lt;seconds&gt;</pre> <p>Specify the times in the format yyyy-mm-ddThh:mm:ss. For example, 2017-02-08T10:10:10.</p> <p>As an alternative to <b>end-time</b>, specify the duration to run the subscriber vTap, in seconds from <b>start-time</b>.</p> <p>Use the following command to start the subscriber vTap immediately:</p> <pre>an3000-mcm-slot7cpu1(config)# vtap start &lt;name&gt;</pre> <p>Use the following command to stop the subscriber vTap immediately or disable a scheduled subscriber vTap:</p> <pre>an3000-mcm-slot7cpu1(config)# vtap stop &lt;name&gt;</pre> <p>Use the following command to view the <b>start-time</b> and <b>end-time</b> of scheduled subscriber vTaps:</p> <pre>an3000-mcm-slot7cpu1(config)# show vtap subscriber-tap-schedule</pre> <pre>vtap subscriber-tap-schedule VTAP SUBSCRIBER TAP NAME      STARTTIME      ENDTIME      DURATION  VTAPSTATE ----- tap1          2017-03-01T23:02:36  2017-03-01T23:04:36  0         Running tap2          2017-03-01T23:05:36  2017-03-01T23:08:36  0         Scheduled tap3          2017-03-01T23:09:36  2017-03-01T23:12:36  0         Idle tap4          2017-03-01T23:13:36  2017-03-01T23:16:36  0         Idle</pre> |

| Feature                                                                                                                    | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Added support for Real Time Troubleshooting (RTT) EDRs<br>See <a href="#">Configuring Real Time Troubleshooting EDRs</a> . | <p>Added support for the Real Time Troubleshooting (RTT) EDRs on the SGW, PGW, ePDG, and WAG. The EDR records provide visibility into the network and subscriber events to help diagnose networking and subscriber issues.</p> <p>RTT EDR supports the following features:</p> <ul style="list-style-type: none"> <li>■ Generates real time event records for: <ul style="list-style-type: none"> <li>– GTP events over the S4/S5/S8/S11/S2a/S2b/Gn/Gp interfaces</li> <li>– Diameter events over Gx/Gy/S6b/SWm interfaces</li> <li>– IKE events over the SWu interface</li> </ul> </li> <li>■ Supports enabling/disabling RTT EDRs on a per-interface basis.</li> <li>■ Supports multiple simultaneous EDR servers, each of which can sample different records or overlapping data.</li> <li>■ Supports redundant (primary and secondary) TCP connections from MCC to the EDR Server, with keepalives for failover.</li> <li>■ Maintains EDR statistics per server for the number of EDR send successes/failures for each interface.</li> <li>■ Provides connection statistics for all connections to the EDR Servers.</li> </ul> |
| <b>Release 7.0.0.0 Features</b>                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| AI-vProbe support for standalone SGW deployments                                                                           | Added EDR support for standalone SGW deployments. The EDRs contain only the data record fields meaningful on the SGW.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Release 6.5.0.0 Features</b>                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| AI-vProbe support for streaming EDRs to a standalone Mediation Server over TCP                                             | <p>AI-vProbe streams EDRs to a standalone Mediation Server over TCP. The Mediation Server converts the files into CSV file format.</p> <p>Mediation Server supports the following features:</p> <ul style="list-style-type: none"> <li>■ Receives, processes, and store EDR streams from the AI-vProbe</li> <li>■ Allows Redundant (primary and secondary) TCP connections from MCC to Mediation Server</li> <li>■ Supports configuration of CSV file generation in Mediation Server. The Operator configures which EDR fields to map to the CSV file and which fields to ignore.</li> </ul> <p>In addition, MCC supports configuration of different subsets of EDR data to send to separate destinations.</p>                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Added AI-vProbe function for subscriber traffic mirroring                                                                  | <p>Added support for configuring a subscriber vTap to replicate subscriber traffic for mirroring. The subscriber vTap configures:</p> <ul style="list-style-type: none"> <li>■ One or more associated subscriber filter(s) that define the set of subscribers whose traffic is to be replicated. With multiple filters, a packet is considered to match the vTap object if it belongs to one of the sessions that match the configured filters.</li> <li>■ One or more associated sink(s) that define whether to stream the replicated packets or log them to a file. For multiple sinks, the matching packet is replicated and sent to each sink.</li> <li>■ One or more associated reference point(s) that specify where to take a snapshot of the received subscriber packet. With multiple reference points, a copy of the packet is taken and sent out for each reference point.</li> </ul>                                                                                                                                                                                                                                   |

| Feature                                                               | Description                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AI-vProbe support for conversion of HTTP EDRs to syslog event records | AI-vProbe/Mediation Server supports the following features: <ul style="list-style-type: none"><li>■ Conversion of HTTP EDRs to syslog event records</li><li>■ Configuration of CSV file generation in Mediation Server. The Operator configures which EDR fields to map to the CSV file and which fields to ignore.</li></ul>                                                             |
| AI-vProbe EDR enhancements                                            | Added support in the Flow EDR Profile to enable sending the following record information to the EDR server: <ul style="list-style-type: none"><li>■ dpi-info: Enable/Disable flow deep packet inspection information</li><li>■ subscriber-info: Enable/Disable flow subscriber information</li><li>■ tcp-retransmission-info: Enable/Disable flow TCP retransmission statistics</li></ul> |

## Conventions

This guide uses the following conventions, when applicable:

| Description                                                                                | Convention                                   | Examples                                                                                                   |
|--------------------------------------------------------------------------------------------|----------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Command names, keywords, dialog boxes, buttons, icons, menus                               | <b>Times New Roman bold font</b>             | Use <b>configure_ha</b> to configure...<br>Enter <b>y</b> .<br>To continue, press <b>Enter</b> .           |
| Selecting a menu item                                                                      | <b>Menu =&gt; Option</b>                     | Select <b>Security =&gt; Group Management</b> .                                                            |
| Variable parameters for which you supply values                                            | <b>&lt;courier bold italic blue font&gt;</b> | <b>configure_dr convert slave --masterhost &lt;host IP&gt;</b>                                             |
| User input                                                                                 | <b>Courier bold blue font</b>                | <b>cd /opt/Affirmed/NMS/bin</b>                                                                            |
| Screen messages, system output, sample source code                                         | Courier regular blue font                    | Checking for License...                                                                                    |
| Required keywords and arguments in square brackets                                         | <b>[ ]</b>                                   | <b>chassis [chassis] clearcrashes</b>                                                                      |
| Optional keywords and arguments in curly brackets                                          | <b>{ }</b>                                   | <b>{cdr-file-name}</b>                                                                                     |
| Keywords separated by the pipe symbol. Requires you to specify only one item from the set. | <b> </b>                                     | Is this correct? [y n]                                                                                     |
| Keywords representing a single element in angle brackets                                   | <b>&lt; &gt;</b>                             | <b>&lt;host name&gt;</b>                                                                                   |
| Asterisk after a CLI object indicates a required object                                    | <b>*</b>                                     | <b>name*</b>                                                                                               |
| Variables, book and chapter titles, file path names, new terms, and emphasized text        | <i>Times New Roman italic font</i>           | See the <i>Acuitas User's Guide</i> .<br><i>Workflow</i> is an application running on the network element. |



**ESD:** Static sensitive equipment. Wear a properly grounded wrist strap.



**CAUTION:** Proceed carefully to avoid possible equipment damage or data loss.



---

**WARNING:** Proceed carefully to avoid possible personal injury.

---

---

**Note:** Additional information that may apply to the subject text.

---

---

**Tip:** Helpful suggestions.

---

## How to Obtain Product Documentation

Documentation is available for currently supported product releases. Documentation is available online in Adobe PDF format. You can view PDFs online using the latest version of Adobe Reader<sup>®</sup>. To download the latest version of the Adobe Reader software from the Adobe web site, go to <http://get.adobe.com/reader/>

You can access Affirmed Networks documentation from the SFTP server. Contact Affirmed Networks for instructions on how to access the SFTP server.

See the *Affirmed Networks Documentation Library* for a complete list of Affirmed Networks product documentation.

## Technical Assistance Center

Affirmed Networks Technical Assistance Center (TAC) provides technical support 24 hours a day, 7 days a week (24 x 7) for any customer with an active maintenance contract.

To contact the TAC:



E-mail: [prodsupport@affirmednetworks.com](mailto:prodsupport@affirmednetworks.com)



24 x 7 Telephone support:

Local: 1-978-268-0871

Toll free: 1-888-283-2838



## Affirmed Active Intelligent AI-vProbe Overview

Affirmed Active Intelligent vProbe (AI-vProbe) is an intelligent tool that provides the following features:

- **Event Data Record (EDR) Generation:** Data reporting mechanism that provides timely activity and statistics information about PDN sessions and subscriber requests. EDRs provide visibility into session, bearer, flow (data plane connection), and transaction state. See [Event Data Record \(EDR\) Generation](#).
- **Control Plane Traffic Mirroring:** Protocol logging with a GRE-based sink to replicate control plane traffic. vProbe adds a GRE header and sends the encapsulated packet to a remote destination. See [Control Plane Traffic Mirroring](#).
- **Data Plane Traffic Mirroring:** Subscriber data plane traffic mirroring. A configure vTap object replicates subscriber traffic and either streams the replicated packets or logs them to a file. See [Data Plane Traffic Mirroring](#).

---

**Note:** vProbe **is not** supported on subscriber-unaware GiGW.  
vProbe **is** supported on subscriber-aware GiGW.

---

## Event Data Record (EDR) Generation

The information contained in the EDRs can be used to isolate network issues and to derive traffic analytics. The information provides network intelligence for:

- Troubleshooting and proactive fault detection: Operators can isolate specific failures in the network.
- Security analysis: Operators can discover and isolate security concerns.
- Market analysis: Operators can derive traffic and usage analytics.
- Network capacity planning

The AI-vProbe generates EDRs from various MCC software components and delivers them to multiple external EDR servers as a serialized protocol buffer stream over multiple TCP connections.

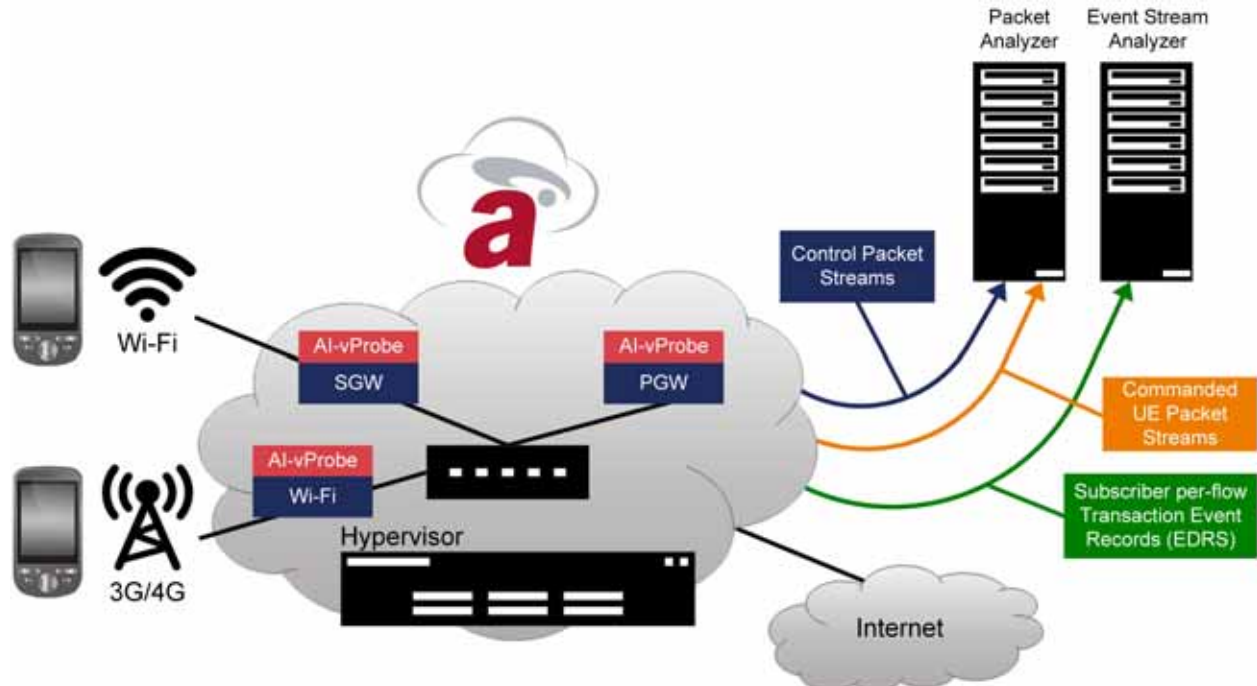
The AI-vProbe functions as a self-probe that allows Operators to tap into both standards-based interfaces (such as Gn and Gi) and internal interfaces. This architecture allows each network element to be programmed to act as a self-probe.

The EDR records supported provide activity data related to PDN sessions, the bearers contained within those sessions, individual data flows established within a bearer, and the various transactions initiated by a subscriber on those flows.

Each EDR contains a set of data fields that can be individually enabled, allowing the Operator to specify which fields and which records to be delivered to the EDR server. The EDR server is an external application where the event records can be processed directly. An example of such an application is the Affirmed Mediation Service Function (MSF), which receives and processes EDRs from the AI-vProbe. For more information, see the *Affirmed Networks Mediation Service Function Administration Guide*.

The following figure illustrates the position and role of the AI-vProbe with the MCC in a network deployment.

**Figure 1-1 AI-vProbe Network Deployment**





## EDR Record Types

The AI-vProbe supports the following EDR record types:

- **SESSION Record** — Similar to Call Detail Records (CDRs), SESSION EDRs convey information related to the PDN session lifecycle. Supported SESSION EDRs include:

- **START**: generated when a PDN session is established
- **STOP**: generated when an established PDN session is terminated
- **ATTEMPT**: generated when a PDN session fails to establish
- **INTERIM**: generated when an event occurs on an established session

When enabled, the INTERIM EDRs are generated for all of the following triggers:

- qos-change
- serving-node-change
- serving-node-plmn-change
- user-location-change
- user-csg-change
- rat-type-change
- ue-time-zone-change
- time-limit (optional)

SESSION EDRs are generated by the TOS Gateway application.

- **BEARER Record** — BEARER EDRs convey information for each bearer within an MCC session. Supported BEARER EDR types include:

- **START** – generated when a bearer is established within a session
- **STOP** – generated when a bearer is deleted
- **ATTEMPT** – generated when a bearer fails to establish
- **INTERIM** – generated when some ‘meaningful’ event occurs on an established bearer

When enabled, the INTERIM EDRs are generated for all of the following triggers:

- qos-change
- serving-node-change
- serving-node-plmn-change
- user-location-change
- user-csg-change
- rat-type-change
- ue-time-zone-change

— time-limit (optional)

BEARER EDRs are generated by the TOS Gateway application.

- **FLOW Record** — FLOW EDRs convey information related to individual data flows within the MCC. Supported FLOW EDR types include:
  - **START** – generated when a flow is created within the workflow. For short-lived flows, only a STOP EDR is generated, but contains information from the START EDR, to avoid multiple, separate records.
  - **STOP** – generated when a workflow connection is deleted, specifying cumulative statistics for that flow.
  - **INTERIM** – for long-lived flows, an interim EDR may be generated to provide near real-time statistics for a flow
- **HTTP Transaction Record** — HTTP transaction EDRs contain information from the existing HTDRs and are generated by the HTTP proxy when an HTTP transaction completes.
- **Pilot Record** — Pilot Packets provides key information about a subscriber session, including Subscriber Identity (MSISDN, IMSI, NAI), Subscriber IP address, and Subscriber NAT IP address and port range. This information can be used to enable new services, such as Subscriber Analytics. Supported Pilot EDR types include:
  - **START** – generated on session setup
  - **STOP** – generated on session tear-down.
  - **NAT Binding** – generated when a NAT address/port portion allocation/de-allocation occurs
  - **RAT Type Change** – generated when a RAT Type change occurs
- **Real Time Troubleshooting (RTT) Record** — The RTT EDR records provide visibility into the network and subscriber events to help diagnose networking and subscriber issues. Supported RTT EDR types include:
  - EDR\_RTT\_GTP\_S5\_RECORD
  - EDR\_RTT\_GTP\_S8\_RECORD
  - EDR\_RTT\_GTP\_S11\_RECORD
  - EDR\_RTT\_GTP\_S4\_RECORD
  - EDR\_RTT\_GTP\_S2A\_RECORD
  - EDR\_RTT\_GTP\_S2B\_RECORD
  - EDR\_RTT\_GTP\_GN\_RECORD
  - EDR\_RTT\_GTP\_GP\_RECORD
  - EDR\_RTT\_DIAMETER\_GX\_RECORD
  - EDR\_RTT\_DIAMETER\_GY\_RECORD
  - EDR\_RTT\_DIAMETER\_S6B\_RECORD

- EDR\_RTT\_DIAMETER\_SWM\_RECORD
- EDR\_RTT\_IKE\_SWU\_RECORD

RTT EDRs are generated by the TOS Gateway application.

## EDR Server Application Redundancy

For SESSION, BEARER, FLOW, and HTTP Transaction EDRs, MCC supports redundant EDR server applications. MCC establishes concurrent connections to a primary and a secondary EDR server application, allowing the MCC to immediately switch from primary to secondary if the primary connection fails. The primary connection is preferred, if available.

For each EDR record to be sent to a primary/secondary pair, if the connection to the primary is in the established state, MCC sends the record to the primary. If the primary connection fails, the MCC sends the record to the secondary.

If the **edr-server-profile keep-alive** value is non-zero, MCC uses EDR status records to elicit a timely reply from the server. Upon not receiving several replies in a row, the MCC closes the connection and attempts to reconnect. In this scenario, the connection is no longer in the established state, so the traffic is sent to the secondary.

The critical `anEdrServerConnectionFailed` alarm is generated when a connection with the EDR server fails. The alarm is cleared when the connection is reestablished.

## EDR Interface Configuration on Geographic Redundancy Clusters

If Geographic Redundancy is configured, you must provision a separate loopback IP address on each Geographic Redundancy cluster for the EDR interface. In this approach, the loopback IP address is private to each cluster and is not associated with a Geographic Redundancy Group.

For the loopback used by the EDR interface, you must configure the same name on the two redundant clusters, with each cluster configured with a different loopback IP address.

## Control Plane Traffic Mirroring

The AI-vProbe supports protocol logging with a GRE-based sink to replicate control plane traffic. When this protocol log is configured, instead of logging all matching packets to a file, vProbe adds a GRE header and sends the encapsulated packet to a remote destination.

## **Data Plane Traffic Mirroring**

The AI-vProbe supports subscriber data plane traffic mirroring. Operators can configure a subscriber vTap to replicate subscriber traffic for monitoring purposes. The subscriber vTap configures:

- One or more associated subscriber filter(s) that define the set of subscribers whose traffic is to be replicated. With multiple filters, a packet is considered to match the vTap object if it belongs to one of the sessions that match the configured filters.
- One or more associated sink(s) that define whether to stream the replicated packets or log them to a file. For multiple sinks, the matching packet is replicated and sent to each sink.
- One or more associated reference point(s) that specify where to take a snapshot of the received subscriber packet. With multiple reference points, a copy of the packet is taken and sent out for each reference point.

## AI-vProbe Configuration

This chapter provides information on configuring AI-vProbe to generate EDRs. See the *Affirmed Networks CLI Reference Guide* for a complete listing of CLI commands used in this section.

The following topics describe how to configure AI-vProbe to generate EDRs:

- [Generating SESSION, BEARER, FLOW, and HTTP Transaction EDRs](#)
- [Configuring Subscriber Data Plane Mirroring](#)
- [Configuring Control Plane Packet Mirroring](#)
- [Configuring UDP Server to Receive Pilot Packets](#)
- [Configuring Real Time Troubleshooting EDRs](#)

### Generating SESSION, BEARER, FLOW, and HTTP Transaction EDRs

This section describes how to configure the system to generate SESSION, BEARER, FLOW, and HTTP Transaction EDRs.

**Use the following sequence to configure the system to generate SESSION, BEARER, FLOW, and HTTP Transaction EDRs:**

- 1 Create an EDR Server Profile (maximum 8).  
`an3000-mcm-slot7cpu1(config)# service-construct edr-server-profile <name>`

The following table describes the configurable parameters.

| Parameter                | Description                                                                                                                        |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <code>admin-state</code> | Admin state of the profile: <ul style="list-style-type: none"><li>■ <code>enabled</code></li><li>■ <code>disabled</code></li></ul> |

| Parameter               | Description                                                               |
|-------------------------|---------------------------------------------------------------------------|
| <b>ip-address</b>       | IP address of the EDR server                                              |
| <b>port</b>             | TCP port of the EDR server                                                |
| <b>network-context</b>  | Network context of the EDR server                                         |
| <b>edr-source-ip</b>    | Source IP address to use to reach the EDR server                          |
| <b>transport</b>        | Transport type:<br>■ <b>tcp</b>                                           |
| <b>encoding</b>         | Encoding format:<br>■ <b>protobuf</b><br>■ <b>tlv</b> (type-length-value) |
| <b>keep-alive</b>       | EDR server keep alive interval                                            |
| <b>tcp-user-timeout</b> | Number of seconds for TCP user data retransmission timeout                |

- 2 If desired, enable EDR session volume measurement reporting. This profile contains configuration required for session-level data statistics record.

```
an3000-mcm-slot7cpu1(config)# service-construct  
edr-session-data-stats-profile <name>
```

The following table describes the configurable parameters.

| Parameter                              | Description                                                                                                                                                                                                                   |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>measurement-layer</b>               | TCP/IP layer to be used for bucket volume calculation:<br>■ <b>layer3</b> : counts the entire IP datagram<br>■ <b>layer4</b> : excludes the IP header<br>■ <b>layer7</b> : excludes the IP header and the UDP/TCP/SCTP header |
| <b>include-icmp</b>                    | Include/exclude ICMP traffic in peak throughput and activity duration calculation:<br>■ <b>true</b><br>■ <b>false</b>                                                                                                         |
| <b>activity-duration-time-interval</b> | Time interval to be used for session activity duration calculation in uplink/downlink direction                                                                                                                               |
| <b>peak-throughput-time-interval</b>   | Time interval to be used for session peak throughput calculation in uplink/downlink direction                                                                                                                                 |
| <b>bucket-volume-time-interval</b>     | Time interval to be used for session volume throughput calculation in uplink/downlink direction                                                                                                                               |

- 3 If desired, configure the EDR flow network stats profile. This profile contains configuration required for flow-level network-stats-info record.

```
an3000-mcm-slot7cpu1(config)# service-construct  
edr-flow-network-stats-profile <name>
```

The following table describes the configurable parameters.

| Parameter                              | Description                                                                                                                                                                                                                                                                           |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>measurement-layer</b>               | TCP/IP layer to be used for bucket volume calculation: <ul style="list-style-type: none"> <li>■ <b>layer3</b>: counts the entire IP datagram</li> <li>■ <b>layer4</b>: excludes the IP header</li> <li>■ <b>layer7</b>: excludes the IP header and the UDP/TCP/SCTP header</li> </ul> |
| <b>include-icmp</b>                    | Include/exclude ICMP traffic in peak throughput and activity duration calculation: <ul style="list-style-type: none"> <li>■ <b>true</b></li> <li>■ <b>false</b></li> </ul>                                                                                                            |
| <b>activity-duration-time-interval</b> | Time interval to be used for session activity duration calculation in uplink/downlink direction                                                                                                                                                                                       |
| <b>peak-throughput-time-interval</b>   | Time interval to be used for session peak throughput calculation in uplink/downlink direction                                                                                                                                                                                         |

- 4 Configure the EDR Interface Profile (maximum 256).

```
an3000-mcm-slot7cpu1(config)# service-construct edr-interface-profile  
<name>
```

The following table describes the configurable parameters.

| Parameter                          | Description                                                                                                               |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>admin-state</b>                 | Admin state of the profile: <ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul> |
| <b>session-statistics-interval</b> | Time interval, in seconds, between Session and Bearer Interim event data records. Default 0 (disabled)                    |
| <b>flow-statistics-interval</b>    | Time interval, in seconds, between Flow Interim event data records. Default: 600 (enabled)                                |
| <b>flow-min-lifetime</b>           | Number of seconds to wait before generating initial Flow event data record following flow creation. Default: 60 (enabled) |

| Parameter                         | Description                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>max-url-length</b>             | <p>Length to which the URL field should be truncated within the HTTP clientUrl, serverUrl, and/or refererURL parameters to reduce EDR size</p> <p>You must explicitly configure the max-url-length parameter to truncate the URL.</p> <p>The command truncates the record length to the configured value but maintains the last 5 characters of the record (...RPT0=) to retain some file type information visibility.</p> |
| <b>session-data-stats-profile</b> | <p>EDR session profile to be applied on attributes reported via session data statistics.</p> <p>If edr-session-data-stats-profile is configured and mapped to edr-interface-profile, only session-related attributes are reported in EDR</p>                                                                                                                                                                               |
| <b>flow-network-stats-profile</b> | <p>EDR session profile to be applied on attributes reported via flow network statistics.</p> <p>If no edr-flow-network-stats-profile is mapped to edr-interface-profile, default values for the parameters are used to calculate the peak throughput and activity duration.</p>                                                                                                                                            |

- 5 Configure the EDR Service Profile (maximum 8).

```
an3000-mcm-slot7cpu1(config)# service-construct edr-interface-profile
<name> edr-service-profile <name>
```

The following table describes the configurable parameters.

| Parameter               | Description                                                                                                                      |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>admin-state</b>      | <p>Admin state of the profile:</p> <ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul> |
| <b>primary-server</b>   | Primary EDR server for this EDR service                                                                                          |
| <b>secondary-server</b> | Secondary EDR server for this EDR service                                                                                        |

- 6 Configure the EDR Profile(s) and specify the EDR record types (SESSION, BEARER, FLOW, HTTP Transaction) and records (START, STOP, INTERIM, ATTEMPT) to send to the EDR server.

- a Configure the EDR record types:

```
an3000-mcm-slot7cpu1(config)# service-construct
edr-interface-profile <name> edr-service-profile <name>
[session|bearer|flow|http]-edr-profile
```

where [session|bearer|flow|http] specifies the record type.



The following table describes the configurable parameters.

| Parameter          | Description                                                                                                               |
|--------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>admin-state</b> | Admin state of the profile: <ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul> |

- b** For SESSION, BEARER, and FLOW EDR records, specify which record types to send to the EDR server. (*Skip this step for HTTP Transaction EDR records*):

```
an3000-mcm-slot7cpu1(config)# service-construct
edr-interface-profile <name> edr-service-profile <name>
[session|bearer|flow|http]-edr-profile record-type
[start|stop|attempt|interim]
```

The following table describes the configurable parameters.

| Parameter              | Description                                                                                                                                                                             |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>attempt-records</b> | Enable/Disable ATTEMPT records: <ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul> <i>Applies to SESSION and BEARER records only.</i>        |
| <b>interim-records</b> | Enable/Disable INTERIM records: <ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul> <i>Applies to SESSION, BEARER, and FLOW records only.</i> |
| <b>start-records</b>   | Enable/Disable START records: <ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>– <b>disabled</b></li> </ul> <i>Applies to SESSION, BEARER, and FLOW records only.</i>   |
| <b>stop-records</b>    | Enable/Disable STOP records: <ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul> <i>Applies to SESSION, BEARER, and FLOW records only.</i>    |

- c** Specify the record information to send to the EDR server:

```
an3000-mcm-slot7cpu1(config)# service-construct
edr-interface-profile <name> edr-service-profile <name>
[session|bearer|flow|http]-edr-profile record-info <field name>
```

where **<field name>** varies by EDR profile type.

The following table describes the configurable parameters.

| Parameter                      | Description                                                                                                                                              |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Bearer EDR Profile:</b>     |                                                                                                                                                          |
| <b>bearer-info</b>             | Enable/Disable bearer information:<br><ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>                      |
| <b>bearer-qos-information</b>  | Enable/Disable bearer QoS information:<br><ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>                  |
| <b>data-statistics</b>         | Enable/Disable data statistics:<br><ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>                         |
| <b>edr-cause-codes</b>         | Enable/Disable EDR cause codes:<br><ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>                         |
| <b>Flow EDR Profile:</b>       |                                                                                                                                                          |
| <b>data-statistics</b>         | Enable/Disable data statistics:<br><ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>                         |
| <b>dpi-info</b>                | Enable/Disable flow deep packet inspection information:<br><ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul> |
| <b>subscriber-info</b>         | Enable/Disable flow subscriber information:<br><ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>             |
| <b>tcp-retransmission-info</b> | Enable/Disable flow TCP retransmission statistics:<br><ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>      |
| <b>network-perf-info</b>       | Enable/Disable flow network performance statistics:<br><ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>     |
| <b>network-stats-info</b>      | Enable/Disable flow network statistics:<br><ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>                 |

| Parameter                       | Description                                                                                                                                                     |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>nat-binding-info</b>         | Enable/Disable reporting NAT IP Address and NAT port information: <ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul> |
| <b>HTTP EDR Profile:</b>        |                                                                                                                                                                 |
| <b>cache-info</b>               | Enable/Disable HTTP cache information: <ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>                            |
| <b>compression-info</b>         | Enable/Disable HTTP compression information: <ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>                      |
| <b>connection-info</b>          | Enable/Disable HTTP connection information: <ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>                       |
| <b>dpi-info</b>                 | Enable/Disable deep packet inspection information: <ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>                |
| <b>entitlement-info</b>         | Enable/Disable entitlement information: <ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>                           |
| <b>filter-info</b>              | Enable/Disable content filter information: <ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>                        |
| <b>http-subscriber-info</b>     | Enable/Disable HTTP subscriber information: <ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>                       |
| <b>manifest-info</b>            | Enable/Disable HTTP manifest information: <ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>                         |
| <b>minification-information</b> | Enable/Disable HTTP minification information: <ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>                     |
| <b>preempt-dns-info</b>         | Enable/Disable HTTP preempt DNS information: <ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>                      |

| Parameter                         | Description                                                                                                                                               |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>request-info</b>               | Enable/Disable HTTP request information:<br><ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>                 |
| <b>response-info</b>              | Enable/Disable HTTP response information:<br><ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>                |
| <b>tcp-client-connection-info</b> | Enable/Disable HTTP TCP client connection information:<br><ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>   |
| <b>tcp-server-connection-info</b> | Enable/Disable HTTP TCP server connection information:<br><ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>   |
| <b>tcp-splicing-info</b>          | Enable/Disable HTTP TCP splicing connection information:<br><ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul> |
| <b>time-info</b>                  | Enable/Disable HTTP time information:<br><ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>                    |
| <b>udp-proxy-info</b>             | Enable/Disable UDP Proxy information:<br><ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>                    |
| <b>video-info</b>                 | Enable/Disable HTTP video information:<br><ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>                   |
| <b>web-image-opt-info</b>         | Enable/Disable HTTP web image optimization information:<br><ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>  |
| <b>Session EDR Profile:</b>       |                                                                                                                                                           |
| <b>data-statistics</b>            | Enable/Disable data statistics:<br><ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>                          |

| Parameter                      | Description                                                                                                                                  |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>edr-cause-codes</b>         | Enable/Disable EDR cause codes: <ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>                |
| <b>gateway-info</b>            | Enable/Disable gateway information: <ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>            |
| <b>network-stats-info</b>      | Enable/Disable network statistics: <ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>             |
| <b>serving-network-info</b>    | Enable/Disable serving network information: <ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>    |
| <b>session-qos-info</b>        | Enable/Disable session QoS information: <ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>        |
| <b>session-subscriber-info</b> | Enable/Disable session subscriber information: <ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul> |
| <b>session-user-csg-info</b>   | Enable/Disable session user csg information: <ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>   |

- 7 For PGW, GGSN, WAG, and ePDG, associate the EDR Interface Profile with a Workflow Control Profile:

```
an3000-mcm-slot7cpu1(config)# workflow control-profile <name>
edr-interface <name>
```

**Note:** The AF-EDR-Interface-Name vendor-specific attribute (VSA) can override the EDR Interface Profile configured in the Workflow Control Profile.

When the GGSN/PGW receives the AF-EDR-Interface-Name VSA in the RADIUS Access-Accept message, it replaces the EDR Interface Profile configured in the Workflow Control Profile with the EDR interface received in the AF-EDR-Interface-Name VSA.

The EDR interface received in the AF-EDR-Interface-Name VSA must match a pre-configured EDR interface on the MCC. If the received EDR interface name does not match an EDR interface configured on the MCC, the configured EDR Interface Profile from the Workflow Control Profile will be used.

The EDR interface received in the AF-EDR-Interface-Name VSA (within the Access-Accept message) takes precedence over any configured value for the EDR interface. For example, if the Workflow Control Profile or Subscriber Analyzer is switched, the EDR interface received in the Access-Accept message will still take precedence over the EDR Interface Profile configured in the new Workflow Control Profile.

- 8 For SGW, associate the EDR Interface Profile with the SGW object:

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway sgw <sgw-name>
edr-interface <name>
```

## Configuring Subscriber Data Plane Mirroring

This section describes how to configure a subscriber vTap to replicate subscriber traffic. The vTap specifies the subscriber session(s) that require a data tap. When a subscriber vTap is configured, all workflow sessions for the subscriber(s) are marked for tapping, and all the data traffic processed for that session are tapped.

In a CUPS deployment, perform these steps on the User Plane Function (UPF) MCC.

**Use the following sequence to configure a subscriber vTap:**

1. Configure a GRE tunnel that is used to send the replicated packets.

```
an3000-mcm-slot7cpu1(config)# network-context <name> tunnel <name>
type GRE local-address <IP address> remote-ip-address <IP address>
include-checksum [yes|no] tunnel-key <key>
```

The following table describes the configurable parameters.

| Parameter                | Description                                                                                                      |
|--------------------------|------------------------------------------------------------------------------------------------------------------|
| <b>type</b>              | Tunnel type:<br>■ <b>GRE</b>                                                                                     |
| <b>local-address</b>     | Local IP address of the tunnel (that is, the source IP address in the outer packet). Must be a loopback address. |
| <b>remote-ip-address</b> | Remote IP address (destination) of the tunnel.                                                                   |
| <b>include-checksum</b>  | Specifies whether to include the checksum field in the tunnel packets:<br>■ yes<br>■ no                          |
| <b>tunnel-key</b>        | Optional key used in the GRE packets                                                                             |

2. Configure a vTap sink object to specify whether to stream the replicated packets.

```
an3000-mcm-slot7cpu1(config)# vtap sink <name> type [gre] tunnel
<name> packet-metadata [include-ebi|include-imsi|include-timestamp]
```

The following table describes the configurable parameters.

| Parameter              | Description                                        |
|------------------------|----------------------------------------------------|
| <b>type</b>            | Sink type:<br>■ <b>GRE</b>                         |
| <b>network-context</b> | Network-Context of the tunnel interface            |
| <b>tunnel</b>          | Tunnel name configured in <a href="#">step 1</a> . |

| Parameter              | Description                                                                                                                                                                                                             |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>packet-metadata</b> | List of fields to be included in the packet metadata sent to the destination: <ul style="list-style-type: none"> <li>■ <b>include-ebi</b></li> <li>■ <b>include-imsi</b></li> <li>■ <b>include-timestamp</b></li> </ul> |

- 3 Configure a subscriber filter that defines the set of subscribers whose traffic is to be replicated.

```
an3000-mcm-slot7cpu1(config)# vtap subscriber-filter <name> key <name>
```

The following table describes the configurable parameters.

| Parameter       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>key</b>      | List of analyzed keys. A key can have multiple attributes and all attributes specified in the key must match for a key to be considered matching. If any one key matches, the subscriber-filter is considered matching: <ul style="list-style-type: none"> <li>■ <b>apn-name</b></li> <li>■ <b>imei-range</b></li> <li>■ <b>imsi-range</b></li> <li>■ <b>imsi-list</b></li> <li>■ <b>ue-ip-address-range</b></li> <li>■ <b>mac-address-list</b> - List of MAC addresses in common notation (with or without - or :)</li> <li>■ <b>nai-list</b> - List of NAIs (Network Address Identifiers). Matching is performed after stripping any decoration (preceding a ! or :) from the received NAI</li> <li>■ <b>nai-realm-list</b> - Matches only the realm portion (after the @) of received NAIs. Provides a group matching capability like the IMSI range filter</li> </ul> |
| <b>apn-name</b> | APN name                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>high</b>     | User equipment identifier or IP address end value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>low</b>      | User equipment identifier or IP address start value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |



- 4 Configure a subscriber vTap that specifies one or more subscriber filters, sinks, and reference points.
- When multiple subscriber filters are configured, a packet matches the subscriber vTap if it belongs to one of the sessions that match the configured filters.
  - When multiple sinks are configured, the system replicates the matching packet and sends it to each sink.
  - When multiple reference points are configured, the system copies the packet and sends it for each reference point.

```
an3000-mcm-slot7cpu1(config)# vtap subscriber-tap <name> admin-state
[enabled|disabled] requires-external-activation [yes|no]
subscriber-filter-set <name> reference-point
[access|core|encrypted-access|intermediate] sink <name>
```

The following table describes the configurable parameters.

| Parameter                           | Description                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>admin-state</b>                  | Admin state of the subscriber vTap: <ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>                                                                                                                                                                                                |
| <b>requires-external-activation</b> | Specifies whether the subscriber vTap can be externally activated or deactivated by the <b>vtap start</b> and <b>vtap stop</b> commands: <ul style="list-style-type: none"> <li>■ <b>yes</b></li> <li>■ <b>no</b></li> </ul> <p>The <b>admin-state</b> of a subscriber vTap must be enabled for this setting to take effect.</p> |

| Parameter                    | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>reference-point</b>       | <p>Reference point that specifies where to take a snapshot of the received subscriber packet:</p> <ul style="list-style-type: none"> <li>■ <b>access</b>: Corresponds to the following interfaces: <ul style="list-style-type: none"> <li>– S1-U interface in collated SGW/PGW deployments</li> <li>– S5/S8 interface for standalone PGW deployment</li> <li>– Gn interface for GGSN deployments</li> </ul> </li> <li>■ <b>core</b>: Allows you to specify reference points at which the MCC should take a snapshot of the received subscriber packets before they are decrypted on ingress and after they are encrypted on egress. <p>Corresponds to the following interfaces:</p> <ul style="list-style-type: none"> <li>– Sgi interface for standalone PGW or collocated SGW/PGW deployments</li> <li>– Gi interface for GGSN deployments</li> </ul> </li> <li>■ <b>encrypted-access</b>: Specifies that MCC should take a snapshot of the received subscriber packets before they are decrypted on ingress and after they are encrypted on egress. Corresponds to the following interfaces: <ul style="list-style-type: none"> <li>– SWu interface in ePDG deployments</li> </ul> </li> <li>■ <b>intermediate</b>: Corresponds to the interface between gateway and proxy in deployments that use value-added services such as HTTP proxy or content filtering.</li> </ul> |
| <b>sink</b>                  | vTap sink object                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>subscriber-filter-set</b> | vTap subscriber-filter object                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

- 5 If desired, you can manage when the subscriber vTap is started and stopped. The commands require the **requires-external-activation** of a subscriber vTap to be enabled.

- You can configure the subscriber vTap to be started at a future time. Either specify a start time and end time or specify a start time and a duration, in seconds:  

```
an3000-mcm-slot7cpu1(config)# vtap start <name> start-time <start time> end-time <end time> duration <seconds>
```

The following table describes the configurable parameters.

| Parameter         | Description                                                                                                      |
|-------------------|------------------------------------------------------------------------------------------------------------------|
| <b>start-time</b> | Date and time to start the subscriber vTap, in the format yyyy-mm-ddThh:mm:ss. For example, 2017-02-08T10:10:10. |

| Parameter       | Description                                                                                                                                |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| <b>end-time</b> | Date and time to end the subscriber vTap, in the format <code>yyyy-mm-ddThh:mm:ss</code> . For example, <code>2017-02-08T10:10:20</code> . |
| <b>duration</b> | As an alternative to <b>end-time</b> , specifies the duration to run the subscriber vTap, in seconds from <b>start-time</b> .              |

- You can start the subscriber vTap immediately:  
`an3000-mcm-slot7cpu1(config)# vtap start <name>`
- You can stop the subscriber vTap immediately or disable a scheduled subscriber vTap:  
`an3000-mcm-slot7cpu1(config)# vtap stop <name>`
- To view the **start-time** and **end-time** of scheduled subscriber vTaps:  
`an3000-mcm-slot7cpu1(config)# show vtap subscriber-tap-schedule`  
`vtap subscriber-tap-schedule`  

```

VTAP
SUBSCRIBER
TAP NAME      STARTTIME      ENDTIME      DURATION  VTAPSTATE
-----
tap1          2017-03-01T23:02:36  2017-03-01T23:04:36  0         Running
tap2          2017-03-01T23:05:36  2017-03-01T23:08:36  0         Scheduled
tap3          2017-03-01T23:09:36  2017-03-01T23:12:36  0         Idle
tap4          2017-03-01T23:13:36  2017-03-01T23:16:36  0         Idle

```

## Metadata

The fields included in the packet metadata sent to the destination are:

- IMSI – Value encoded as a 8-byte entity in the network-byte order.
- EBI – EPS Bearer Identifier encoded as a 1-byte entity.
- Timestamp – Value encoded as a 8-byte timestamp since epoch:
  - The first four bytes store the time value in seconds in network-byte order.
  - The second four bytes store the time value in microseconds in network-byte order.

## Configuring Control Plane Packet Mirroring

This section describes how to configure protocol logging with a GRE-based sink to replicate control plane traffic. When this protocol log is configured, instead of logging all matching packets to a file, vProbe adds a GRE header and sends the encapsulated packet to a remote destination.

In a CUPS deployment, perform these steps on the Control Plane Function (CPF) MCC.

**Use the following sequence to configure control plane packet mirroring:**

- 1 Configure a GRE tunnel that is used to send the replicated packets.

```
an3000-mcm-slot7cpu1(config)# network-context <name> tunnel <name>
type GRE local-address <IP address> remote-ip-address <IP address>
include-checksum [yes|no] tunnel-key <key>
```

The following table describes the configurable parameters.

| Parameter                | Description                                                                                                                                         |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>type</b>              | Tunnel type:<br>■ <b>GRE</b>                                                                                                                        |
| <b>local-address</b>     | Local IP address of the tunnel (that is, the source IP address in the outer packet). Must be a loopback address.                                    |
| <b>remote-ip-address</b> | Remote IP address (destination) of the tunnel.                                                                                                      |
| <b>include-checksum</b>  | Specifies whether to include the checksum field in the tunnel packets:<br>■ yes<br>■ no<br>This field is not supported for control plane mirroring. |
| <b>tunnel-key</b>        | Optional key used in the GRE packets                                                                                                                |

- 2 Configure a protocol log to replicate control packets for a specific protocol.

```
an3000-mcm-slot7cpu1(config)# infrastructure logging event-log policy
<name> log-type protocol protocol-logging-filter-field protocol
filter-value <value> filter-field <field> filter-value <value>
```

The following table describes the configurable parameters.

| Object                               | Parameter           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>policy</b>                        | <b>name</b>         | Policy name                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                      | <b>admin-state</b>  | Enable/Disable the policy: <ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                    |
|                                      | <b>log-type</b>     | Log type to collect for this policy: <ul style="list-style-type: none"> <li>■ <b>protocol</b>: captures control packets</li> </ul>                                                                                                                                                                                                                                                                                                                                                          |
| <b>protocol-logging-filter-field</b> | <b>filter-field</b> | Field on which to filter: <ul style="list-style-type: none"> <li>■ <b>destination-ip-address</b></li> <li>■ <b>destination-portnumber</b></li> <li>■ <b>imei</b></li> <li>■ <b>imsi</b></li> <li>■ <b>ip-address-pair</b></li> <li>■ <b>ip-protocol</b></li> <li>■ <b>ipaddress</b></li> <li>■ <b>msisdn</b></li> <li>■ <b>packet-direction</b></li> <li>■ <b>packet-type</b></li> <li>■ <b>protocol</b></li> <li>■ <b>source-ip-address</b></li> <li>■ <b>source-portnumber</b></li> </ul> |
| <b>filter-field</b>                  | <b>filter-value</b> | Value on which to filter. For example, when <b>protocol-logging-filter-field</b> is set to <b>protocol</b> , the values are: <ul style="list-style-type: none"> <li>■ <b>radius</b></li> <li>■ <b>diameter</b></li> <li>■ <b>gtpv0</b></li> <li>■ <b>gtpv1</b></li> <li>■ <b>gtpv2</b></li> <li>■ <b>gtp</b></li> <li>■ <b>dhcp</b></li> <li>■ <b>isakmp</b></li> <li>■ <b>pfc</b></li> </ul>                                                                                               |
|                                      | <b>exclude</b>      | Specifies whether to exclude logs with the filter value: <ul style="list-style-type: none"> <li>■ <b>false</b></li> <li>■ <b>true</b></li> </ul>                                                                                                                                                                                                                                                                                                                                            |

| Object              | Parameter              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>filter-field</b> | <b>interface</b>       | <p>Value on which to filter. For example, when <b>filter-field</b> is set to <b>radius</b>, the values are:</p> <ul style="list-style-type: none"> <li>■ <b>accounting</b></li> <li>■ <b>authentication</b></li> </ul> <p>When <b>filter-field</b> is set to <b>diameter</b>, the values are:</p> <ul style="list-style-type: none"> <li>■ <b>Gx</b></li> <li>■ <b>Gy</b></li> <li>■ <b>Gz</b></li> <li>■ Rx</li> <li>■ <b>S6b</b></li> <li>■ STa</li> <li>■ SWm</li> </ul> <p>When <b>filter-field</b> is set to <b>gtpv2</b>, the values are:</p> <ul style="list-style-type: none"> <li>■ <b>s4</b></li> <li>■ <b>s5</b></li> <li>■ <b>s8</b></li> <li>■ <b>s11</b></li> </ul> |
|                     | <b>exclude</b>         | <p>Specifies whether to exclude logs with the filter value:</p> <ul style="list-style-type: none"> <li>■ <b>false</b></li> <li>■ <b>true</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>sinks</b>        | <b>name</b>            | Sink name                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                     | <b>type</b>            | <p>Sink Type:</p> <ul style="list-style-type: none"> <li>■ <b>gre-tunnel</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                     | <b>network-context</b> | Network-Context of the tunnel interface                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                     | <b>gre-tunnel</b>      | Tunnel name configured in <a href="#">step 1</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                     | <b>metadata</b>        | <p>Specifies whether to send packet metadata to the destination:</p> <ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

## Metadata

The metadata packet format is as follows:

|                                 |
|---------------------------------|
| <b>L2 Header</b>                |
| <b>L3 Header</b>                |
| <b>GRE Header</b>               |
| <b>Metadata Length (1 byte)</b> |
| <b>Metadata in TLV Format</b>   |
| <b>Replicated Packet Data</b>   |

The metadata contains type/length/value (TLV) fields, where control mirrored packets send the following:

**Table 2-1 Metadata Information (in TLV format)**

| Type | Fieldname | Length  | Description                                                                                                                                                                                                                              |
|------|-----------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | IMSI      | 8 bytes | IMSI value will be filled in this field.<br>The IMSI value is included in the metadata only for interfaces s4/s5/s8/s11/Ga/Gx/Gy/Gz/S6b and Radius accounting and authentication.                                                        |
| 2    | EBI       | 1 byte  | EPS Bearer ID will be filled in this field.                                                                                                                                                                                              |
| 3    | Timestamp | 8 bytes | Epoch time: <ul style="list-style-type: none"> <li>■ The first four bytes store the time value in seconds in network-byte order.</li> <li>■ The second four bytes store the time value in microseconds in network-byte order.</li> </ul> |
| 4    | IMEI      | 8 bytes | IMEI value will be filled in case of emergency calls.<br>The IMEI value is included in the metadata only for interfaces s4/s5/s8/s11/Ga/Gx/Gy/Gz/S6b and Radius accounting and authentication.                                           |

All values are sent in network (big-endian) order.

For example:

```
17010800011b043d636471020105030858dd6e280005fa7d
17: Metadata length (0x17 == 23 bytes)
01: Type 1 (IMSI)                                1
08: Length 8 bytes                                1
00011b043d636471: IMSI value                      8
```

## *AI-vProbe Configuration*

|                                   |       |
|-----------------------------------|-------|
| 02: Type 2 (EBI)                  | 1     |
| 01: Length 1 byte                 | 1     |
| 05: Bearer ID 5                   | 1     |
| 03: Type 3 (timestamp)            | 1     |
| 08: length 8                      | 1     |
| 58dd6e280005fa7d: timestamp value | 8     |
|                                   | ----- |
|                                   | 23    |

The first four bytes of the timestamp (58dd6e28H or 1490906664D) is the epoch time in seconds, which converts to 3/30/2017 20:44:24 GMT. The latter four bytes is the microsecond portion, which is 0005fa7dH or 391805 microseconds. The complete timestamp is 3/30/2017 20:44:24.391805.



## Sample Protocol Logging

The following sample configuration illustrates protocol logging with a GRE-based sink to replicate control plane traffic:

```
an3000-mcm-slot7cpu1# show running-config infrastructure logging
event-log policy control_vtap
```

```
infrastructure logging event-log policy control_vtap
```

```
admin-state enabled
```

```
log-type protocol
```

```
protocol-logging-filter-field imsi
```

```
filter-value 226041000000001
```

```
!
```

```
!
```

```
protocol-logging-filter-field protocol
```

```
filter-value gtpv2
```

```
filter-field interface
```

```
filter-value s5
```

```
exclude true
```

```
!
```

```
filter-value s8
```

```
exclude true
```

```
!
```

```
filter-value s11
```

```
exclude false
```

```
!
```

```
!
```

```
!
```

```
!
```

```
sinks sink1
```

```
type gre-tunnel
```

```
network-context BOS
```

```
gre-tunnel GRE_tunnel
```

```
metadata disabled
```

```
!
```

```
!
```

```
an3000-mcm-slot7cpu1# show running-config network-context BOS tunnel
GRE_tunnel
```

```
network-context BOS
```

```
tunnel GRE_tunnel
```

```
type GRE
```

```
local-address gre_tunnel_loopback
```

```
remote-ip-address 179.1.1.2
```

## *AI-vProbe Configuration*

```
include-checksum no
tunnel-key      1234
!
!
```

## Configuring UDP Server to Receive Pilot Packets

AI-vProbe supports sending Pilot Packets to multiple destinations on Session setup/tear-down, RAT-type change, and NAT address/port portion allocation/de-allocation. The Pilot Packet provides key information about a subscriber session, including Subscriber Identity (MSISDN, IMSI, NAI), Subscriber IP address, and Subscriber NAT IP address and port range. This information can be used to enable new services, such as Subscriber Analytics.

A Pilot Packet contains a number of attributes in a string of type-length-values that are sent by UDP transport to multiple destinations.

**Use the following sequence to configure the UDP server:**

- 1 Configure the UDP server to receive Pilot Packets in the type-length-value encoding format.

```
an3000-mcm-slot7cpu1(config)# service-construct edr-server-profile
<name>
```

The following table describes the configurable parameters.

| Parameter               | Description                                                                                                               |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>admin-state</b>      | Admin state of the profile: <ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul> |
| <b>ip-address</b>       | IP address of the EDR server                                                                                              |
| <b>port</b>             | TCP port of the EDR server                                                                                                |
| <b>network-context</b>  | Network context of the EDR server                                                                                         |
| <b>edr-source-ip</b>    | Source IP address to use to reach the EDR server                                                                          |
| <b>transport</b>        | Transport type: <ul style="list-style-type: none"> <li>■ <b>udp</b></li> </ul>                                            |
| <b>encoding</b>         | Encoding format: <ul style="list-style-type: none"> <li>■ <b>tlv</b> (type-length-value)</li> </ul>                       |
| <b>keep-alive</b>       | EDR server keep alive interval                                                                                            |
| <b>tcp-user-timeout</b> | Number of seconds for TCP user data retransmission timeout                                                                |

- 2 Configure the edr-interface-profile to enable/disable the Pilot Packet subtypes.

```
an3000-mcm-slot7cpu1(config)# service-construct edr-interface-profile
<name> edr-service-profile <name>
```

The following table describes the configurable parameters.

| Parameter               | Description                                                                                                                                                    |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>admin-state</b>      | Admin state of the profile:<br><ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>                                   |
| <b>primary-server</b>   | Primary EDR server for this EDR service                                                                                                                        |
| <b>secondary-server</b> | <i><b>Note:</b> Because the Pilot Packets are sent by connectionless UDP, the secondary-server is not used, even if configured in the edr-service-profile.</i> |

- 3 Configure the Pilot EDR Profile and specify the Pilot EDR record type (START, STOP, NAT Binding, RAT Type Change) and records to send to the UDP server.

- a Configure the Pilot EDR profile:

```
an3000-mcm-slot7cpu1(config)# service-construct
edr-interface-profile <name> edr-service-profile <name>
pilot-edr-profile
```

The following table describes the configurable parameters.

| Parameter          | Description                                                                                                                  |
|--------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>admin-state</b> | Admin state of the profile:<br><ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul> |

- b Specify which record types to send to the UDP server:

```
an3000-mcm-slot7cpu1(config)# service-construct
edr-interface-profile <name> edr-service-profile <name>
pilot-edr-profile record-type [start|stop|nat-binding|rat-change]
```

The following table describes the configurable parameters.

| Parameter                  | Description                                                                                                                          |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>start-records</b>       | Enable/Disable START records:<br><ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>       |
| <b>stop-records</b>        | Enable/Disable STOP records:<br><ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>        |
| <b>nat-binding-records</b> | Enable/Disable NAT Binding records:<br><ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul> |

| Parameter                 | Description                                                                                                                      |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>rat-change-records</b> | Enable/Disable RAT Change records: <ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul> |

- c Specify the record information to send to the EDR server:

```
an3000-mcm-slot7cpu1(config)# service-construct
edr-interface-profile <name> edr-service-profile <name>
pilot-edr-profile record-info <field name>
```

The following table describes the configurable parameters.

| Parameter                   | Description                                                                                                                                               |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>serving-network-info</b> | Enable/Disable pilot serving network information reporting: <ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul> |
| <b>subscriber-info</b>      | Enable/Disable pilot subscriber information reporting: <ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>      |

- 4 For PGW, GGSN, WAG, and ePDG, associate the Pilot EDR Interface Profile with a Workflow Control Profile:

```
an3000-mcm-slot7cpu1(config)# workflow control-profile <name>
edr-interface <name>
```

**Note:** The AF-EDR-Interface-Name vendor-specific attribute (VSA) can override the EDR Interface Profile configured in the Workflow Control Profile.

When the GGSN/PGW receives the AF-EDR-Interface-Name VSA in the RADIUS Access-Accept message, it replaces the EDR Interface Profile configured in the Workflow Control Profile with the EDR interface received in the AF-EDR-Interface-Name VSA.

The EDR interface received in the AF-EDR-Interface-Name VSA must match a pre-configured EDR interface on the MCC. If the received EDR interface name does not match an EDR interface configured on the MCC, the configured EDR Interface Profile from the Workflow Control Profile will be used.

The EDR interface received in the AF-EDR-Interface-Name VSA (within the Access-Accept message) takes precedence over any configured value for the EDR interface. For example, if the Workflow Control Profile or Subscriber Analyzer is switched, the EDR interface received in the Access-Accept message will still take precedence over the EDR Interface Profile configured in the new Workflow Control Profile.

- 5 For SGW, associate the Pilot EDR Interface Profile with the SGW object:  
`an3000-mcm-slot7cpu1(config)# zone <name> gateway sgw <sgw-name>  
edr-interface <name>`

## Configuring Real Time Troubleshooting EDRs

AI-vProbe supports Real Time Troubleshooting (RTT) EDRs on the SGW, PGW, ePDG, and WAG. The EDR records provide visibility into the network and subscriber events to help diagnose networking and subscriber issues.

RTT EDR supports the following features:

- Generates real time event records for:
  - GTP events over the S4/S5/S8/S11/S2a/S2b/Gn/Gp interfaces
  - Diameter events over Gx/Gy/S6b/SWm interfaces
  - IKE events over the SWu interface
- Supports enabling/disabling RTT EDRs on a per-interface basis.
- Supports multiple simultaneous EDR servers, each of which can sample different records or overlapping data.
- Supports redundant (primary and secondary) TCP connections from MCC to the EDR Server, with keepalives for failover.
- Maintains EDR statistics per server for the number of EDR send successes/failures for each interface.
- Provides connection statistics for all connections to the EDR Servers.

**Use the following sequence to configure RTT EDRs:**

- 1 Create an EDR Server Profile (maximum 8).

```
an3000-mcm-slot7cpu1(config)# service-construct edr-server-profile
<name>
```

The following table describes the configurable parameters.

| Parameter              | Description                                                                                                               |
|------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>admin-state</b>     | Admin state of the profile: <ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul> |
| <b>ip-address</b>      | IP address of the EDR server                                                                                              |
| <b>port</b>            | TCP port of the EDR server                                                                                                |
| <b>network-context</b> | Network context of the EDR server                                                                                         |
| <b>edr-source-ip</b>   | Source IP address to use to reach the EDR server                                                                          |
| <b>transport</b>       | Transport type: <ul style="list-style-type: none"> <li>■ <b>tcp</b></li> </ul>                                            |

| Parameter               | Description                                                                                                                       |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>encoding</b>         | Encoding format:<br><ul style="list-style-type: none"> <li>■ <b>protobuf</b></li> <li>■ <b>tlv</b> (type-length-value)</li> </ul> |
| <b>keep-alive</b>       | EDR server keep alive interval                                                                                                    |
| <b>tcp-user-timeout</b> | Number of seconds for TCP user data retransmission timeout                                                                        |

- 2 Configure the EDR Interface Profile (maximum 256).

```
an3000-mcm-slot7cpu1(config)# service-construct edr-interface-profile <name>
```

The following table describes the configurable parameters.

| Parameter          | Description                                                                                                                  |
|--------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>admin-state</b> | Admin state of the profile:<br><ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul> |

- 3 Configure the EDR Service Profile (maximum 8).

```
an3000-mcm-slot7cpu1(config)# service-construct edr-interface-profile <name> edr-service-profile <name>
```

The following table describes the configurable parameters.

| Parameter               | Description                                                                                                                  |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>admin-state</b>      | Admin state of the profile:<br><ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul> |
| <b>primary-server</b>   | Primary EDR server for this EDR service                                                                                      |
| <b>secondary-server</b> | Secondary EDR server for this EDR service                                                                                    |

- 4 Configure the RTT EDR Profile and specify the RTT EDR record type (GTP S5, GTP S8, GTP S11, GTP S4, GTP S2a, GTP S2b, GTP Gn, GTP Gp, Diameter Gx, Diameter Gy, Diameter S6b, Diameter SWm, IKE SWu).

- a Configure the RTT EDR profile:

```
an3000-mcm-slot7cpu1(config)# service-construct edr-interface-profile <name> edr-service-profile <name> rtt-edr-profile
```

The following table describes the configurable parameters.



| Parameter          | Description                                                                                                                  |
|--------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>admin-state</b> | Admin state of the profile:<br><ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul> |

**b** Specify which record types:

```
an3000-mcm-slot7cpu1(config)# service-construct
edr-interface-profile <name> edr-service-profile <name>
rtt-edr-profile record-type
[gtp-s5-records|gtp-s8-records|gtp-s11-records|gtp-s4-records|gtp-
s2a-records|gtp-s2b-records|gtp-gn-records|gtp-gp-records|diameter
-gx-records|diameter-gy-records|diameter-s6b-records|diameter-swm-
records|ike-swu-records]
```

The following table describes the configurable parameters.

| Parameter              | Description                                                                                                                      |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>gtp-s5-records</b>  | Enable/Disable GTP S5 records:<br><ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>  |
| <b>gtp-s8-records</b>  | Enable/Disable GTP S8 records:<br><ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>  |
| <b>gtp-s11-records</b> | Enable/Disable GTP S11 records:<br><ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul> |
| <b>gtp-s4-records</b>  | Enable/Disable GTP S4 records:<br><ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>  |
| <b>gtp-s2a-records</b> | Enable/Disable GTP S2a records:<br><ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul> |
| <b>gtp-s2b-records</b> | Enable/Disable GTP S2b records:<br><ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul> |
| <b>gtp-gn-records</b>  | Enable/Disable GTP Gn records:<br><ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>  |

| Parameter                   | Description                                                                                                                           |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>gtp-gp-records</b>       | Enable/Disable GTP Gp records:<br><ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>       |
| <b>diameter-gx-records</b>  | Enable/Disable Diameter Gx records:<br><ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>  |
| <b>diameter-gy-records</b>  | Enable/Disable Diameter Gy records:<br><ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>  |
| <b>diameter-s6b-records</b> | Enable/Disable Diameter S6b records:<br><ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul> |
| <b>diameter-swm-records</b> | Enable/Disable Diameter SWm records:<br><ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul> |
| <b>ike-swu-records</b>      | Enable/Disable IKE SWu records:<br><ul style="list-style-type: none"> <li>■ <b>enabled</b></li> <li>■ <b>disabled</b></li> </ul>      |

- 5 Associate the RTT EDR Interface Profile with a Workflow Control Profile:  

```
an3000-mcm-slot7cpu1(config)# workflow control-profile <name>
edr-interface <name>
```

**Note:** The AF-EDR-Interface-Name vendor-specific attribute (VSA) can override the EDR Interface Profile configured in the Workflow Control Profile.

When the GGSN/PGW receives the AF-EDR-Interface-Name VSA in the RADIUS Access-Accept message, it replaces the EDR Interface Profile configured in the Workflow Control Profile with the EDR interface received in the AF-EDR-Interface-Name VSA.

The EDR interface received in the AF-EDR-Interface-Name VSA must match a pre-configured EDR interface on the MCC. If the received EDR interface name does not match an EDR interface configured on the MCC, the configured EDR Interface Profile from the Workflow Control Profile will be used.

The EDR interface received in the AF-EDR-Interface-Name VSA (within the Access-Accept message) takes precedence over any configured value for the EDR interface. For example, if the Workflow Control Profile or Subscriber Analyzer is switched, the EDR interface received in the Access-Accept message will still take precedence over the EDR Interface Profile configured in the new Workflow Control Profile.

## Performance Statistics

### EDR Record Statistics

Use the following CLI command to display statistics about EDR records:

```
an3000-mcm-slot7cpu1# show infrastructure edr
edr-[bearer|flow|http|pilot|rtt|server|session]-stats
```

where:

- **edr-bearer-stats** — EDR Bearer Statistics
- **edr-flow-stats** — EDR Flow Statistics
- **edr-http-stats** — EDR HTTP Statistics
- **edr-pilot-stats** — EDR Pilot Packet Statistics
- **edr-rtt-stats** — EDR RTT Statistics
- **edr-server-stats** — EDR Server Statistics
- **edr-session-stats** — EDR Session Statistics

The system displays output similar to the following:

```
an3000-mcm-slot8cpu1# show infrastructure edr edr-http-stats
NETWORK          HTTP    HTTP    DISCONTINUITY
CONTEXT  NAME      SUCCESS  FAILED  TIMESTAMP
-----
BOS      ESP      37848    0        1452803037
BOS      ESP_NEW    0         0        1452803037
Other    ESP_OTHER  0         0        1452803037
```

## Performance Statistics

an3000-mcm-slot8cpu1# show infrastructureedr

infrastructureedr edr-server-stats

| NETWORK |           | RECORDS | BYTES    | RECORDS | BYTES  | DISCONTINUITY |            |
|---------|-----------|---------|----------|---------|--------|---------------|------------|
| CONTEXT | NAME      | SUCCESS | SUCCESS  | FAILED  | FAILED | CONNECTS      | TIMESTAMP  |
| BOS     | ESP       | 43451   | 17323658 | 0       | 0      | 1             | 1452803018 |
| BOS     | ESP_NEW   | 0       | 0        | 0       | 0      | 1             | 1452803018 |
| Other   | ESP_OTHER | 0       | 0        | 0       | 0      | 0             | 1452803018 |

infrastructureedr edr-session-stats

| NETWORK |           | SESSION | SESSION | SESSION | SESSION | SESSION | SESSION | SESSION | SESSION | DISCONTINUITY |
|---------|-----------|---------|---------|---------|---------|---------|---------|---------|---------|---------------|
| CONTEXT | NAME      | START   | STOP    | ATTEMPT | INTERIM | START   | STOP    | ATTEMPT | INTERIM | TIMESTAMP     |
| BOS     | ESP       | 100     | 75      | 0       | 683     | 0       | 0       | 0       | 0       | 1452803024    |
| BOS     | ESP_NEW   | 0       | 0       | 0       | 0       | 0       | 0       | 0       | 0       | 1452803024    |
| Other   | ESP_OTHER | 0       | 0       | 0       | 0       | 0       | 0       | 0       | 0       | 1452803024    |

infrastructureedr edr-bearer-stats

| NETWORK |           | BEARER | BEARER | BEARER  | BEARER  | BEARER | BEARER | BEARER  | BEARER  | DISCONTINUITY |
|---------|-----------|--------|--------|---------|---------|--------|--------|---------|---------|---------------|
| CONTEXT | NAME      | START  | STOP   | ATTEMPT | INTERIM | START  | STOP   | ATTEMPT | INTERIM | TIMESTAMP     |
| BOS     | ESP       | 200    | 150    | 0       | 1366    | 0      | 0      | 0       | 0       | 1452803028    |
| BOS     | ESP_NEW   | 0      | 0      | 0       | 0       | 0      | 0      | 0       | 0       | 1452803028    |
| Other   | ESP_OTHER | 0      | 0      | 0       | 0       | 0      | 0      | 0       | 0       | 1452803028    |

infrastructureedr edr-flow-stats

| NETWORK |           | FLOW  | FLOW  | FLOW    | FLOW  | FLOW | FLOW    | DISCONTINUITY |  |
|---------|-----------|-------|-------|---------|-------|------|---------|---------------|--|
| CONTEXT | NAME      | START | STOP  | INTERIM | START | STOP | INTERIM | TIMESTAMP     |  |
| BOS     | ESP       | 46653 | 65110 | 0       | 0     | 0    | 0       | 1453137478    |  |
| BOS     | ESP_NEW   | 0     | 0     | 0       | 0     | 0    | 0       | 1453137478    |  |
| Other   | ESP_OTHER | 0     | 0     | 0       | 0     | 0    | 0       | 1453137478    |  |

infrastructureedr edr-pilot-stats

| NETWORK |           | PILOT | PILOT | PILOT | PILOT | PILOT | PILOT | PILOT | PILOT | DISCONTINUITY |
|---------|-----------|-------|-------|-------|-------|-------|-------|-------|-------|---------------|
| CONTEXT | NAME      | START | STOP  | NAT   | NAT   | NAT   | NAT   | RAT   | RAT   | TIMESTAMP     |
| BOS     | ESP       | 200   | 150   | 0     | 1366  | 0     | 0     | 0     | 0     | 1452803028    |
| BOS     | ESP_NEW   | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1452803028    |
| Other   | ESP_OTHER | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1452803028    |

```

infrastructure edr edr-rtt-stats
rtt GTP S5 success      0
rtt GTP S8 success      0
rtt GTP S11 success     0
rtt GTP S4 success      0
rtt GTP S2A success     0
rtt GTP S2B success     0
rtt GTP GN success      0
rtt GTP GP success      0
rtt diameter gx success 4
rtt diameter gy success 3
rtt diameter S6B success 0
rtt GTP S5 failed       0
rtt GTP S8 failed       0
rtt GTP S11 failed      0
rtt GTP S4 failed       0
rtt GTP S2A failed      0
rtt GTP S2B failed      0
rtt GTP GN failed       0
rtt GTP GP failed       0
rtt diameter gx failed  0
rtt diameter gy failed  0
rtt diameter S6B failed 0
Discontinuity Timestamp 1480623624

```

EDR record statistics are also written to .csv files in the following directory:

*/var/log/psm/*

The filename is:

EDR\_[BEARER | FLOW | HTTP | PILOT | RTT | SERVER | SESSION]\_STATS-timestamp.csv

## EDR Client and Server Statistics

Use the following CLI commands to display statistics on the EDRs:

- Client Statistics: EDR's sent by that client on that node to an EDR agent.
- Server Statistics: EDR's sent by that EDR agent on that node to an EDR server.

```
an3000-mcm-slot8cpu1# show cluster node edr-bearer-client-statistics
```

```
an3000-mcm-slot8cpu1# show cluster node edr-flow-client-statistics
```

```
an3000-mcm-slot8cpu1# show cluster node edr-http-client-statistics
```

```
an3000-mcm-slot8cpu1# show cluster node edr-pilot-client-statistics
```

```
an3000-mcm-slot8cpu1# show cluster node edr-rtt-client-statistics
```

```
an3000-mcm-slot8cpu1# show cluster node edr-server-statistics
```

```
an3000-mcm-slot8cpu1# show cluster node edr-session-client-statistics
```

where:

- **edr-bearer-client-statistics** — EDR Bearer Client Statistics
- **edr-flow-client-statistics** — EDR Flow Client Statistics
- **edr-http-client-statistics** — EDR HTTP Client Statistics
- **edr-pilot-client-statistics** — EDR Pilot Client Statistics
- **edr-rtt-client-statistics** — EDR RTT Client Statistics
- **edr-server-statistics** — EDR Server Client Statistics
- **edr-session-client-statistics** — EDR Session Client Statistics

The system displays output similar to the following:

```
an3000-mcm-slot8cpu1# show cluster node edr-server-statistics
```

|       |    |      |           | RECORDS | BYTES    | RECORDS | BYTES  |          |
|-------|----|------|-----------|---------|----------|---------|--------|----------|
| ID    | ID | TASK | NAME      | SUCCESS | SUCCESS  | FAILED  | FAILED | CONNECTS |
| ----- |    |      |           |         |          |         |        |          |
| 43    | 1  |      | ESP       | 42250   | 17074272 | 0       | 0      | 1        |
|       |    |      | ESP_NEW   | 0       | 0        | 0       | 0      | 1        |
|       |    |      | ESP_OTHER | 0       | 0        | 0       | 0      | 0        |
|       | 2  |      | ESP       | 1350    | 285959   | 0       | 0      | 1        |
|       |    |      | ESP_NEW   | 0       | 0        | 0       | 0      | 1        |
|       |    |      | ESP_OTHER | 0       | 0        | 0       | 0      | 0        |
| 5     |    |      |           |         |          |         |        |          |

```
an3000-mcm-slot8cpu1# show cluster node edr-session-client-statistics
```

|    |    |                          | SESSION | SESSION | SESSION | SESSION | SESSION | SESSION | SESSION |         |
|----|----|--------------------------|---------|---------|---------|---------|---------|---------|---------|---------|
|    |    |                          | START   | STOP    | ATTEMPT | INTERIM | START   | STOP    | ATTEMPT | INTERIM |
| ID | ID | TASK                     | SUCCESS | SUCCESS | SUCCESS | SUCCESS | FAILED  | FAILED  | FAILED  | FAILED  |
| 43 | 1  | PDNGW_43_1_1_0_PDNGW_0   | 6       | 6       | 0       | 42      | 0       | 0       | 0       | 0       |
|    |    | PDNGW_43_1_1_10_PDNGW_20 | 7       | 7       | 0       | 49      | 0       | 0       | 0       | 0       |
|    |    | PDNGW_43_1_1_12_PDNGW_24 | 6       | 6       | 0       | 42      | 0       | 0       | 0       | 0       |
|    |    | PDNGW_43_1_1_14_PDNGW_28 | 6       | 6       | 0       | 42      | 0       | 0       | 0       | 0       |
|    |    | PDNGW_43_1_1_2_PDNGW_4   | 6       | 6       | 0       | 42      | 0       | 0       | 0       | 0       |
|    |    | PDNGW_43_1_1_4_PDNGW_8   | 6       | 6       | 0       | 42      | 0       | 0       | 0       | 0       |
|    |    | PDNGW_43_1_1_6_PDNGW_12  | 6       | 6       | 0       | 42      | 0       | 0       | 0       | 0       |
|    |    | PDNGW_43_1_1_8_PDNGW_16  | 7       | 7       | 0       | 49      | 0       | 0       | 0       | 0       |
|    | 2  | PDNGW_43_2_1_11_PDNGW_22 | 7       | 7       | 0       | 49      | 0       | 0       | 0       | 0       |
|    |    | PDNGW_43_2_1_13_PDNGW_26 | 6       | 6       | 0       | 42      | 0       | 0       | 0       | 0       |
|    |    | PDNGW_43_2_1_15_PDNGW_30 | 6       | 6       | 0       | 42      | 0       | 0       | 0       | 0       |
|    |    | PDNGW_43_2_1_1_PDNGW_2   | 6       | 6       | 0       | 42      | 0       | 0       | 0       | 0       |
|    |    | PDNGW_43_2_1_3_PDNGW_6   | 6       | 6       | 0       | 42      | 0       | 0       | 0       | 0       |
|    |    | PDNGW_43_2_1_5_PDNGW_10  | 6       | 6       | 0       | 42      | 0       | 0       | 0       | 0       |
|    |    | PDNGW_43_2_1_7_PDNGW_14  | 6       | 6       | 0       | 42      | 0       | 0       | 0       | 0       |
|    |    | PDNGW_43_2_1_9_PDNGW_18  | 7       | 7       | 0       | 49      | 0       | 0       | 0       | 0       |
| 5  |    |                          |         |         |         |         |         |         |         |         |

## Performance Statistics

an3000-mcm-slot8cpu1# show cluster node edr-bearer-client-statistics  
cluster

| ID | ID | TASK                     | BEARER<br>START | BEARER<br>STOP | BEARER<br>ATTEMPT | BEARER<br>INTERIM | BEARER<br>START | BEARER<br>STOP | BEARER<br>ATTEMPT | BEARER<br>INTERIM |
|----|----|--------------------------|-----------------|----------------|-------------------|-------------------|-----------------|----------------|-------------------|-------------------|
|    |    |                          | SUCCESS         | SUCCESS        | SUCCESS           | SUCCESS           | FAILED          | FAILED         | FAILED            | FAILED            |
| 43 | 1  | PDNGW_43_1_1_0_PDNGW_0   | 12              | 12             | 0                 | 84                | 0               | 0              | 0                 | 0                 |
|    |    | PDNGW_43_1_1_10_PDNGW_20 | 14              | 14             | 0                 | 98                | 0               | 0              | 0                 | 0                 |
|    |    | PDNGW_43_1_1_12_PDNGW_24 | 12              | 12             | 0                 | 84                | 0               | 0              | 0                 | 0                 |
|    |    | PDNGW_43_1_1_14_PDNGW_28 | 12              | 12             | 0                 | 84                | 0               | 0              | 0                 | 0                 |
|    |    | PDNGW_43_1_1_2_PDNGW_4   | 12              | 12             | 0                 | 84                | 0               | 0              | 0                 | 0                 |
|    |    | PDNGW_43_1_1_4_PDNGW_8   | 12              | 12             | 0                 | 84                | 0               | 0              | 0                 | 0                 |
|    |    | PDNGW_43_1_1_6_PDNGW_12  | 12              | 12             | 0                 | 84                | 0               | 0              | 0                 | 0                 |
|    |    | PDNGW_43_1_1_8_PDNGW_16  | 14              | 14             | 0                 | 98                | 0               | 0              | 0                 | 0                 |
|    | 2  | PDNGW_43_2_1_11_PDNGW_22 | 14              | 14             | 0                 | 98                | 0               | 0              | 0                 | 0                 |
|    |    | PDNGW_43_2_1_13_PDNGW_26 | 12              | 12             | 0                 | 84                | 0               | 0              | 0                 | 0                 |
|    |    | PDNGW_43_2_1_15_PDNGW_30 | 12              | 12             | 0                 | 84                | 0               | 0              | 0                 | 0                 |
|    |    | PDNGW_43_2_1_1_PDNGW_2   | 12              | 12             | 0                 | 84                | 0               | 0              | 0                 | 0                 |
|    |    | PDNGW_43_2_1_3_PDNGW_6   | 12              | 12             | 0                 | 84                | 0               | 0              | 0                 | 0                 |
|    |    | PDNGW_43_2_1_5_PDNGW_10  | 12              | 12             | 0                 | 84                | 0               | 0              | 0                 | 0                 |
|    |    | PDNGW_43_2_1_7_PDNGW_14  | 12              | 12             | 0                 | 84                | 0               | 0              | 0                 | 0                 |
|    |    | PDNGW_43_2_1_9_PDNGW_18  | 14              | 14             | 0                 | 98                | 0               | 0              | 0                 | 0                 |

5

an3000-mcm-slot7cpu1# show cluster node edr-flow-client-statistics  
cluster

| ID | ID | TASK                             | FLOW<br>START | FLOW<br>STOP | FLOW<br>INTERIM | FLOW<br>START | FLOW<br>STOP | FLOW<br>INTERIM |
|----|----|----------------------------------|---------------|--------------|-----------------|---------------|--------------|-----------------|
|    |    |                                  | SUCCESS       | SUCCESS      | SUCCESS         | FAILED        | FAILED       | FAILED          |
| 43 | 5  | FPexecMaint_43_5_1_FPexecMaint_0 | 1             | 1            | 2               | 0             | 0            | 0               |

6

an3000-mcm-slot8cpu1# show cluster node edr-http-client-statistics

| ID | ID | TASK                             | HTTP<br>SUCCESS | HTTP<br>FAILED |
|----|----|----------------------------------|-----------------|----------------|
|    |    |                                  | SUCCESS         | FAILED         |
| 43 | 1  | HttpProxy_43_1_1_0_HttpProxy_0   | 4071            | 0              |
|    |    | HttpProxy_43_1_1_10_HttpProxy_10 | 3899            | 0              |
|    |    | HttpProxy_43_1_1_12_HttpProxy_12 | 3928            | 0              |
|    |    | HttpProxy_43_1_1_14_HttpProxy_14 | 3911            | 0              |
|    |    | HttpProxy_43_1_1_16_HttpProxy_16 | 4018            | 0              |
|    |    | HttpProxy_43_1_1_18_HttpProxy_18 | 5196            | 0              |
|    |    | HttpProxy_43_1_1_2_HttpProxy_2   | 3948            | 0              |
|    |    | HttpProxy_43_1_1_4_HttpProxy_4   | 3949            | 0              |
|    |    | HttpProxy_43_1_1_6_HttpProxy_6   | 3908            | 0              |
|    |    | HttpProxy_43_1_1_8_HttpProxy_8   | 4072            | 0              |

2



```
an3000-mcm-slot8cpu1# show cluster nodeedr-pilot-client-statistics
cluster
```

| ID | ID | TASK                     | PILOT<br>START<br>SUCCESS | PILOT<br>STOP<br>SUCCESS | PILOT<br>NAT                | PILOT<br>NAT               | PILOT<br>NAT               | PILOT<br>NAT              | PILOT<br>RAT              | PILOT<br>RAT             |
|----|----|--------------------------|---------------------------|--------------------------|-----------------------------|----------------------------|----------------------------|---------------------------|---------------------------|--------------------------|
|    |    |                          |                           |                          | BINDING<br>START<br>SUCCESS | BINDING<br>STOP<br>SUCCESS | BINDING<br>START<br>FAILED | BINDING<br>STOP<br>FAILED | TYPE<br>CHANGE<br>SUCCESS | TYPE<br>CHANGE<br>FAILED |
| 43 | 1  | PDNGW_43_1_1_0_PDNGW_0   | 12                        | 12                       | 0                           | 84                         | 0                          | 0                         | 0                         | 0                        |
|    |    | PDNGW_43_1_1_10_PDNGW_20 | 14                        | 14                       | 0                           | 98                         | 0                          | 0                         | 0                         | 0                        |
|    |    | PDNGW_43_1_1_12_PDNGW_24 | 12                        | 12                       | 0                           | 84                         | 0                          | 0                         | 0                         | 0                        |
|    |    | PDNGW_43_1_1_14_PDNGW_28 | 12                        | 12                       | 0                           | 84                         | 0                          | 0                         | 0                         | 0                        |
|    |    | PDNGW_43_1_1_2_PDNGW_4   | 12                        | 12                       | 0                           | 84                         | 0                          | 0                         | 0                         | 0                        |
|    |    | PDNGW_43_1_1_4_PDNGW_8   | 12                        | 12                       | 0                           | 84                         | 0                          | 0                         | 0                         | 0                        |
|    |    | PDNGW_43_1_1_6_PDNGW_12  | 12                        | 12                       | 0                           | 84                         | 0                          | 0                         | 0                         | 0                        |
|    |    | PDNGW_43_1_1_8_PDNGW_16  | 14                        | 14                       | 0                           | 98                         | 0                          | 0                         | 0                         | 0                        |
|    | 2  | PDNGW_43_2_1_11_PDNGW_22 | 14                        | 14                       | 0                           | 98                         | 0                          | 0                         | 0                         | 0                        |
|    |    | PDNGW_43_2_1_13_PDNGW_26 | 12                        | 12                       | 0                           | 84                         | 0                          | 0                         | 0                         | 0                        |
|    |    | PDNGW_43_2_1_15_PDNGW_30 | 12                        | 12                       | 0                           | 84                         | 0                          | 0                         | 0                         | 0                        |

```
an3000-mcm-slot8cpu1# show cluster nodeedr-rtt-client-statistics
cluster
```

| node    |                | S5      | S8      | S11     | S4      | S2A     | S2B     | GN      | GP      | GX      |
|---------|----------------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| ID      | TASK           | SUCCESS | SUCCESS | SUCCESS | SUCCESS | SUCCESS | SUCCESS | SUCCESS | SUCCESS | SUCCESS |
| 1       | PDNGW_35_1_1_0 | 0       | 0       | 0       | 0       | 0       | 0       | 0       | 0       | 4       |
| GY      | S6B            | S5      | S8      | S11     | S4      | S2A     | S2B     | GN      | GP      | GX      |
| SUCCESS | SUCCESS        | FAILED  | FAILED  | FAILED  | FAILED  | FAILED  | FAILED  | FAILED  | FAILED  | FAILED  |
| 3       | 0              | 0       | 0       | 0       | 0       | 0       | 0       | 0       | 0       | 0       |

## Procedure Statistics

Use the following CLI command to display statistics of success/failure during the Workflow Control Profile EDR interface ID change handling:

```
an3000-mcm-slot7cpu1# show zone <name> gateway statistics
[cluster|grid]-level procedure [ggsn|pgw|saegw-pgw]
```

The system displays the following statistics:

- **edr-intf-id-change-success:** Total number of Control Profile EDR Interface ID change success
- **edr-intf-id-change-fail:** Total number of Control Profile EDR Interface ID change failed

## Subscriber vTap Statistics

Use the following CLI commands to display vTap statistics.

To display the number of packets/bytes replicated for each subscriber vTap:

```
an3000-mcm-slot8cpu1# show vtap subscriber-tap <name>
```

To display the number of packets/bytes replicated for each vTap sink:

```
an3000-mcm-slot8cpu1# show vtap sink <name>
```

The system displays output similar to the following:

```
vtap sink vtap_sink_2
vtap-sink-stats vtap_sink_1
numingresspkts 0
numingressbytes 0
numegresspkts 0
numegressbytes 0
numingresserrpkts 0
numingresserrbytes 0
numegresserrpkts 0
numegresserrbytes 0
vtap-sink-stats vtap_sink_2
numingresspkts 10274
numingressbytes 684665
numegresspkts 10216
numegressbytes 2358713
numingresserrpkts 0
numingresserrbytes 0
numegresserrpkts 0
numegresserrbytes 0
vtap-sink-stats vtap_sink_3
numingresspkts 0
numingressbytes 0
numegresspkts 0
numegressbytes 0
numingresserrpkts 0
numingresserrbytes 0
numegresserrpkts 0
numegresserrbytes 0
```

## EDR Record Format

This section lists the EDR data record fields and data types.

---

**Note:** For the GiGW, the EDRs contain only the data record fields meaningful on the GiGW. Data record fields that **are not** supported on the GiGW are identified in Notes in the individual data record field descriptions in this chapter.

---

### Base Event Data Record

This file contains the message definition for the base event data record (EDR) message.

The following table lists the name, data type, and description for each base field.

**Table 4-1**    **EventDataRecord Fields**

| Name              | Data Type    | Description                                                                                                                                                                                                                                                                                                                                              |
|-------------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| timestamp         | EdrTimeInfo  | Timestamp expressed by two fields: <ul style="list-style-type: none"><li>■ 1 = Epoch timestamp (uint32)</li><li>■ 2 = Fraction of the current second in microseconds (uint32)</li></ul>                                                                                                                                                                  |
| sourceId          | uint32       | The sourcing entity identifier string or value                                                                                                                                                                                                                                                                                                           |
| sourceSequenceNum | uint32       | Sequence number relative to the sourcing entity                                                                                                                                                                                                                                                                                                          |
| recordType        | EdrType      | Type of record included as one of the optional record instances. Only one of the set of record types is included. The following values are defined: <ul style="list-style-type: none"><li>■ 1 = EDR_STATUS_RECORD</li><li>■ 2 = EDR_SESSION_RECORD</li><li>■ 3 = EDR_BEARER_RECORD</li><li>■ 4 = EDR_FLOW_RECORD</li><li>■ 5 = EDR_HTTP_RECORD</li></ul> |
| statusRecord      | StatusRecord | For details, see <a href="#">STATUS Event Data Record</a> .                                                                                                                                                                                                                                                                                              |

**Table 4-1    EventDataRecord Fields**

| Name          | Data Type     | Description                                                           |
|---------------|---------------|-----------------------------------------------------------------------|
| sessionRecord | SessionRecord | For details, see <a href="#">SESSION Event Data Record</a> .          |
| bearerRecord  | BearerRecord  | For details, see <a href="#">BEARER Event Data Record</a> .           |
| flowRecord    | FlowRecord    | For details, see <a href="#">FLOW Event Data Record</a> .             |
| httpRecord    | HttpRecord    | For details, see <a href="#">HTTP Transaction Event Data Record</a> . |

## STATUS Event Data Record

This file contains the message definition for the STATUS event data record (EDR) message. The STATUS EDR is intended to be used to convey application status and version information between client and server endpoints of an EDR interface. The STATUS EDR is also intended to be used as a keep-alive indicator between endpoints.

The following table lists the name, data type, and description for each STATUS field.

**Table 4-2    StatusRecord Fields**

| Name               | Data Type | Description                                                                                                                                                                                                                                                    |
|--------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Status             |           | <p>The following values are defined:</p> <ul style="list-style-type: none"> <li>■ 1 = STATUS_NORMAL</li> <li>■ 2 = STATUS_ERROR</li> <li>■ 3 = STATUS_CONGESTED</li> </ul>                                                                                     |
| majorVersion       | uint32    | <p>Major version of the interface</p> <p>Current value = 1</p>                                                                                                                                                                                                 |
| minorVersion       | uint32    | <p>Minor version of the interface.</p> <p>This value increments whenever the protobufs change. For example:</p> <ul style="list-style-type: none"> <li>■ For Release 6.5 = 16</li> <li>■ For Release 6.7 or 7.0 = 9</li> <li>■ For Release 7.2 = 10</li> </ul> |
| status             | Status    | Current status of this sending application                                                                                                                                                                                                                     |
| vendorId           | bytes     | Optional vendor identifier field                                                                                                                                                                                                                               |
| keepAliveFrequency | uint32    | Number of seconds between keep alive events on an idle connection used to inform the other endpoint of the keep alive interval                                                                                                                                 |

## SESSION Event Data Record

This file contains the message definition for the SESSION EDR message. The SESSION EDR contains status and statistics information for a PDN session event.

### SessionRecord

The following table lists the name, data type, and description for each top-level SessionRecord field. Several top-level SessionRecord fields contain a set of subfields.

**Table 4-3 SessionRecord Fields**

| Name               | Data Type                             | Description                                                                                                                                                                                                                                                                          |
|--------------------|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| keys               | <a href="#">SessionRecordKeys</a>     | Mandatory key - PDN session identifier. Contains subfields.                                                                                                                                                                                                                          |
| sessionRecordType  | EdrSubType                            | Session record subtype. The following values are defined: <ul style="list-style-type: none"> <li>■ 1 = EDR_START_RECORD</li> <li>■ 2 = EDR_STOP_RECORD</li> <li>■ 3 = EDR_ATTEMPT_RECORD</li> <li>■ 4 = EDR_INTERIM_RECORD</li> </ul>                                                |
| gwType             | EdrGatewayType                        | Gateway type. The following values are defined: <ul style="list-style-type: none"> <li>■ 1 = EDR_GW_TYPE_PGW</li> <li>■ 2 = EDR_GW_TYPE_SGW</li> <li>■ 3 = EDR_GW_TYPE_GGSN</li> <li>■ 4 = EDR_GW_TYPE_MCC</li> <li>■ 5 = EDR_GW_TYPE_WAG</li> <li>■ 6 = EDR_GW_TYPE_EPDG</li> </ul> |
| subscriberInfo     | <a href="#">SessionSubscriberInfo</a> | Subscriber information. Contains subfields.                                                                                                                                                                                                                                          |
| servingNetworkInfo | <a href="#">ServingNetworkInfo</a>    | Serving network information. Contains subfields.                                                                                                                                                                                                                                     |
| gatewayInfo        | <a href="#">GatewayInfo</a>           | Gateway information. Contains subfields.                                                                                                                                                                                                                                             |

**Table 4-3     SessionRecord Fields**

| Name           | Data Type                             | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dataStatistics | DataStatistics                        | <p>Data statistics (uint64). Protocol reporting level for counters with the Octets suffix: The counters are reported at a layer-3 reporting level and include IP header and IP payload. The counters are computed after stripping off GTP-U header for the packets that are received on either S1-U or S5/S8 interfaces.</p> <p>The following values are defined:</p> <ul style="list-style-type: none"> <li>■ 1 = upLinkOctets</li> <li>■ 2 = upLinkPackets</li> <li>■ 3 = downLinkOctets</li> <li>■ 4 = downLinkPackets</li> <li>■ 5 = uplinkSessionActivityDuration</li> <li>■ 6 = downlinkSessionActivityDuration</li> <li>■ 7 = uplinkSessionPeakThroughput (bytes per second)</li> <li>■ 8 = downlinkSessionPeakThroughput (bytes per second)</li> <li>■ 9 = edrSessionBucket</li> </ul> <p>edrSessionBucket is a repeated message of data type <a href="#">EdrSessionBucketInfo</a>. Contains subfields.</p> |
| causeCodes     | <a href="#">EdrCauseCode</a>          | EDR cause codes information. Contains subfields.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| qosInfo        | <a href="#">SessionQosInformation</a> | Quality of service information. Contains subfields.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| userCsgInfo    | <a href="#">SessionUserCSGInfo</a>    | <p>Closed Subscriber Group information. Contains subfields.</p> <p><b>Note:</b> This data record may not be available on GiGW unless it is received in some RADIUS/Diameter message.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

## SessionRecordKeys

The following table lists the name, data type, and description for each subfield contained in the SessionRecordKeys data record.

**Table 4-4 SessionRecordKeys Subfields**

| Name            | Data Type | Description                                                                 |
|-----------------|-----------|-----------------------------------------------------------------------------|
| sessionId       | uint64    | PDN session identifier                                                      |
| uPlaneSessionId | uint64    | Session identifier on U-Plane<br><i>Note: Valid only in a CUPS network.</i> |

## SessionSubscriberInfo

The following table lists the name, data type, and description for each subfield contained in the SessionSubscriberInfo data record.

**Table 4-5 SessionSubscriberInfo Subfields**

| Name              | Data Type                             | Description                                                                                                                                                                                                                                   |
|-------------------|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| imsi              | bytes                                 | International Mobile Subscriber Identity                                                                                                                                                                                                      |
| msisdn            | bytes                                 | Mobile Station International Subscriber Directory Number                                                                                                                                                                                      |
| imeisv            | bytes                                 | International Mobile Station Equipment Identity Software Version                                                                                                                                                                              |
| meid              | bytes                                 | Mobile Equipment Identifier, unique identifier for the mobile device                                                                                                                                                                          |
| mnnai             | bytes                                 | Subscription ID in NAI format                                                                                                                                                                                                                 |
| pdpAddress        | bytes                                 | Subscriber IP address                                                                                                                                                                                                                         |
| pdpType           | EdrIpType                             | Subscriber IP address type. The following values are defined: <ul style="list-style-type: none"> <li>0 = PDP_TYPE_INV</li> <li>1 = PDP_TYPE_IPV4</li> <li>2 = PDP_TYPE_IPV6</li> <li>3 = PDP_TYPE_IPV4V6</li> <li>4 = PDP_TYPE_PPP</li> </ul> |
| msTimezone        | <a href="#">EdrSubscriberTimezone</a> | Subscriber timezone. Contains subfields.                                                                                                                                                                                                      |
| userLocationInfo  | <a href="#">EdrUserLocationInfo</a>   | Subscriber ULI - User Location Info. Contains subfields.                                                                                                                                                                                      |
| pdnConnectionId   | uint64                                | PDN connection ID                                                                                                                                                                                                                             |
| accessOutCtrlTeid | <a href="#">EdrFullQualTeid</a>       | Remote Access Control Teid. Contains subfields.<br><i>Note: TEID information is not valid on the GiGW.</i>                                                                                                                                    |

**Table 4-5     SessionSubscriberInfo Subfields**

| Name               | Data Type                       | Description                                                                                                 |
|--------------------|---------------------------------|-------------------------------------------------------------------------------------------------------------|
| accessInCtrlTeid   | <a href="#">EdrFullQualTeid</a> | Local Access Control Teid. Contains subfields.<br><i>Note: TEID information is not valid on the GiGW.</i>   |
| networkOutCtrlTeid | <a href="#">EdrFullQualTeid</a> | Remote Network Control Teid. Contains subfields.<br><i>Note: TEID information is not valid on the GiGW.</i> |
| networkInCtrlTeid  | <a href="#">EdrFullQualTeid</a> | Local Network Control Teid. Contains subfields.<br><i>Note: TEID information is not valid on the GiGW.</i>  |
| startTime          | uint64                          | Session start time                                                                                          |
| stopTime           | uint64                          | Session end time                                                                                            |
| realApn            | string                          | Final access point name in the Create Session Request.                                                      |
| incomingApn        | string                          | Incoming access point name in the Create Session Request.                                                   |

## ServingNetworkInfo

The following table lists the name, data type, and description for each subfield contained in the ServingNetworkInfo data record.

**Table 4-6     ServingNetworkInfo Subfields**

| Name           | Data Type | Description                                                                                                                                                                                                                                               |
|----------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| nodeAddress    | bytes     | Serving node address                                                                                                                                                                                                                                      |
| nodeIpAddrType | EdrIpType | Serving node IP Address type. The following values are defined: <ul style="list-style-type: none"> <li>■ 0 = PDP_TYPE_INV</li> <li>■ 1 = PDP_TYPE_IPV4</li> <li>■ 2 = PDP_TYPE_IPV6</li> <li>■ 3 = PDP_TYPE_IPV4V6</li> <li>■ 4 = PDP_TYPE_PPP</li> </ul> |



**Table 4-6    ServingNetworkInfo Subfields**

| Name       | Data Type          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| nodeType   | EdrServingNodeType | <p>Serving Node Type. The following values are defined:</p> <ul style="list-style-type: none"> <li>■ 0 = SERVING_NODE_TYPE_SGSN</li> <li>■ 1 = SERVING_NODE_TYPE_PMIPSGW</li> <li>■ 2 = SERVING_NODE_TYPE_GTPSGW</li> <li>■ 3 = SERVING_NODE_TYPE_ePDG</li> <li>■ 4 = SERVING_NODE_TYPE_hSGW</li> <li>■ 5 = SERVING_NODE_TYPE_MME</li> <li>■ 6 = SERVING_NODE_TYPE_TWAN</li> <li>■ 7 = SERVING_NODE_TYPE_GGSN</li> <li>■ 8 = SERVING_NODE_TYPE_GTPPGW</li> </ul> |
| nodePlmnId | EdrPlmnIdType      | Serving network public land mobile network ID. Contains subfields.                                                                                                                                                                                                                                                                                                                                                                                               |
| apnId      | string             | Serving network access point name ID                                                                                                                                                                                                                                                                                                                                                                                                                             |
| ratType    | RatType            | <p>Serving network radio access type.</p> <p>The following values are defined:</p> <ul style="list-style-type: none"> <li>■ 0 = RAT_TYPE_INVALID</li> <li>■ 1 = RAT_TYPE_UTRAN</li> <li>■ 2 = RAT_TYPE_GERAN</li> <li>■ 3 = RAT_TYPE_WLAN</li> <li>■ 4 = RAT_TYPE_GAN</li> <li>■ 5 = RAT_TYPE_HSPA</li> <li>■ 6 = RAT_TYPE_EUTRAN</li> <li>■ 7 = RAT_TYPE_NBIOT_EUTRAN</li> <li>■ 8 = RAT_TYPE_VIRTUAL</li> </ul>                                                |

## GatewayInfo

The following table lists the name, data type, and description for each subfield contained in the GatewayInfo data record.

**Table 4-7 GatewayInfo Subfields**

| Name           | Data Type                     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| gwAddress      | bytes                         | Gateway Address<br><i>Note: This field is not filled on the GiGW.</i>                                                                                                                                                                                                                                                                                                                                                                                      |
| gwIpAddrType   | EdrIpType                     | Gateway IP Address type. The following values are defined: <ul style="list-style-type: none"> <li>■ 0 = PDP_TYPE_INV</li> <li>■ 1 = PDP_TYPE_IPV4</li> <li>■ 2 = PDP_TYPE_IPV6</li> <li>■ 3 = PDP_TYPE_IPV4V6</li> <li>■ 4 = PDP_TYPE_PPP</li> </ul> <i>Note: This field is not filled on the GiGW.</i>                                                                                                                                                    |
| gwNodeID       | string                        | Gateway Node ID                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| gwPlmnId       | <a href="#">EdrPlmnIdType</a> | Gateway public land mobile network ID. Contains subfields.                                                                                                                                                                                                                                                                                                                                                                                                 |
| nodeAddress    | bytes                         | PGW/GGSN node address                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| nodeIpAddrType | EdrIpType                     | PGW/GGSN node IP Address type. The following values are defined: <ul style="list-style-type: none"> <li>■ 0 = PDP_TYPE_INV</li> <li>■ 1 = PDP_TYPE_IPV4</li> <li>■ 2 = PDP_TYPE_IPV6</li> <li>■ 3 = PDP_TYPE_IPV4V6</li> <li>■ 4 = PDP_TYPE_PPP</li> </ul>                                                                                                                                                                                                 |
| nodeType       | EdrServingNode Type           | PGW/GGSN Node Type. The following values are defined: <ul style="list-style-type: none"> <li>■ 0 = SERVING_NODE_TYPE_SGSN</li> <li>■ 1 = SERVING_NODE_TYPE_PMIPSGW</li> <li>■ 2 = SERVING_NODE_TYPE_GTPSGW</li> <li>■ 3 = SERVING_NODE_TYPE_ePDG</li> <li>■ 4 = SERVING_NODE_TYPE_hSGW</li> <li>■ 5 = SERVING_NODE_TYPE_MME</li> <li>■ 6 = SERVING_NODE_TYPE_TWAN</li> <li>■ 7 = SERVING_NODE_TYPE_GGSN</li> <li>■ 8 = SERVING_NODE_TYPE_GTPPGW</li> </ul> |
| nodePlmnId     | <a href="#">EdrPlmnIdType</a> | PGW/GGSN network public land mobile network ID. Contains subfields.                                                                                                                                                                                                                                                                                                                                                                                        |

## EdrSessionBucketInfo

The following table lists the name, data type, and description for each subfield contained in the EdrSessionBucketInfo data record.

**Table 4-8 EdrSessionBucketInfo Subfields**

| Name       | Data Type | Description                                                                                   |
|------------|-----------|-----------------------------------------------------------------------------------------------|
| bucketName | string    | Name of the uplink/downlink bucket                                                            |
| min        | uint32    | Bucket minimum value (in bytes) [default = 0]                                                 |
| max        | uint32    | Bucket maximum value (in bytes) [default = 4294967295]                                        |
| direction  | string    | Bucket direction: <ul style="list-style-type: none"><li>■ uplink</li><li>■ downlink</li></ul> |
| time       | uint32    | Accumulated time in bucket                                                                    |

## EdrCauseCode

The following table lists the name, data type, and description for each subfield contained in the EdrCauseCode data record.

**Table 4-9 EdrCauseCode Subfields**

| Name                    | Data Type                           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| recordTriggerCause      | EdrTriggerCauseCode                 | <p>EDR record creation trigger code. The following values are defined:</p> <ul style="list-style-type: none"> <li>■ 0 = CAUSE_CODE_NORMAL_RELEASE</li> <li>■ 1 = CAUSE_CODE_ABNORMAL_RELEASE</li> <li>■ 2 = CAUSE_CODE_QOS_CHANGE</li> <li>■ 3 = CAUSE_CODE_VOLUME_LIMIT</li> <li>■ 4 = CAUSE_CODE_TIME_LIMIT</li> <li>■ 5 = CAUSE_CODE_SERVING_NODE_CHANGE</li> <li>■ 6 = CAUSE_CODE_SGSN_PLMNID_CHANGE</li> <li>■ 7 = CAUSE_CODE_ULI_CHANGE</li> <li>■ 8 = CAUSE_CODE_RAT_CHANGE</li> <li>■ 9 = CAUSE_CODE_MS_TIMEZONE_CHANGE</li> <li>■ 10 = CAUSE_CODE_TARIFF_TIME_CHANGE</li> <li>■ 11 = CAUSE_CODE_SERVICE_IDLE_OUT</li> <li>■ 12 = CAUSE_CODE_SERVSPEC_UNIT_LIMIT</li> <li>■ 13 = CAUSE_CODE_MAX_CHANGE_CONDITION</li> <li>■ 14 = CAUSE_CODE_CGI_SAI_CHANGE</li> <li>■ 15 = CAUSE_CODE_RAI_CHANGE</li> <li>■ 16 = CAUSE_CODE_ECGI_CHANGE</li> <li>■ 17 = CAUSE_CODE_TAI_CHANGE</li> <li>■ 18 = CAUSE_CODE_SERV_DATA_VOLUME_LIMIT</li> <li>■ 19 = CAUSE_CODE_SERV_DATA_TIME_LIMIT</li> <li>■ 20 = CAUSE_CODE_MANAGEMENT_INTERVENE</li> <li>■ 21 = CAUSE_CODE_SERVICE_STOP</li> <li>■ 22 = CAUSE_CODE_USER_CSG_CHANGE</li> <li>■ 23 = CAUSE_CODE_USER_SGW_CHANGE</li> </ul> |
| closeInfo               | <a href="#">EdrCloseInformation</a> | Session close information. Contains subfields.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| gtpCauseCodeForAttempts | bytes                               | GTP cause codes for session/bearer setup                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

## SessionQosInformation

The following table lists the name, data type, and description for each subfield contained in the SessionQosInformation data record.

**Table 4-10 SessionQosInformation Subfields**

| Name            | Data Type | Description                             |
|-----------------|-----------|-----------------------------------------|
| apnAmbrUplink   | uint64    | Uplink APN Aggregate maximum Bit-Rate   |
| apnAmbrDownlink | uint64    | Downlink APN Aggregate maximum Bit-Rate |

## SessionUserCSGInfo

The following table lists the name, data type, and description for each subfield contained in the SessionUserCSGInfo data record.

**Note:** This data record may not be available on GiGW unless it is received in some RADIUS/Diameter message.

**Table 4-11 SessionUserCSGInfo Subfields**

| Name                 | Data Type | Description                                   |
|----------------------|-----------|-----------------------------------------------|
| csgId                | uint64    | Closed subscriber group identity              |
| accessMode           | uint32    | Closed subscriber group access mode           |
| membershipIndication | uint32    | Closed subscriber group membership indication |

## EdrSubscriberTimezone

The following table lists the name, data type, and description for each subfield contained in the EdrSubscriberTimezone data record.

**Table 4-12 EdrSubscriberTimezone Subfields**

| Name                | Data Type | Description      |
|---------------------|-----------|------------------|
| timezone            | uint32    | MS timezone      |
| daylightSavingsTime | uint32    | Daylight savings |

## EdrUserLocationInfo

The following table lists the name, data type, and description for each subfield contained in the EdrUserLocationInfo data record.

**Table 4-13 EdrUserLocationInfo Subfields**

| Name | Data Type                                   | Description                                      |
|------|---------------------------------------------|--------------------------------------------------|
| cgi  | <a href="#">EdrCellGlobalIdentity</a>       | Cell Global Identity. Contains subfields.        |
| sai  | <a href="#">EdrServiceAreaIdentity</a>      | Service Area Identity. Contains subfields.       |
| rai  | <a href="#">EdrRoutingAreaIdentity</a>      | Routing Area Identity. Contains subfields.       |
| tai  | <a href="#">EdrTrackingAreaIdentity</a>     | Tracking Area Identity. Contains subfields.      |
| ecgi | <a href="#">EdrEUTRANCellGlobalIdentity</a> | Eutran Cell Global Identity. Contains subfields. |
| lai  | <a href="#">EdrLocationAreaIdentity</a>     | Location Area Identity. Contains subfields.      |

## EdrFullQualTeid

The following table lists the name, data type, and description for each subfield contained in the EdrFullQualTeid data record.

**Note:** This data record is not be valid on the GiGW.

**Table 4-14 EdrFullQualTeid Subfields**

| Name     | Data Type           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| intfType | EdrGtpInterfaceType | Interface Type. The following values are defined: <ul style="list-style-type: none"> <li>■ 0 = EGTP_S1U_ENB_GTPU_INTF</li> <li>■ 1 = EGTP_S1U_SGW_GTPU_INTF</li> <li>■ 2 = EGTP_S12_RNC_GTPU_INTF</li> <li>■ 3 = EGTP_S12_SGW_GTPU_INTF</li> <li>■ 4 = EGTP_S5S8_SGW_GTPU_INTF</li> <li>■ 5 = EGTP_S5S8_PGW_GTPU_INTF</li> <li>■ 6 = EGTP_S5S8_SGW_GTPC_INTF</li> <li>■ 7 = EGTP_S5S8_PGW_GTPC_INTF</li> <li>■ 8 = EGTP_S5S8_SGW_PMIPv6_INTF</li> <li>■ 9 = EGTP_S5S8_PGW_PMIPv6_INTF</li> <li>■ 10 = EGTP_S11_MME_GTPC_INTF</li> <li>■ 11 = EGTP_S11S4_SGW_GTPC_INTF</li> <li>■ 12 = EGTP_S10_MME_GTPC_INTF</li> <li>■ 13 = EGTP_S3_MME_GTPC_INTF</li> <li>■ 14 = EGTP_S3_SGSN_GTPU_INTF</li> </ul> |

**Table 4-14** EdrFullQualTeid Subfields

| Name                        | Data Type           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| intfType <i>(Continued)</i> | EdrGtpInterfaceType | <ul style="list-style-type: none"> <li>■ 15 = EGTP_S4_SGSN_GTPU_INTF</li> <li>■ 16 = EGTP_S4_SGW_GTPU_INTF</li> <li>■ 17 = EGTP_S4_SGSN_GTPC_INTF</li> <li>■ 18 = EGTP_S16_SGSN_GTPC_INTF</li> <li>■ 19 = EGTP_DLDF_ENB_GTPU_INTF</li> <li>■ 20 = EGTP_ULDF_ENB_GTPU_INTF</li> <li>■ 21 = EGTP_DF_RNC_GTPU_INTF</li> <li>■ 22 = EGTP_DF_SGSN_GTPU_INTF</li> <li>■ 23 = EGTP_DLDF_SGW_GTPU_INTF</li> <li>■ 28 = EGTP_ULDF_SGW_GTPU_INTF</li> <li>■ 30 = EGTP_S2B_EPDG_GTPC_INTF</li> <li>■ 31 = EGTP_S2B_EPDG_GTPU_INTF</li> <li>■ 32 = EGTP_S2B_PGW_GTPC_INTF</li> <li>■ 33 = EGTP_S2B_PGW_GTPU_INTF</li> <li>■ 34 = EGTP_S2A_TWAN_GTPU_INTF</li> <li>■ 35 = EGTP_S2A_TWAN_GTPC_INTF</li> <li>■ 36 = EGTP_S2A_PGW_GTPC_INTF</li> <li>■ 37 = EGTP_S2A_PGW_GTPU_INTF</li> <li>■ 200 = EGTP_GNGP_GGSN_GTPU_INTF</li> <li>■ 201 = EGTP_GNGP_SGSN_GTPU_INTF</li> <li>■ 202 = EGTP_GNGP_GGSN_GTPC_INTF</li> <li>■ 203 = EGTP_GNGP_SGSN_GTPC_INT</li> <li>■ 255 = EGTP_INVALID_INTF</li> </ul> |
| ipAddress                   | bytes               | IP address                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| teId                        | uint32              | TEID                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

## EdrPlmnIdType

The following table lists the name, data type, and description for each subfield contained in the EdrPlmnIdType data record.

**Table 4-15** EdrPlmnIdType Subfields

| Name | Data Type | Description         |
|------|-----------|---------------------|
| mcc  | uint32    | Mobile Country Code |
| mnc  | uint32    | Mobile Network Code |

## EdrCloseInformation

The following table lists the name, data type, and description for each subfield contained in the EdrCloseInformation data record.

**Table 4-16 EdrCloseInformation Subfields**

| Name                     | Data Type      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| closeInitiator           | CloseInitiator | Session close Initiator. The following values are defined: <ul style="list-style-type: none"> <li>■ 0 = CLOSE_INITIATOR_PCRF</li> <li>■ 1 = CLOSE_INITIATOR_OCS</li> <li>■ 2 = CLOSE_INITIATOR_PGW</li> <li>■ 3 = CLOSE_INITIATOR_SGW</li> <li>■ 4 = CLOSE_INITIATOR_OFCS</li> <li>■ 5 = CLOSE_INITIATOR_RADIUS_AUTH</li> <li>■ 6 = CLOSE_INITIATOR_RADIUS_ACCT</li> <li>■ 7 = CLOSE_INITIATOR_LDAP</li> <li>■ 8 = CLOSE_INITIATOR_S6B</li> <li>■ 9 = CLOSE_INITIATOR_DHCP</li> <li>■ 255 = CLOSE_INITIATOR_INVALID</li> </ul> |
| localExternalCloseReason | uint32         | Session close or session setup failure due to local or external normal or failure reason. The following values are defined: <ul style="list-style-type: none"> <li>■ 0 = CLOSE_REASON_INVALID</li> <li>■ 1 = LOCAL_CLOSE_NORMAL</li> <li>■ 2 = LOCAL_CLOSE_FAILURE</li> <li>■ 3 = EXTERNAL_CLOSE_NORMAL</li> <li>■ 4 = EXTERNAL_CLOSE_FAILURE</li> </ul>                                                                                                                                                                       |

## EdrCellGlobalIdentity

The following table lists the name, data type, and description for each subfield contained in the EdrCellGlobalIdentity data record.

**Table 4-17 EdrCellGlobalIdentity Subfields**

| Name | Data Type | Description         |
|------|-----------|---------------------|
| mcc  | uint32    | Mobile Country Code |
| mnc  | uint32    | Mobile Network Code |
| lac  | uint32    | Location Area Code  |
| ci   | uint32    | Cell Identity       |



## EdrServiceAreaIdentity

The following table lists the name, data type, and description for each subfield contained in the EdrServiceAreaIdentity data record.

**Table 4-18 EdrServiceAreaIdentity Subfields**

| Name | Data Type | Description         |
|------|-----------|---------------------|
| mcc  | uint32    | Mobile Country Code |
| mnc  | uint32    | Mobile Network Code |
| lac  | uint32    | Location Area Code  |
| sac  | uint32    | Service Area Code   |

## EdrRoutingAreaIdentity

The following table lists the name, data type, and description for each subfield contained in the EdrRoutingAreaIdentity data record.

**Table 4-19 EdrRoutingAreaIdentity Subfields**

| Name | Data Type | Description         |
|------|-----------|---------------------|
| mcc  | uint32    | Mobile Country Code |
| mnc  | uint32    | Mobile Network Code |
| lac  | uint32    | Location Area Code  |
| rac  | uint32    | Routing Area Code   |

## EdrTrackingAreaIdentity

The following table lists the name, data type, and description for each subfield contained in the EdrTrackingAreaIdentity data record.

**Table 4-20 EdrTrackingAreaIdentity Subfields**

| Name | Data Type | Description         |
|------|-----------|---------------------|
| mcc  | uint32    | Mobile Country Code |
| mnc  | uint32    | Mobile Network Code |
| tac  | uint32    | Tracking Area Code  |

## EdrEUTRANCellGlobalIdentity

The following table lists the name, data type, and description for each subfield contained in the EdrEUTRANCellGlobalIdentity data record.

**Table 4-21 EdrEUTRANCellGlobalIdentity Subfields**

| Name | Data Type | Description             |
|------|-----------|-------------------------|
| mcc  | uint32    | Mobile Country Code     |
| mnc  | uint32    | Mobile Network Code     |
| eci  | uint32    | E-UTRAN Cell Identifier |

## EdrLocationAreaIdentity

The following table lists the name, data type, and description for each subfield contained in the EdrLocationAreaIdentity data record.

**Table 4-22 EdrLocationAreaIdentity Subfields**

| Name       | Data Type | Description                       |
|------------|-----------|-----------------------------------|
| mcc        | uint32    | Mobile Country Code               |
| mnc        | uint32    | Mobile Network Code               |
| lac        | uint32    | Location Area Code                |
| mnc length | uint32    | Length of the Mobile Network Code |

## BEARER Event Data Record

This file contains the message definition for the BEARER event data record (EDR) message. The BEARER EDR contains status and statistics information for a PDN bearer connectivity event.

### BearerRecord

The following table lists the name, data type, and description for each top-level BearerRecord field. Several top-level BearerRecord fields contains a set of subfields.

**Table 4-23 BearerRecord Subfields**

| Name             | Data Type                            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| keys             | <a href="#">BearerRecordKeys</a>     | Bearer Record Keys. Contains subfields.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| bearerRecordType | EdrSubType                           | Bearer record subtype. The following values are defined: <ul style="list-style-type: none"> <li>■ 1 = EDR_START_RECORD</li> <li>■ 2 = EDR_STOP_RECORD</li> <li>■ 3 = EDR_ATTEMPT_RECORD</li> <li>■ 4 = EDR_INTERIM_RECORD</li> </ul>                                                                                                                                                                                                                                                                                                    |
| bearerInfo       | <a href="#">BearerInfo</a>           | Bearer information. Contains subfields.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| causeCodes       | <a href="#">EdrCauseCode</a>         | EDR cause codes information. Contains subfields.<br><i><b>Note:</b> For BearerRecord, the only causeCode set is recordTriggerCause. The other causeCode fields (closeInfo and gtpCauseCodeForAttempts) are not set in bearer EDRs.</i>                                                                                                                                                                                                                                                                                                  |
| qosInfo          | <a href="#">BearerQosInformation</a> | Quality of service information. Contains subfields.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| dataStatistics   | DataStatistics                       | Data statistics (uint64). The following values are defined: <ul style="list-style-type: none"> <li>■ 1 = upLinkOctets</li> <li>■ 2 = upLinkPackets</li> <li>■ 3 = downLinkOctets</li> <li>■ 4 = downLinkPackets</li> </ul> <p><b>Protocol reporting level counters with the Octets suffix:</b></p> <p>The counters are reported at a layer-3 reporting level and include IP header and IP payload. The counters are computed after stripping off GTP-U header for the packets that are received on either S1-U or S5/S8 interfaces.</p> |

## BearerRecordKeys

The following table lists the name, data type, and description for each subfield contained in the BearerRecordKeys data record.

**Table 4-24 BearerRecordKeys Subfields**

| Name            | Data Type | Description                                                                 |
|-----------------|-----------|-----------------------------------------------------------------------------|
| sessionId       | uint64    | PDN session identifier                                                      |
| bearerId        | uint64    | Logical Bearer ID                                                           |
| uPlaneSessionId | uint64    | Session identifier on U-Plane<br><i>Note: Valid only in a CUPS network.</i> |

## BearerInfo

The following table lists the name, data type, and description for each subfield contained in the BearerInfo data record.

**Table 4-25 BearerInfo Subfields**

| Name               | Data Type                       | Description                                                                                              |
|--------------------|---------------------------------|----------------------------------------------------------------------------------------------------------|
| chargingId         | uint64                          | Charging ID                                                                                              |
| epsBearerId        | uint32                          | EPS Bearer ID                                                                                            |
| isDefault          | bool                            | True if this is a default bearer, else False                                                             |
| accessOutUserTeid  | <a href="#">EdrFullQualTeid</a> | Remote Access User Teid. Contains subfields.<br><i>Note: TEID information is not valid on the GiGW.</i>  |
| accessInUserTeid   | <a href="#">EdrFullQualTeid</a> | Local Access User Teid. Contains subfields.<br><i>Note: TEID information is not valid on the GiGW.</i>   |
| networkOutUserTeid | <a href="#">EdrFullQualTeid</a> | Remote Network User Teid. Contains subfields.<br><i>Note: TEID information is not valid on the GiGW.</i> |
| networkInUserTeid  | <a href="#">EdrFullQualTeid</a> | Local Network User Teid. Contains subfields.<br><i>Note: TEID information is not valid on the GiGW.</i>  |
| startTime          | uint64                          | Bearer setup time. Contains subfields.                                                                   |
| stopTime           | uint64                          | Bearer stop time. Contains subfields                                                                     |
| imsi               | uint64                          | International Mobile Subscriber Identity                                                                 |

## BearerQosInformation

The following table lists the name, data type, and description for each subfield contained in the BearerQosInformation data record.

**Table 4-26 BearerQosInformation Subfields**

| Name                   | Data Type                              | Description                                        |
|------------------------|----------------------------------------|----------------------------------------------------|
| qosClassId             | uint32                                 | QoS class ID                                       |
| mbrUplink              | uint64                                 | Uplink maximum bit-rate in kbps                    |
| mbrDownlink            | uint64                                 | Downlink maximum bit-rate in kbps                  |
| gbrUplink              | uint64                                 | Uplink guaranteed bit-rate in kbps                 |
| gbrDownlink            | uint64                                 | Downlink guaranteed bit-rate in kbps               |
| allocRetentionPriority | <a href="#">AllocRetentionPriority</a> | Allocation-retention-priority. Contains subfields. |

## AllocRetentionPriority

The following table lists the name, data type, and description for each subfield contained in the AllocRetentionPriority data record.

**Table 4-27 AllocRetentionPriority Subfields**

| Name                        | Data Type | Description                   |
|-----------------------------|-----------|-------------------------------|
| allocRetentionPriorityLevel | uint32    | Allocation Retention Priority |
| preemptionCapability        | uint32    | Mobile Network Code           |
| preemptionVulnerability     | uint32    | Preemption Vulnerability      |

## FLOW Event Data Record

This file contains the message definition for the FLOW event data record (EDR) message. The FLOW EDR contains status and statistics information for an internal data flow within a PDN session and bearer.

### FlowRecord

The following table lists the name, data type, and description for each top-level FlowRecord field. Several top-level FlowRecord fields contains a set of subfields.

**Table 4-28 FlowRecord Subfields**

| Name           | Data Type                      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| keys           | <a href="#">FlowRecordKeys</a> | Flow record key information. Contains subfields.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| flowRecordType | EdrSubType                     | Type of flow record. The following values are defined: <ul style="list-style-type: none"> <li>■ 1 = EDR_START_RECORD</li> <li>■ 2 = EDR_STOP_RECORD</li> <li>■ 3 = EDR_ATTEMPT_RECORD</li> <li>■ 4 = EDR_INTERIM_RECORD</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| ipTuple        | <a href="#">ipTuple</a>        | IP flow 5-tuple information. Contains subfields.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| creationtime   | EdrTimeInfo                    | Timestamp when the flow was created: <ul style="list-style-type: none"> <li>■ 1 = Epoch timestamp (uint32)</li> <li>■ 2 = Fraction of the current second in microseconds</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| lastaccesstime | EdrTimeInfo                    | Timestamp when data was last present on the flow: <ul style="list-style-type: none"> <li>■ 1 = Epoch timestamp (uint32)</li> <li>■ 2 = Fraction of the current second in microseconds</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| dataStats      | DataStatistics                 | Data statistics (uint64). The following values are defined: <ul style="list-style-type: none"> <li>■ 1 = upLinkOctets</li> <li>■ 2 = upLinkPackets</li> <li>■ 3 = downLinkOctets</li> <li>■ 4 = downLinkPackets</li> <li>■ 5 = upLinkDropOctets</li> <li>■ 6 = upLinkDropPackets</li> <li>■ 7 = downLinkDropOctets</li> <li>■ 8 = downLinkDropPackets</li> </ul> <p><b>Protocol reporting level for counters with the Octets suffix:</b></p> <p>The counters are reported at a layer-3 reporting level and include IP header and IP payload. The counters are computed after stripping off GTP-U header for the packets that are received on either S1-U or S5/S8 interfaces.</p> |

Table 4-28 FlowRecord Subfields

| Name               | Data Type                          | Description                                                                                                                                                                                                                                                                                                                                           |
|--------------------|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| subscriberInfo     | <a href="#">FlowSubscriberInfo</a> | Subscriber information. Contains subfields.                                                                                                                                                                                                                                                                                                           |
| tcpRetransInfo     | tcpRetransInfo                     | TCP retransmission statistics (uint64). The following values are defined: <ul style="list-style-type: none"> <li>■ 1 = uplinkRetransPackets</li> <li>■ 2 = uplinkRetransBytes</li> <li>■ 3 = downlinkRetransPackets</li> <li>■ 4 = downlinkRetransBytes</li> </ul>                                                                                    |
| networkPerfInfo    | <a href="#">NetworkPerfInfo</a>    | Network performance information. Contains subfields.                                                                                                                                                                                                                                                                                                  |
| networkStatsInfo   | <a href="#">NetworkStatsInfo</a>   | Network statistics information. Contains subfields.                                                                                                                                                                                                                                                                                                   |
| dpiStringInfo      | string                             | DPI information in string form for the flow.                                                                                                                                                                                                                                                                                                          |
| tetheredFlow       | bool                               | Indicates whether the flow is tethered. The following values are defined: <ul style="list-style-type: none"> <li>■ 0 = False</li> <li>■ 1 = True</li> </ul>                                                                                                                                                                                           |
| tetheringEntitled  | bool                               | Indicates whether the subscriber is entitled to perform tethering. The following values are defined: <ul style="list-style-type: none"> <li>■ 0 = null</li> <li>■ 1 = True</li> </ul> <p><b>Note:</b> Is displayed in flow record only if the flow is detected as tethered and the flow is tethering entitled.</p>                                    |
| tetheringAlgorithm | string                             | Tethering detection algorithm used to detect the flow as tethered. The following values are defined: <ul style="list-style-type: none"> <li>■ 1 = TTL</li> <li>■ 2 = USER-AGENT</li> <li>■ 3 = TCP-FINGERPRINTING</li> <li>■ 4 = IP-FINGERPRINTING</li> <li>■ 5 = GQUIC-USER-AGENT</li> </ul>                                                         |
| qualityIndex       | uint64                             | Reported percentage value for Quality Index (Confidence Index) to mark connection as tethered.<br><br>A connection is marked as tethered only if this reported qualityIndex is greater than or equal to the detection algorithm QI value (configured using the <b>data-path dpi tethering-quality-index value &lt;QI&gt;</b> command [default = 70]). |
| natBindingInfo     | <a href="#">NatBindingInfo</a>     | NAT binding information. Contains subfields.                                                                                                                                                                                                                                                                                                          |

**Table 4-28** FlowRecord Subfields

| Name        | Data Type | Description                                                                                                                                                                       |
|-------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proxyBypass | bool      | Indicates whether the connection is marked as proxy bypassed.<br>The following values are defined: <ul style="list-style-type: none"><li>■ 0 = False</li><li>■ 1 = True</li></ul> |



## FlowRecordKeys

The following table lists the name, data type, and description for each subfield contained in the FlowRecordKeys data record.

**Table 4-29 FlowRecordKeys Subfields**

| Name            | Data Type | Description                   |
|-----------------|-----------|-------------------------------|
| sessionId       | uint64    | PDN session identifier        |
| bearerId        | uint64    | PDN bearer identifier         |
| flowId          | uint64    | Connection flow identifier    |
| cPlaneSessionId | uint64    | Session identifier on C-Plane |

## ipTuple

The following table lists the name, data type, and description for each subfield contained in the ipTuple data record.

**Table 4-30 ipTuple Subfields**

| Name             | Data Type | Description                                                                |
|------------------|-----------|----------------------------------------------------------------------------|
| ueIpAddress      | bytes     | IP address of the UE that is either the source or destination of the flow. |
| uePort           | uint62    | Transport layer port number used by the UE.                                |
| networkIpAddress | bytes     | IP address of the destination that UE is communicating with.               |
| networkPort      | uint32    | Transport layer port number used by the destination.                       |
| protocol         | uint32    | Transport layer protocol ID.                                               |

## FlowSubscriberInfo

The following table lists the name, data type, and description for each subfield contained in the FlowSubscriberInfo data record.

**Table 4-31 FlowSubscriberInfo Subfields**

| Name | Data Type | Description                              |
|------|-----------|------------------------------------------|
| Imsi | uint64    | International Mobile Subscriber Identity |

## NetworkPerfInfo

The following table lists the name, data type, and description for each subfield contained in the NetworkPerfInfo data record.

**Table 4-32 NetworkPerfInfo Subfields**

| Name         | Data Type   | Description                                                                                                                                         |
|--------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| initialRTT   | EdrTimeInfo | Initial Round Trip Time: Measure of the time elapsed between UE TCP SYN send and ACK received, signaling that the TCP session has been established. |
| HttpTtfbTime | EdrTimeInfo | Time to First Byte: Measure of the time elapsed between the UE TCP SYN send and ACK for first TCP data packet received.                             |

## NetworkStatsInfo

The following table lists the name, data type, and description for each subfield contained in the NetworkStatsInfo data record.

**Table 4-33 NetworkStatsInfo Subfields**

| Name               | Data Type   | Description                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| initialuplinkRTT   | EdrTimeInfo | Initial Round Trip Time in uplink direction. It is the time interval between TCP SYN and TCP SYN/ACK packet. It applies only to TCP connections.<br><br>In case of Proxy, it is calculated between UE and Proxy server.                                                                                                                                                                                              |
| initialdownlinkRTT | EdrTimeInfo | Initial Round Trip Time in downLink direction. It is the time interval between TCP SYN/ACK and TCP ACK packet. It applies only to TCP connections.<br><br>In case of Proxy, it is calculated between UE and Proxy server.                                                                                                                                                                                            |
| activitydurationUL | EdrTimeInfo | Duration between first and last packet received in the connection in uplink minus inactivity periods (period with no traffic during more than 500ms). Frames without payload (TCP Syn, TCP Ack) are not included in the activity duration calculation. This field is available in Flow EDRs (EDR_INTERIM_RECORD and EDR_STOP_RECORD). It has the value 0 in EDR_START_RECORD. It applied to TCP and UDP connections. |

**Table 4-33 NetworkStatsInfo Subfields**

| Name               | Data Type   | Description                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| activitydurationDL | EdrTimeInfo | Duration between first and last packet received in the connection in downlink minus inactivity periods (period with no traffic during more than 500ms). Frames without payload (TCP Syn, TCP Ack) are not included in the activity duration calculation. This field is available in Flow EDRs (EDR_INTERIM_RECORD and EDR_STOP_RECORD). It has the value 0 in EDR_START_RECORD. It applies to TCP and UDP connections. |
| peakthroughputUL   | uint64      | Peak Throughput (in Bps) for uplink with interval 1 second. This field is available in Flow EDRs (EDR_INTERIM_RECORD and EDR_STOP_RECORD). It has the value 0 in EDR_START_RECORD. It applies to TCP and UDP connections.                                                                                                                                                                                              |
| peakthroughputDL   | uint64      | Peak Throughput (in Bps) for downlink with interval 1 second. This field is available in Flow EDRs (EDR_INTERIM_RECORD and EDR_STOP_RECORD). It has the value 0 in EDR_START_RECORD. It applies to TCP and UDP connections.                                                                                                                                                                                            |

## NatBindingInfo

The following table lists the name, data type, and description for each subfield contained in the NatBindingInfo data record.

**Table 4-34 NatBindingInfo Subfields**

| Name         | Data Type | Description                             |
|--------------|-----------|-----------------------------------------|
| natIpAddress | bytes     | Connection's NAT IP Address information |
| natPort      | uint32    | Connection's NAT port information       |

**Note:** If an SSM failure occurs, the session is migrated to the backup SSM and new NAT resources are allocated to ongoing connections. Therefore, a single connection may appear with different NAT resources (ip/port) in subsequent flow EDRs.

## HTTP Transaction Event Data Record

This file contains the message definition for the HTTP Transaction EDR message. HTTP Transaction Data Records provides information about each HTTP Transaction.

### HTTPTransactionRecord

The following table lists the name, data type, and description for each top-level HTTPTransactionRecord field. The top-level HTTPTransactionRecord fields contain a set of subfields.

**Table 4-35 HTTPTransactionRecord Fields**

| Name                    | Description                                                |
|-------------------------|------------------------------------------------------------|
| timeInfo                | For details, see <a href="#">timeInfo</a> .                |
| subscriberInfo          | For details, see <a href="#">subscriberInfo</a> .          |
| connectionInfo          | For details, see <a href="#">connectionInfo</a> .          |
| requestInfo             | For details, see <a href="#">requestInfo</a> .             |
| responseInfo            | For details, see <a href="#">responseInfo</a> .            |
| compressionInfo         | For details, see <a href="#">compressionInfo</a> .         |
| webImageOptInfo         | For details, see <a href="#">webImageOptInfo</a> .         |
| videoInfo               | For details, see <a href="#">videoInfo</a> .               |
| minifyInfo              | For details, see <a href="#">minifyInfo</a> .              |
| preemptDnsInfo          | For details, see <a href="#">preemptDnsInfo</a> .          |
| cacheInfo               | For details, see <a href="#">cacheInfo</a> .               |
| tcpSplicingInfo         | For details, see <a href="#">tcpSplicingInfo</a> .         |
| manifestInfo            | For details, see <a href="#">manifestInfo</a> .            |
| tcpClientConnectionInfo | For details, see <a href="#">tcpClientConnectionInfo</a> . |
| filterInfo              | For details, see <a href="#">filterInfo</a> .              |
| tcpServerConnectionInfo | For details, see <a href="#">tcpServerConnectionInfo</a> . |
| entitlementInfo         | For details, see <a href="#">entitlementInfo</a> .         |
| udpProxyInfo            | For details, see <a href="#">udpProxyInfo</a> .            |
| DpiInformation          | For details, see <a href="#">DpiInformation</a> .          |
| SHIPInformation         | For details, see <a href="#">SHIPInformation</a> .         |

## timeInfo

The following table lists the name, data type, and description for each subfield contained in the timeInfo data record.

**Table 4-36** timeInfo Subfields

| Name                      | Data Type | Description                                                                                                                                                                                                                                                                    |
|---------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| transactionTimestamp      | int64     | The timestamp indicating when the HTTP transaction starts, in Epoch time (seconds).                                                                                                                                                                                            |
| requestLatency            | uint64    | Indicates the amount of time, in milliseconds, it takes the HTTP proxy to forward an HTTP request to the server. (Starts counting from the moment the HTTP proxy receives the first byte of data from the client to when it sends the last byte of data to the server.)        |
| responseLatency           | uint64    | Indicates the amount of time, in milliseconds, it takes the HTTP proxy to forward an HTTP response to the client. (Starts counting from the moment the HTTP proxy receives the first byte of data from the server to when it sends the last byte of data to the client.)       |
| transactionDuration       | uint64    | Indicates the total amount of time of the transaction, in milliseconds.                                                                                                                                                                                                        |
| transactionTimeSubSeconds | int32     | The millisecond portion of the timestamp, indicating when the HTTP transaction starts.<br><br><i><b>Note:</b> Field 1 and Field 5 together provide the date and time of the transaction to the millisecond level.</i>                                                          |
| transactionId             | int32     | Proxy Transaction ID                                                                                                                                                                                                                                                           |
| recordEntity              | string    | Proxy Process Name<br><br>If the vm-name is configured for a cluster node, the vm-name is included in this field. If the vm-name is not configured for a cluster node, a generic string, such as node_personality-node_number, is used. For example, ISM-5, CSM-1, ASM-2, etc. |

## subscriberInfo

The following table lists the name, data type, and description for each subfield contained in the subscriberInfo data record.

**Table 4-37 subscriberInfo Subfields**

| Name          | Data Type | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| imsi          | uint64    | Indicates the International Mobile Subscriber Identity (IMSI) of the User Equipment (UE) that originated the request.                                                                                                                                                                                                                                                                                                                                                                        |
| imei          | uint64    | Indicates the International Mobile Equipment Identity (IMEI) of the UE that originated the request.                                                                                                                                                                                                                                                                                                                                                                                          |
| msisdn        | uint64    | Indicates the primary Mobile Station (MS) ISDN number (MSISDN) of the subscriber.                                                                                                                                                                                                                                                                                                                                                                                                            |
| sessionId     | uint32    | Identifies the Mobile Content Cloud's session ID for the subscriber's PDN connection.                                                                                                                                                                                                                                                                                                                                                                                                        |
| servingMcc    | uint32    | Indicates the Mobile Country Code (MCC) of the serving network.                                                                                                                                                                                                                                                                                                                                                                                                                              |
| servingMnc    | uint32    | Indicates the Mobile Network Code (MNC) of the serving network.                                                                                                                                                                                                                                                                                                                                                                                                                              |
| plmnIdLength  | uint32    | Indicates the length of the home Public Land Mobile Network (PLMN) Identifier.                                                                                                                                                                                                                                                                                                                                                                                                               |
| ratType       | uint32    | Indicates the Radio Access Technology (RAT) type currently used by the Mobile Station, when available. The following values are defined: <ul style="list-style-type: none"> <li>■ 0 = Reserved</li> <li>■ 1 = UTRAN</li> <li>■ 2 = GERAN</li> <li>■ 3 = WLAN</li> <li>■ 4 = GAN</li> <li>■ 5 = HSPA-Evolution</li> <li>■ 6 = EUTRAN</li> <li>■ 101 = IEEE 802.16e</li> <li>■ 102 = 3GPP2-eHRPD</li> <li>■ 103 = 3GPP2-HRPD</li> <li>■ 104 = 3GPP2-1xRTT</li> <li>■ 105 = 3GPP-EPS</li> </ul> |
| roamingStatus | bool      | Indicates whether the subscriber is roaming. The following values are defined: <ul style="list-style-type: none"> <li>■ 0 = False</li> <li>■ 1 = True</li> </ul>                                                                                                                                                                                                                                                                                                                             |
| sgwIp         | bytes     | Indicates the IP address of the SGW used (can be an IPv4 or IPv6 address).                                                                                                                                                                                                                                                                                                                                                                                                                   |
| realApn       | string    | Indicates the logical name of the mapped access point to the external packet data network.                                                                                                                                                                                                                                                                                                                                                                                                   |

**Table 4-37 subscriberInfo Subfields**

| Name                  | Data Type                           | Description                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| originalApn           | string                              | Indicates the logical name of the connected access point to the external packet data network.                                                                                                                                                                                                                                                                                                                     |
| deviceType            | enum                                | Indicates the subscriber device type. The following values are defined: <ul style="list-style-type: none"> <li>■ 1 = Handheld</li> <li>■ 2 = FeaturePhone</li> <li>■ 3 = SmartPhone</li> <li>■ 4 = Modem</li> <li>■ 5 = Dongle</li> <li>■ 6 = Tablet</li> <li>■ 7 = Vehicle</li> <li>■ 8 = ConnectedDevice</li> <li>■ 9 = Router</li> <li>■ 10 = Module</li> <li>■ 11 = eBook</li> <li>■ 12 = FemotoAp</li> </ul> |
| deviceMake            | string                              | Indicates the subscriber device make (brand)                                                                                                                                                                                                                                                                                                                                                                      |
| deviceModel           | string                              | Indicates the subscriber device model                                                                                                                                                                                                                                                                                                                                                                             |
| deviceScreenWidth     | uint32                              | Indicates the subscriber device screen width                                                                                                                                                                                                                                                                                                                                                                      |
| deviceScreenHeight    | uint32                              | Indicates the subscriber device screen height                                                                                                                                                                                                                                                                                                                                                                     |
| cPlaneSessionId       | uint64                              | Session identifier on C-Plane                                                                                                                                                                                                                                                                                                                                                                                     |
| proxyInstanceName     | string                              | Proxy Service Configuration Name                                                                                                                                                                                                                                                                                                                                                                                  |
| sessOperatorSpecific1 | string                              | Operator-specific data                                                                                                                                                                                                                                                                                                                                                                                            |
| userLocationInfo      | bytes                               | User location information                                                                                                                                                                                                                                                                                                                                                                                         |
| userLocationInfoMsg   | <a href="#">EdrUserLocationInfo</a> | User location information message                                                                                                                                                                                                                                                                                                                                                                                 |

## connectionInfo

The following table lists the name, data type, and description for each subfield contained in the connectionInfo data record.

**Table 4-38 connectionInfo Subfields**

| Name            | Data Type | Description                                                                                                                                                                                                                                                                                                                                            |
|-----------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| giNc            | string    | Indicates the network context of the Gi interface.                                                                                                                                                                                                                                                                                                     |
| sourceIp        | byte      | Indicates the source IP address of the IP packets issued by the UE (can be an IPv4 or IPv6 address).                                                                                                                                                                                                                                                   |
| sourcePort      | uint32    | Indicates the source port of the IP packets received from the UE.                                                                                                                                                                                                                                                                                      |
| destinationIp   | byte      | Indicates the destination IP address of the IP packets issued by the UE (can be an IPv4 or IPv6 address).                                                                                                                                                                                                                                              |
| destinationPort | uint32    | Indicates the destination port of the IP packets received from the UE.                                                                                                                                                                                                                                                                                 |
| ipTransparency  | bool      | Indicates whether IP-transparency is enabled or disabled. With IP-transparency, HTTP requests forwarded by the HTTP proxy to origin servers contain the UE's original IP address as the source TCP/IP address in the IP packet. The following values are defined: <ul style="list-style-type: none"><li>■ 0 = Disabled</li><li>■ 1 = Enabled</li></ul> |



## requestInfo

The following table lists the name, data type, and description for each subfield contained in the requestInfo data record.

**Table 4-39 requestInfo Subfields**

| Name           | Data Type          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| requestMethod  | RequestMethod      | <p>Indicates the reason for issuing the HTTP request. The following values are defined:</p> <ul style="list-style-type: none"> <li>■ 0 = Unknown</li> <li>■ 1 = OPTIONS</li> <li>■ 2 = GET</li> <li>■ 3 = HEAD</li> <li>■ 4 = POST</li> <li>■ 5 = PUT</li> <li>■ 6 = DELETE</li> <li>■ 7 = TRACE</li> <li>■ 8 = CONNECT</li> <li>■ 9 = MKCOL</li> <li>■ 10 = COPY</li> <li>■ 11 = MOVE</li> <li>■ 12 = PROPFIND</li> <li>■ 13 = PROPATCH</li> <li>■ 14 = LOCK</li> <li>■ 15 = UNLOCK</li> <li>■ 16 = PATCH</li> </ul> |
| clientUrl      | string             | Indicates the URL the client (UE) used to make the request.                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| serverUrl      | string             | <p>Indicates the URL sent to the server.</p> <p><b>Note:</b> This field is filled only if the HTTP proxy updates the URL sent to the server.</p>                                                                                                                                                                                                                                                                                                                                                                      |
| clientProtocol | ConnectionProtocol | <p>Indicates the protocol used by the client to communicate with the server. The following values are defined:</p> <ul style="list-style-type: none"> <li>■ 0 = TCP</li> <li>■ 1 = TLS</li> <li>■ 2 = WAP 1.x Connectionless</li> <li>■ 3 = WAP 1.x Connection Oriented</li> <li>■ 4 = WAP 1.x Connectionless with Security</li> <li>■ 5 = WAP 1.x Connection Oriented with Security</li> </ul>                                                                                                                       |

**Table 4-39 requestInfo Subfields**

| Name             | Data Type          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| serverUrl        | ConnectionProtocol | <p>Indicates the protocol used by the server to communicate with the client. The following values are defined:</p> <ul style="list-style-type: none"> <li>■ 0 = TCP</li> <li>■ 1 = TLS</li> <li>■ 2 = WAP 1.x Connectionless</li> <li>■ 3 = WAP 1.x Connection Oriented</li> <li>■ 4 = WAP 1.x Connectionless with Security</li> <li>■ 5 = WAP 1.x Connection Oriented with Security</li> </ul>                                                                                                                                                                                                                                                                                                         |
| userAgent        | string             | Identifies the client software originating the request as received by the HTTP proxy.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| contentType      | string             | <p>Indicates the kind of content contained in the HTTP request. Only applicable for the HTTP POST request method. The following values are defined:</p> <ul style="list-style-type: none"> <li>■ 0 = Not Applicable</li> <li>■ 1 = video/x-flv</li> <li>■ 2 = video/mp4</li> <li>■ 3 = video/webm</li> <li>■ 4 = video/other</li> <li>■ 5 = app/octetstream</li> <li>■ 6 = app/javascript</li> <li>■ 7 = text/javascript</li> <li>■ 8 = text/plain</li> <li>■ 9 = text/html</li> <li>■ 10 = text/css</li> <li>■ 11 = image/bmp</li> <li>■ 12 = image/gif</li> <li>■ 13 = image/jpeg</li> <li>■ 14 = image/png</li> <li>■ 15 = video/DASH</li> <li>■ 16 = video/HLS</li> <li>■ 17 = video/HDS</li> </ul> |
| contentLength    | uint64             | Indicates the amount of data (in bytes) in the Content-Length Header of the HTTP request from the client.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| clientHeaderSize | uint64             | Indicates the size of the headers in the HTTP request from the client, in bytes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| clientBodySize   | uint64             | Indicates the size of the body in the HTTP request from the client, in bytes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| serverHeaderSize | uint64             | Indicates the size of the Header in the HTTP request sent from the HTTP proxy to the server, in bytes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

**Table 4-39 requestInfo Subfields**

| Name                 | Data Type | Description                                                                                                                                                                                                                                                                                        |
|----------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| serverBodySize       | uint64    | Indicates the size of the body in the HTTP request sent from the HTTP proxy to the sever, in bytes.                                                                                                                                                                                                |
| rangeRequest         | bool      | Indicates whether the HTTP request is a Range Request. Range Requests are requests for a subset of a larger file based on one or more ranges. The following values are defined: <ul style="list-style-type: none"> <li>■ 0 = False</li> <li>■ 1 = True</li> </ul>                                  |
| requestSuccessful    | bool      | Indicates whether the HTTP request is successful. The following values are defined: <ul style="list-style-type: none"> <li>■ 0 = False</li> <li>■ 1 = True</li> </ul>                                                                                                                              |
| chunked              | bool      | Indicates whether the HTTP request is a chunked. The following values are defined: <ul style="list-style-type: none"> <li>■ 0 = False</li> <li>■ 1 = True</li> </ul>                                                                                                                               |
| explicitProxyRequest | bool      | Indicates whether the HTTP request is an explicit proxy request. With explicit proxy, the client's mobile equipment is configured to send requests directly to the HTTP proxy. The following values are defined: <ul style="list-style-type: none"> <li>■ 0 = False</li> <li>■ 1 = True</li> </ul> |
| dnsResolutionApplied | bool      | Indicates whether Domain Name System (DNS) resolution was used. The following values are defined: <ul style="list-style-type: none"> <li>■ 0 = False</li> <li>■ 1 = True</li> </ul>                                                                                                                |
| localIp              | bytes     | Indicates the HTTP proxy local loopback address used (can be an IPv4 or IPv6 address).                                                                                                                                                                                                             |
| localPort            | uint32    | Indicates the HTTP proxy local port.                                                                                                                                                                                                                                                               |
| serverIp             | bytes     | Indicates the IP address of the Origin Server (can be an IPv4 or IPv6 address).                                                                                                                                                                                                                    |
| serverPort           | uint32    | Indicates the Origin Server port.                                                                                                                                                                                                                                                                  |
| befApplied           | bool      | Indicates whether Back-End Forwarding (BEF) was used. The following values are defined: <ul style="list-style-type: none"> <li>■ 0 = False</li> <li>■ 1 = True</li> </ul>                                                                                                                          |

**Table 4-39 requestInfo Subfields**

| Name              | Data Type              | Description                                                                                                                                                                                                                                                                                                                               |
|-------------------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proxyNc           | string                 | Indicates the network context of the HTTP proxy.<br><br><i><b>Note:</b> The value for this field might differ from the value in the Gi Network Context subfield (Client Connection Information) if the server-connection network context field provisioned in the HTTP Proxy Instance is different from the incoming network context.</i> |
| refererUrl        | string                 | Percent-encoded URL string that refers to the original domain that originated a particular request.                                                                                                                                                                                                                                       |
| appliedPolicyName | percent encoded string | Name of the optimization policy that was matched and applied to the request. For example, if the request is video, MCC first evaluates about its eligibility for optimizations. If MCC finds it to be eligible, optimizations such as video pacing or rate limiting are applied to it.                                                    |
| clientRangeStart  | uint64                 | Indicates the start of an HTTP Range Request from the client. Range Requests are requests for a subset of a larger file based on one or more ranges.                                                                                                                                                                                      |
| clientRangeEnd    | uint64                 | Indicates the end of an HTTP Range Request from the client.                                                                                                                                                                                                                                                                               |

**Table 4-39 requestInfo Subfields**

| Name          | Data Type      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| failureReason | HttpSvcErrCode | <p>HTTP failure reason</p> <p>The following values are defined:</p> <ul style="list-style-type: none"> <li>■ 0 = HTTP_SVC_ERR_INIT</li> <li>■ 1 = HTTP_SVC_ERR_CLIENT_CLOSE_INCOMPLETE_REQ</li> <li>■ 2 = HTTP_SVC_ERR_CLIENT_CLOSE_INCOMPLETE_RESP</li> <li>■ 3 = HTTP_SVC_ERR_CLIENT_CLOSE_PIPELINE_DROP</li> <li>■ 4 = HTTP_SVC_ERR_REQ_OVERDUE</li> <li>■ 5 = HTTP_SVC_ERR_INVALID_REQ_NON_HTTP</li> <li>■ 6 = HTTP_SVC_ERR_INVALID_REQ_TOO_LARGE</li> <li>■ 7 = HTTP_SVC_ERR_INVALID_REQ_LINE_PARSE_FAILURE</li> <li>■ 8 = HTTP_SVC_ERR_INVALID_REQ_HEADERS_PARSE_FAILURE</li> <li>■ 9 = HTTP_SVC_ERR_INVALID_REQ_HEADER_BAD_ATTR</li> <li>■ 10 = HTTP_SVC_ERR_INVALID_REQ_HEADER_DUP_ATTR</li> <li>■ 11 = HTTP_SVC_ERR_INVALID_REQ_HEADER_BAD_HOSTNAME</li> <li>■ 12 = HTTP_SVC_ERR_INVALID_REQ_HEADER_NO_HOST_FOUND</li> <li>■ 13 = HTTP_SVC_ERR_REQ_PROCESS_FAILURE_UNSUPPORTED</li> <li>■ 14 = HTTP_SVC_ERR_REQ_PROCESS_FAILURE_EXPECT</li> <li>■ 15 = HTTP_SVC_ERR_REQ_PROCESS_FAILURE_BODY_READ</li> <li>■ 16 = HTTP_SVC_ERR_REQ_PROCESS_FAILURE_REQ_COMPILE_ERR</li> <li>■ 17 = HTTP_SVC_ERR_REQ_PROCESS_FAILURE_SERVER_INIT</li> </ul> |

**Table 4-39 requestInfo Subfields**

| Name                         | Data Type      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| failureReason<br>(Continued) | HttpSvcErrCode | <ul style="list-style-type: none"> <li>■ 18 = HTTP_SVC_ERR_SERVER_CLOSE_IMCOMPLETE_REQ</li> <li>■ 19 = HTTP_SVC_ERR_SERVER_CLOSE_INCOMPLETE_RESP</li> <li>■ 20 = HTTP_SVC_ERR_INVALID_RESP_HEADER_TOO_LARGE</li> <li>■ 21 = HTTP_SVC_ERR_INVALID_RESP_STATUS_PARSE_FAILURE</li> <li>■ 22 = HTTP_SVC_ERR_INVALID_RESP_HEADERS_PARSE_FAILURE</li> <li>■ 23 = HTTP_SVC_ERR_RESP_PROCESS_FAILURE</li> <li>■ 24 = HTTP_SVC_ERR_CACHED_RESP_READ_FAILURE</li> <li>■ 25 = HTTP_SVC_ERR_CACHED_RESP_PROCESS_FAILURE</li> <li>■ 26 = HTTP_SVC_ERR_FILTER_RESP_PROCESS_FAILURE</li> <li>■ 27 = HTTP_SVC_ERR_PRECONDITION_FAILURE</li> <li>■ 28 = HTTP_SVC_ERR_CLIENT_CLOSE_TIMEOUT</li> <li>■ 29 = HTTP_SVC_ERR_SERVER_CLOSE_TIMEOUT</li> <li>■ 30 = HTTP_SVC_ERR_MAINTENANCE_CONN_CLOSE</li> <li>■ 31 = HTTP_SVC_ERR_LDAP_REJECT</li> <li>■ 32 = HTTP_SVC_ERR_MMS_MAX_MESSAGE_SIZE_EXCEEDED</li> <li>■ 33 = HTTP_SVC_ERR_MMS_MAX_RECIPIENTS_EXCEEDED</li> <li>■ 34 = HTTP_SVC_ERR_CFI_FAILURE</li> <li>■ 35 = HTTP_SVC_ERR_CONNECT_REQ_ERROR</li> <li>■ 36 = HTTP_SVC_ERR_FILTER_REJECT</li> <li>■ 37 = HTTP_SVC_ERR_AOC_REJECT</li> <li>■ 38 = HTTP_SVC_ERR_SERVICE_UNAVAILABLE</li> <li>■ 39 = HTTP_SVC_ERR_SERVICE_RESOURCE_UNAVAILABLE</li> <li>■ 40 = HTTP_SVC_ERR_REDIRECTED</li> <li>■ 41 = HTTP_SVC_ERR_BLOCKED</li> <li>■ 42 = HTTP_SVC_ERR_PLUGIN_SERVER</li> <li>■ 43 = HTTP_SVC_ERR_HOSTIP_SAME_AS_PROXYLOCALIP</li> <li>■ 44 = HTTP_SVC_ERR_CLIENT_HANDOFF_FOR_BYPASS_FAILED</li> <li>■ 45 = HTTP_SVC_ERR_CACHE_INCORRECT_STATUS</li> <li>■ 46 = HTTP_SVC_ERR_REQ_BYPASS</li> <li>■ 47 = HTTP_SVC_ERR_NO_RSP_FROM_SERVER</li> <li>■ 48 = HTTP_SVC_ERR_CONTENTINSERTION_SERVER</li> <li>■ 49 = HTTP_SVC_ERR_UNSUPPORTED_METHOD</li> <li>■ 50 = HTTP_SVC_ERR_REQ_CONNECT_TO_LOCAL_ADDR</li> <li>■ 51 = HTTP_SVC_ERR_OUT_OF_BUFFER</li> </ul> |

## responseInfo

The following table lists the name, data type, and description for each subfield contained in the responseInfo data record.

---

**Note:** The HTTP Response Information subfields are not populated for TCP-spliced transactions.

---

**Table 4-40 responseInfo Subfields**

| Name                | Data Type | Description                                                                                                                                                                                                                           |
|---------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| responseStatus      | uint32    | Indicates the HTTP response status code.                                                                                                                                                                                              |
| serverContentType   | string    | Indicates the type of content contained in the HTTP response sent from the server (if applicable).                                                                                                                                    |
| clientContentType   | string    | Indicates the type of content contained in the HTTP response sent to the client (UE).<br><b>Note:</b> This field is filled only if the HTTP proxy updates the content-type sent to the client.                                        |
| serverContentLength | uint64    | Indicates the amount of data (in bytes) in the body of the HTTP response from the server (if present).                                                                                                                                |
| clientContentLength | uint64    | Indicates the amount of data (in bytes) in the body of the HTTP response sent to the client, if updated by the HTTP proxy.<br><b>Note:</b> This field is filled only if the HTTP proxy updates the content-length sent to the client. |
| serverHeaderSize    | uint64    | Indicates the size of the headers in the HTTP response from the server, in bytes.                                                                                                                                                     |
| serverBodySize      | uint64    | Indicates the size of the body in the HTTP response from the server, in bytes.                                                                                                                                                        |
| clientHeaderSize    | uint64    | Indicates the size of the headers in the HTTP response sent to the client, in bytes.                                                                                                                                                  |
| clientBodySize      | uint64    | Indicates the size of the body in the HTTP response sent to the client, in bytes.                                                                                                                                                     |
| chunked             | bool      | Indicates whether the HTTP response is chunked. The following values are defined: <ul style="list-style-type: none"> <li>■ 0 = False</li> <li>■ 1 = True</li> </ul>                                                                   |

**Table 4-40 responseInfo Subfields**

| Name                | Data Type              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| servedFromCache     | bool                   | Indicates whether the HTTP proxy served the response from the cache. The following values are defined: <ul style="list-style-type: none"> <li>■ 0 = False</li> <li>■ 1 = True</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| cached              | bool                   | Indicates whether the HTTP proxy cached the response. The following values are defined: <ul style="list-style-type: none"> <li>■ 0 = False</li> <li>■ 1 = True</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| detectedContentType | ContentType            | Type of content detected in the HTTP response sent to the UE (if updated). The following values are defined: <ul style="list-style-type: none"> <li>■ 1 = video/x-flv</li> <li>■ 2 = video/mp4</li> <li>■ 3 = video/webm</li> <li>■ 4 = video/other</li> <li>■ 5 = application/octet-stream</li> <li>■ 6 = application /javascript</li> <li>■ 7 = text/javascript</li> <li>■ 8 = text/plain</li> <li>■ 9 = text/html</li> <li>■ 10 = text/css</li> <li>■ 11 = image/bmp</li> <li>■ 12 = image /gif</li> <li>■ 13 = image /jpeg</li> <li>■ 14 = image/png</li> <li>■ 15 = video/dash</li> <li>■ 16 = video/hls</li> <li>■ 17 = video/hds</li> <li>■ 18 = application/xml</li> </ul> <p><b>Note:</b> This field is filled only if the HTTP proxy updates the detected content type sent to the UE.</p> |
| appliedPolicyName   | percent encoded string | Name of the optimization policy that was matched and applied to the request. For example, if the request is video, MCC first evaluates about its eligibility for optimizations. If MCC finds it to be eligible, optimizations such as video pacing or rate limiting are applied to it.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| serverRangeStart    | uint64                 | Indicates the start of an HTTP Range Request to the server from the MCC. Range Requests are requests for a subset of a larger file based on one or more ranges.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| serverRangeEnd      | uint64                 | Indicates the end of an HTTP Range Request to the server from MCC.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |



## compressionInfo

The following table lists the name, data type, and description for each subfield contained in the compressionInfo data record.

**Table 4-41** compressionInfo Subfields

| Name              | Data Type | Description                                                                                                                                                                                                                                           |
|-------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| compressionMethod | ZipType   | Indicates the method used for compressing the HTTP data, if compression is applied. The following values are defined: <ul style="list-style-type: none"> <li>■ 0 = Identity</li> <li>■ 1 = gzip</li> <li>■ 2 = Deflate</li> <li>■ 3 = Last</li> </ul> |
| serverCompress    | bool      | Indicates whether the origin server compressed the HTTP response. The following values are defined: <ul style="list-style-type: none"> <li>■ 0 = False</li> <li>■ 1 = True</li> </ul>                                                                 |
| proxyCompress     | bool      | Indicates whether the HTTP proxy compressed the HTTP response sent to the client. The following values are defined: <ul style="list-style-type: none"> <li>■ 0 = False</li> <li>■ 1 = True</li> </ul>                                                 |
| compressionRatio  | uint32    | Indicates the percentage of bandwidth saved, if compression is applied.                                                                                                                                                                               |
| bytesSaved        | uint64    | Indicates the number of bytes saved, if compression is applied.                                                                                                                                                                                       |

## webImageOptInfo

The following table lists the name, data type, and description for each subfield contained in the webImageOptInfo data record.

**Table 4-42** webImageOptInfo Subfields

| Name              | Data Type | Description                                                                              |
|-------------------|-----------|------------------------------------------------------------------------------------------|
| serverImageType   | string    | Indicates the type of image sent from the server.                                        |
| clientImageType   | string    | Indicates the type of image sent to the client.                                          |
| serverResolution  | string    | Indicates the resolution of the image sent from the server, displayed as width x height. |
| clientResolution  | string    | Indicates the resolution of the image sent to the client, displayed as width x height.   |
| optimizationRatio | uint32    | Indicates the percentage of bandwidth saved, if Web optimization is applied.             |
| bytesSaved        | uint64    | Indicates the number of bytes saved, if Web optimization is applied.                     |

## videoInfo

The following table lists the name, data type, and description for each subfield contained in the videoInfo data record.

**Table 4-43 videoInfo Subfields**

| Name                                                                      | Data Type   | Description                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------------------------------------------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| servedContainerType                                                       | ContentType | Indicates the type of container served to the client. The following values are defined: <ul style="list-style-type: none"> <li>■ 1 = video/x-flv</li> <li>■ 2 = video/mp4</li> <li>■ 3 = video/webm</li> <li>■ 4 = other video types (not supported for transcoding)</li> <li>■ 15 = video/dash</li> <li>■ 16 = video/hls</li> <li>■ 17 = video/hds</li> </ul> |
| OnlineTranscodingApplied<br><i>Not supported in version 8.2 and later</i> | bool        | Indicates whether online transcoding was used. Online transcoding is used to compress video content in real-time when video is playing back from a user's device. The following values are defined: <ul style="list-style-type: none"> <li>■ 0 = False</li> <li>■ 1 = True</li> </ul>                                                                          |
| servedAdaptedVideo<br><i>Not supported in version 8.2 and later</i>       | bool        | Indicates whether the HTTP proxy served a transcoded video from the cache. The following values are defined: <ul style="list-style-type: none"> <li>■ 0 = False</li> <li>■ 1 = True</li> </ul>                                                                                                                                                                 |
| videoPaced                                                                | bool        | Indicates whether the video was paced. The following values are defined: <ul style="list-style-type: none"> <li>■ 0 = False</li> <li>■ 1 = True</li> </ul>                                                                                                                                                                                                     |
| servedQoe<br><i>Not supported in version 8.2 and later</i>                | uint32      | Indicates the Quality of Experience (QoE) grade level (0-5).                                                                                                                                                                                                                                                                                                   |
| pacingBitRate                                                             | uint32      | Indicates the bit rate of the video, in bits per second (bps).                                                                                                                                                                                                                                                                                                 |
| bytesSaved<br><i>Not supported in version 8.2 and later</i>               | uint64      | Indicates the number of bytes saved, if video adaptation is used.                                                                                                                                                                                                                                                                                              |
| source_width<br><i>Not supported in version 8.2 and later</i>             | int32       | Indicates the width of the source media.                                                                                                                                                                                                                                                                                                                       |
| target_width<br><i>Not supported in version 8.2 and later</i>             | int32       | Indicates the width of the target media.                                                                                                                                                                                                                                                                                                                       |

**Table 4-43    videoInfo Subfields**

| <b>Name</b>                                                               | <b>Data Type</b> | <b>Description</b>                                                                                 |
|---------------------------------------------------------------------------|------------------|----------------------------------------------------------------------------------------------------|
| source_height<br><i>Not supported in version 8.2 and later</i>            | int32            | Indicates the height of the source media.                                                          |
| target_height<br><i>Not supported in version 8.2 and later</i>            | int32            | Indicates the height of the target media.                                                          |
| source_bitrate<br><i>Not supported in version 8.2 and later</i>           | int32            | Indicates the bit rate of the source media, in bps.                                                |
| target_bitrate<br><i>Not supported in version 8.2 and later</i>           | int32            | Indicates the bit rate of the target media, in bps.                                                |
| source_framerate<br><i>Not supported in version 8.2 and later</i>         | double           | Indicates the frame rate of the source media, in frames per second (fps).                          |
| target_framerate<br><i>Not supported in version 8.2 and later</i>         | double           | Indicates the frame rate of the target media, in frames per second (fps).                          |
| source_audioSamplingRate<br><i>Not supported in version 8.2 and later</i> | int32            | Indicates the audio sampling rate of the source media, in Hertz (Hz).                              |
| target_audioSamplingRate<br><i>Not supported in version 8.2 and later</i> | int32            | Indicates the audio sampling rate of the target media, in Hertz (Hz).                              |
| source_videoCodec<br><i>Not supported in version 8.2 and later</i>        | string           | Indicates the video codec name of the source media, such as H265 or VP9.                           |
| target_videoCodec<br><i>Not supported in version 8.2 and later</i>        | string           | Indicates the video codec name of the target media, such as H265 or VP9.                           |
| source_audioCodec<br><i>Not supported in version 8.2 and later</i>        | string           | Indicates the audio codec name of the source media, such as Advanced Audio Coding (AAC) or Vorbis. |
| target_audioCodec<br><i>Not supported in version 8.2 and later</i>        | string           | Indicates the audio codec name of the target media, such as Advanced Audio Coding (AAC) or Vorbis. |

**Table 4-43    videoInfo Subfields**

| Name                                                                  | Data Type          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------------------------------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| source_audioChannels<br><i>Not supported in version 8.2 and later</i> | int32              | Indicates the audio channel used for the source media. The following values are defined: <ul style="list-style-type: none"> <li>■ 1 = Monaural</li> <li>■ 2 = Stereo</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| target_audioChannels<br><i>Not supported in version 8.2 and later</i> | int32              | Indicates the audio channel used for the target media. The following values are defined: <ul style="list-style-type: none"> <li>■ 1 = Monaural</li> <li>■ 2 = Stereo</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| videoBypassReason<br><i>Not supported in version 8.2 and later</i>    | Video Bypass Cause | Indicates the reason that the file was bypassed. The following values are defined: <ul style="list-style-type: none"> <li>■ 0 = VIDEO_BYPASS_SUCCESS</li> <li>■ 1 = VIDEO_BYPASS_SRC_FMT_NOT_SUPPORTED</li> <li>■ 2 = VIDEO_BYPASS_SRC_MEDIA_FMT_NOT_STREAMABLE</li> <li>■ 3 = VIDEO_BYPASS_SRC_MEDIA_FMT_ERROR</li> <li>■ 4 = VIDEO_BYPASS_ABOVE_MAX_SRC_OBJECT_SIZE</li> <li>■ 5 = VIDEO_BYPASS_BELOW_MIN_SRC_OBJECT_SIZE</li> <li>■ 6 = VIDEO_BYPASS_ABOVE_MAX_SRC_DURATION</li> <li>■ 7 = VIDEO_BYPASS_BELOW_MIN_SRC_DURATION</li> <li>■ 8 = VIDEO_BYPASS_ABOVE_MAX_SRC_BITRATE</li> <li>■ 9 = VIDEO_BYPASS_ABOVE_MAX_SRC_FRAMERATE</li> <li>■ 10 = VIDEO_BYPASS_OPERATION_CANCELLED</li> <li>■ 11 = VIDEO_BYPASS_IN_ERROR_STATE</li> <li>■ 12 = VIDEO_BYPASS_ABOVE_MAX_PICTURE_DIMS</li> <li>■ 13 = VIDEO_BYPASS_NO_SRC_AUDIO_STREAM</li> <li>■ 14 = VIDEO_BYPASS_NO_SRC_VIDEO_STREAM</li> <li>■ 15 = VIDEO_BYPASS_UNKNOWN_SRC_MEDIA_FORMAT</li> <li>■ 16 = VIDEO_BYPASS_COULD_NOT_FIND_SRC_MEDIA_STREAM_INFO</li> <li>■ 17 = VIDEO_BYPASS_UNSUPPORTED_SRC_AUDIO_CODEC</li> <li>■ 18 = VIDEO_BYPASS_UNSUPPORTED_SRC_VIDEO_CODEC</li> <li>■ 19 = VIDEO_BYPASS_MOOV_ATOM_MISSING</li> <li>■ 20 = VIDEO_BYPASS_MOOV_ATOM_BAD</li> <li>■ 21 = VIDEO_BYPASS_BELOW_BITRATE_SAVINGS</li> <li>■ 22 = VIDEO_BYPASS_MAP_TABLE_ENTRY_IND_BYPASS</li> <li>■ 23 = VIDEO_BYPASS_MAP_TABLE_ENTRY_NO_MATCH</li> <li>■ 24 = VIDEO_BYPASS_OPERATING_MODE_INVALID</li> <li>■ 25 = VIDEO_BYPASS_MAX_QUAL_GRADE_LIST_TRUNCATED</li> </ul> |

**Table 4-43 videoInfo Subfields**

| Name                             | Data Type | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| videoBypassReason<br>(Continued) |           | <ul style="list-style-type: none"> <li>■ 26 = VIDEO_BYPASS_MAX_QUAL_GRADE_OUT_OF_RANGE</li> <li>■ 27 = VIDEO_BYPASS_NO_PROFILE_DB_OR_MAP</li> <li>■ 28 = VIDEO_BYPASS_SPEED_LEVEL_PARM_INVALID</li> <li>■ 29 = VIDEO_BYPASS_MAX_OBJ_SIZE_PARM_INVALID</li> <li>■ 30 = VIDEO_BYPASS_MIN_OBJ_SIZE_PARM_INVALID</li> <li>■ 31 = VIDEO_BYPASS_MAX_DURATION_PARM_INVALID</li> <li>■ 32 = VIDEO_BYPASS_MIN_DURATION_PARM_INVALID</li> <li>■ 33 = VIDEO_BYPASS_MAX_BITRATE_PARM_INVALID</li> <li>■ 34 = VIDEO_BYPASS_MAX_FRAMERATE_PARM_INVALID</li> <li>■ 35 = VIDEO_BYPASS_BITRATE_SAVINGS_PARAM_INVALID</li> <li>■ 36 = VIDEO_BYPASS_SUPPORTED_CNTR_TYPE_NOT_IN_DB</li> <li>■ 37 =<br/>VIDEO_BYPASS_NO_QUAL_GRADES_SUPPORTED_IN_DB</li> <li>■ 38 = VIDEO_BYPASS_MAX_HORZ_PIXELS_NOT_IN_DB</li> <li>■ 39 = VIDEO_BYPASS_ERR_READING_PROPERTY_IN_DB</li> <li>■ 40 = VIDEO_BYPASS_PROP_VAL_SYNTAX_ERR_IN_DB</li> <li>■ 41 = VIDEO_BYPASS_PROP_VAL_RANGE_ERR_IN_DB</li> <li>■ 42 = VIDEO_BYPASS_MAX_VERT_PIXELS_NOT_IN_DB</li> <li>■ 43 =<br/>VIDEO_BYPASS_MATCH_CRITERIA_RANGE_ERR_IN_MAP_TBL</li> <li>■ 44 =<br/>VIDEO_BYPASS_XCODE_PARM_RANGE_ERR_IN_MAP_TBL</li> <li>■ 45 = VIDEO_BYPASS_TGT_FMT_NOT_SUPPORTED</li> <li>■ 46 = VIDEO_BYPASS_TGT_FMT_PARAMETERS_NOT_VALID</li> <li>■ 47 =<br/>VIDEO_BYPASS_TGT_XCODING_PARAMETERS_NOT_VALID</li> <li>■ 48 = VIDEO_BYPASS_UNSUPPORTED_TGT_AUDIO_CODEC</li> <li>■ 49 = VIDEO_BYPASS_UNSUPPORTED_TGT_VIDEO_CODEC</li> <li>■ 50 = VIDEO_BYPASS_SRC_MEDIA_DECODING_FAILURE</li> <li>■ 51 = VIDEO_BYPASS_TGT_MEDIA_TRANSCODING_FAILURE</li> <li>■ 52 = VIDEO_BYPASS_INTERNAL_XCODE_ERROR</li> <li>■ 53 = VIDEO_BYPASS_VIDEO_OPTIMIZATION_DISABLED</li> <li>■ 54 =<br/>VIDEO_BYPASS_VIDEO_OPTIMIZATION_NOT_REQUIRED</li> <li>■ 55 = VIDEO_BYPASS_OPTIMIZATION_OVERLOAD</li> <li>■ 56 = VIDEO_BYPASS_NO_QOE</li> <li>■ 57 = VIDEO_BYPASS_INTERNAL_FAILURE</li> <li>■ 58 = VIDEO_BYPASS_NOT_APPLICABLE_MEDIA</li> </ul> |
| videoPacingBytesSaved            | uint64    | Indicates the number of bytes from video pacing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

**Table 4-43** videoInfo Subfields

| Name                         | Data Type              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| abrRateLimitingApplied       | bool                   | Indicates whether ABR rate limiting was applied. The following values are defined: <ul style="list-style-type: none"> <li>■ 0 = False</li> <li>■ 1 = True</li> </ul>                                                                                                                                                                                                                                                                                                         |
| abrRateLimitingBypass Reason | ABRBypass Cause        | Indicates the reason that ABR rate limiting was bypassed. The following values are defined: <ul style="list-style-type: none"> <li>■ 1 = ABR_SUBSCRIBER_OPTOUT</li> <li>■ 2 = ABR_NO_MATCHING_PROFILE</li> <li>■ 3 = ABR_VIDEO_PACING_APPLIED</li> <li>■ 4 = ABR_SUBSCRIBER_OPTIN_REQ_NOT_MET</li> <li>■ 5 = ABR_CANCEL_TOOSHORT</li> <li>■ 6 = ABR_CP_MANIFEST_NO_MAX_RATE</li> <li>■ 7 = ABR_CP_MANIFEST_INVALID_MAX_RATE</li> <li>■ 8 = ABR_SAVINGS_ESTIMATION</li> </ul> |
| abrRateLimitingBitrate       | double                 | Actual bit rate of video segment, in bits per second (bps).                                                                                                                                                                                                                                                                                                                                                                                                                  |
| abrSessionId                 | uint64                 | Unique Session ID for ABR Session                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| abrBytesSaved                | uint64                 | Total bytes saved by enforcing ABR                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| appliedPolicyName            | percent encoded string | Name of the optimization policy that was matched and applied to the request. For example, if the request is video, MCC first evaluates about its eligibility for optimizations. If MCC finds it to be eligible, optimizations such as video pacing or rate limiting are applied to it.                                                                                                                                                                                       |
| transcodingSessionId         | uint64                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

## minifyInfo

The following table lists the name, data type, and description for each subfield contained in the minifyInfo data record.

**Table 4-44 MinyInfo Subfields**

| Name        | Data Type   | Description                                                                                                                                                                                      |
|-------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| contentType | ContentType | Indicates the type of data minified. The following values are defined: <ul style="list-style-type: none"> <li>■ 7 = text/javascript</li> <li>■ 9 = text/html</li> <li>■ 10 = text/css</li> </ul> |
| bytesSaved  | uint64      | Indicates the number of bytes saved, if minification is used.                                                                                                                                    |

## preemptDnsInfo

The following table lists the name, data type, and description for each subfield contained in the PreemptDnsInfo data record.

**Table 4-45 PreemptDnsInfo Subfields**

| Name              | Data Type | Description                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| isAppliedResponse | uint32    | Indicates whether the HTTP proxy applied preemptive DNS to the HTTP response. Preemptive DNS replaces URLs in the HTTP response with the IP address of one of the Affirmed Networks Gi local loopback addresses and configured DNS Token before sending the response to the client. The following values are defined: <ul style="list-style-type: none"> <li>■ 0 = False</li> <li>■ 1 = True</li> </ul> |
| numUpdatedUrls    | uint32    | Indicates the number of URLs in the Web page for which preemptive DNS was applied.                                                                                                                                                                                                                                                                                                                      |



## cacheInfo

The following table lists the name, data type, and description for each subfield contained in the cacheInfo data record.

**Table 4-46 cacheInfo Subfields**

| Name                           | Data Type            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |
|--------------------------------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------------|---------------|-------------------------|------------|---|------------------------|------------|---|----------------------|------------|---|-----------------------|------------|---|------------------------|------------|----|----------------------------|------------|----|-------------------------------|------------|----|--------------------------------|------------|-----|------------------------------|------------|-----|--------------------|------------|-----|----------------------|------------|------|------------------------|------------|------|-------------------------|------------|------|----------------------|------------|------|-----------------------|------------|-------|------------------------|------------|--------|-------------------------|------------|--------|-------------------------|------------|--------|-----------------------|------------|---------|
| cachedAge                      | int64                | Indicates the amount of time, in seconds, that the fetched item has been in the cache.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |
| requestCacheMaxAge             | int32                | If there is a max-age in the HTTP request header, this field indicates its value, in seconds.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |
| responseCacheMaxAge            | int32                | If there is a max-age in the HTTP response header, this field indicates its value, in seconds.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |
| requestCacheControlFlags       | uint32               | Indicates the different cache-control flags present in the HTTP request header. If multiple cache control flags are present, the bit mask values are added together and the sum is presented as a decimal value in the HTDR. For example, a value of 65538 (0x10002) is equal to CACHE_CONTROL_MAX_AGE + CACHE_CONTROL_NO_CACHE. The following values are defined:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |
|                                |                      | <table> <tr> <th></th><th>Bit Mask Value (Hex)</th><th>Decimal Value</th></tr> <tr> <td>CACHE_CONTROL_NO_HIDDEN</td><td>0x00000001</td><td>1</td></tr> <tr> <td>CACHE_CONTROL_NO_CACHE</td><td>0x00000002</td><td>2</td></tr> <tr> <td>CACHE_CONTROL_PUBLIC</td><td>0x00000004</td><td>4</td></tr> <tr> <td>CACHE_CONTROL_PRIVATE</td><td>0x00000008</td><td>8</td></tr> <tr> <td>CACHE_CONTROL_NO_STORE</td><td>0x00000010</td><td>16</td></tr> <tr> <td>CACHE_CONTROL_NO_TRANSFORM</td><td>0x00000020</td><td>32</td></tr> <tr> <td>CACHE_CONTROL_MUST_REVALIDATE</td><td>0x00000040</td><td>64</td></tr> <tr> <td>CACHE_CONTROL_PROXY_REVALIDATE</td><td>0x00000080</td><td>128</td></tr> <tr> <td>CACHE_CONTROL_ONLY_IF_CACHED</td><td>0x00000100</td><td>256</td></tr> <tr> <td>CACHE_CONTROL_VARY</td><td>0x00000200</td><td>512</td></tr> <tr> <td>CACHE_CONTROL_COOKIE</td><td>0x00000400</td><td>1024</td></tr> <tr> <td>CACHE_CONTROL_MISMATCH</td><td>0x00000800</td><td>2048</td></tr> <tr> <td>CACHE_CONTROL_NO_PRAGMA</td><td>0x00001000</td><td>4096</td></tr> <tr> <td>CACHE_CONTROL_ALWAYS</td><td>0x00002000</td><td>8192</td></tr> <tr> <td>CACHE_CONTROL_MAX_AGE</td><td>0x00010000</td><td>65536</td></tr> <tr> <td>CACHE_CONTROL_S_MAXAGE</td><td>0x00020000</td><td>131072</td></tr> <tr> <td>CACHE_CONTROL_MIN_FRESH</td><td>0x00040000</td><td>262144</td></tr> <tr> <td>CACHE_CONTROL_MAX_STALE</td><td>0x00080000</td><td>524288</td></tr> <tr> <td>CACHE_CONTROL_CHUNKED</td><td>0x00100000</td><td>1048576</td></tr> </table> |  | Bit Mask Value (Hex) | Decimal Value | CACHE_CONTROL_NO_HIDDEN | 0x00000001 | 1 | CACHE_CONTROL_NO_CACHE | 0x00000002 | 2 | CACHE_CONTROL_PUBLIC | 0x00000004 | 4 | CACHE_CONTROL_PRIVATE | 0x00000008 | 8 | CACHE_CONTROL_NO_STORE | 0x00000010 | 16 | CACHE_CONTROL_NO_TRANSFORM | 0x00000020 | 32 | CACHE_CONTROL_MUST_REVALIDATE | 0x00000040 | 64 | CACHE_CONTROL_PROXY_REVALIDATE | 0x00000080 | 128 | CACHE_CONTROL_ONLY_IF_CACHED | 0x00000100 | 256 | CACHE_CONTROL_VARY | 0x00000200 | 512 | CACHE_CONTROL_COOKIE | 0x00000400 | 1024 | CACHE_CONTROL_MISMATCH | 0x00000800 | 2048 | CACHE_CONTROL_NO_PRAGMA | 0x00001000 | 4096 | CACHE_CONTROL_ALWAYS | 0x00002000 | 8192 | CACHE_CONTROL_MAX_AGE | 0x00010000 | 65536 | CACHE_CONTROL_S_MAXAGE | 0x00020000 | 131072 | CACHE_CONTROL_MIN_FRESH | 0x00040000 | 262144 | CACHE_CONTROL_MAX_STALE | 0x00080000 | 524288 | CACHE_CONTROL_CHUNKED | 0x00100000 | 1048576 |
|                                | Bit Mask Value (Hex) | Decimal Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |
| CACHE_CONTROL_NO_HIDDEN        | 0x00000001           | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |
| CACHE_CONTROL_NO_CACHE         | 0x00000002           | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |
| CACHE_CONTROL_PUBLIC           | 0x00000004           | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |
| CACHE_CONTROL_PRIVATE          | 0x00000008           | 8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |
| CACHE_CONTROL_NO_STORE         | 0x00000010           | 16                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |
| CACHE_CONTROL_NO_TRANSFORM     | 0x00000020           | 32                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |
| CACHE_CONTROL_MUST_REVALIDATE  | 0x00000040           | 64                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |
| CACHE_CONTROL_PROXY_REVALIDATE | 0x00000080           | 128                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |
| CACHE_CONTROL_ONLY_IF_CACHED   | 0x00000100           | 256                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |
| CACHE_CONTROL_VARY             | 0x00000200           | 512                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |
| CACHE_CONTROL_COOKIE           | 0x00000400           | 1024                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |
| CACHE_CONTROL_MISMATCH         | 0x00000800           | 2048                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |
| CACHE_CONTROL_NO_PRAGMA        | 0x00001000           | 4096                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |
| CACHE_CONTROL_ALWAYS           | 0x00002000           | 8192                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |
| CACHE_CONTROL_MAX_AGE          | 0x00010000           | 65536                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |
| CACHE_CONTROL_S_MAXAGE         | 0x00020000           | 131072                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |
| CACHE_CONTROL_MIN_FRESH        | 0x00040000           | 262144                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |
| CACHE_CONTROL_MAX_STALE        | 0x00080000           | 524288                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |
| CACHE_CONTROL_CHUNKED          | 0x00100000           | 1048576                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |

**Table 4-46    cacheInfo Subfields**

| Name                           | Data Type            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |
|--------------------------------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------------|---------------|-------------------------|------------|---|------------------------|------------|---|----------------------|------------|---|-----------------------|------------|---|------------------------|------------|----|----------------------------|------------|----|-------------------------------|------------|----|--------------------------------|------------|-----|------------------------------|------------|-----|--------------------|------------|-----|----------------------|------------|------|------------------------|------------|------|-------------------------|------------|------|----------------------|------------|------|-----------------------|------------|-------|------------------------|------------|--------|-------------------------|------------|--------|-------------------------|------------|--------|-----------------------|------------|---------|
| responseCacheControlFlags      | uint32               | Indicates the different cache-control flags present in the HTTP response header. If multiple cache control flags are present, the bit mask values are added together and the sum is presented as a decimal value in the HTDR. For example, a value of 65538 (0x10002) is equal to CACHE_CONTROL_MAX_AGE + CACHE_CONTROL_NO_CACHE. The following values are defined:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |
|                                |                      | <table> <tr> <th></th><th>Bit Mask Value (Hex)</th><th>Decimal Value</th></tr> <tr> <td>CACHE_CONTROL_NO_HIDDEN</td><td>0x00000001</td><td>1</td></tr> <tr> <td>CACHE_CONTROL_NO_CACHE</td><td>0x00000002</td><td>2</td></tr> <tr> <td>CACHE_CONTROL_PUBLIC</td><td>0x00000004</td><td>4</td></tr> <tr> <td>CACHE_CONTROL_PRIVATE</td><td>0x00000008</td><td>8</td></tr> <tr> <td>CACHE_CONTROL_NO_STORE</td><td>0x00000010</td><td>16</td></tr> <tr> <td>CACHE_CONTROL_NO_TRANSFORM</td><td>0x00000020</td><td>32</td></tr> <tr> <td>CACHE_CONTROL_MUST_REVALIDATE</td><td>0x00000040</td><td>64</td></tr> <tr> <td>CACHE_CONTROL_PROXY_REVALIDATE</td><td>0x00000080</td><td>128</td></tr> <tr> <td>CACHE_CONTROL_ONLY_IF_CACHED</td><td>0x00000100</td><td>256</td></tr> <tr> <td>CACHE_CONTROL_VARY</td><td>0x00000200</td><td>512</td></tr> <tr> <td>CACHE_CONTROL_COOKIE</td><td>0x00000400</td><td>1024</td></tr> <tr> <td>CACHE_CONTROL_MISMATCH</td><td>0x00000800</td><td>2048</td></tr> <tr> <td>CACHE_CONTROL_NO_PRAGMA</td><td>0x00001000</td><td>4096</td></tr> <tr> <td>CACHE_CONTROL_ALWAYS</td><td>0x00002000</td><td>8192</td></tr> <tr> <td>CACHE_CONTROL_MAX_AGE</td><td>0x00010000</td><td>65536</td></tr> <tr> <td>CACHE_CONTROL_S_MAXAGE</td><td>0x00020000</td><td>131072</td></tr> <tr> <td>CACHE_CONTROL_MIN_FRESH</td><td>0x00040000</td><td>262144</td></tr> <tr> <td>CACHE_CONTROL_MAX_STALE</td><td>0x00080000</td><td>524288</td></tr> <tr> <td>CACHE_CONTROL_CHUNKED</td><td>0x00100000</td><td>1048576</td></tr> </table> |  | Bit Mask Value (Hex) | Decimal Value | CACHE_CONTROL_NO_HIDDEN | 0x00000001 | 1 | CACHE_CONTROL_NO_CACHE | 0x00000002 | 2 | CACHE_CONTROL_PUBLIC | 0x00000004 | 4 | CACHE_CONTROL_PRIVATE | 0x00000008 | 8 | CACHE_CONTROL_NO_STORE | 0x00000010 | 16 | CACHE_CONTROL_NO_TRANSFORM | 0x00000020 | 32 | CACHE_CONTROL_MUST_REVALIDATE | 0x00000040 | 64 | CACHE_CONTROL_PROXY_REVALIDATE | 0x00000080 | 128 | CACHE_CONTROL_ONLY_IF_CACHED | 0x00000100 | 256 | CACHE_CONTROL_VARY | 0x00000200 | 512 | CACHE_CONTROL_COOKIE | 0x00000400 | 1024 | CACHE_CONTROL_MISMATCH | 0x00000800 | 2048 | CACHE_CONTROL_NO_PRAGMA | 0x00001000 | 4096 | CACHE_CONTROL_ALWAYS | 0x00002000 | 8192 | CACHE_CONTROL_MAX_AGE | 0x00010000 | 65536 | CACHE_CONTROL_S_MAXAGE | 0x00020000 | 131072 | CACHE_CONTROL_MIN_FRESH | 0x00040000 | 262144 | CACHE_CONTROL_MAX_STALE | 0x00080000 | 524288 | CACHE_CONTROL_CHUNKED | 0x00100000 | 1048576 |
|                                | Bit Mask Value (Hex) | Decimal Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |
| CACHE_CONTROL_NO_HIDDEN        | 0x00000001           | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |
| CACHE_CONTROL_NO_CACHE         | 0x00000002           | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |
| CACHE_CONTROL_PUBLIC           | 0x00000004           | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |
| CACHE_CONTROL_PRIVATE          | 0x00000008           | 8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |
| CACHE_CONTROL_NO_STORE         | 0x00000010           | 16                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |
| CACHE_CONTROL_NO_TRANSFORM     | 0x00000020           | 32                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |
| CACHE_CONTROL_MUST_REVALIDATE  | 0x00000040           | 64                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |
| CACHE_CONTROL_PROXY_REVALIDATE | 0x00000080           | 128                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |
| CACHE_CONTROL_ONLY_IF_CACHED   | 0x00000100           | 256                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |
| CACHE_CONTROL_VARY             | 0x00000200           | 512                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |
| CACHE_CONTROL_COOKIE           | 0x00000400           | 1024                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |
| CACHE_CONTROL_MISMATCH         | 0x00000800           | 2048                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |
| CACHE_CONTROL_NO_PRAGMA        | 0x00001000           | 4096                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |
| CACHE_CONTROL_ALWAYS           | 0x00002000           | 8192                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |
| CACHE_CONTROL_MAX_AGE          | 0x00010000           | 65536                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |
| CACHE_CONTROL_S_MAXAGE         | 0x00020000           | 131072                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |
| CACHE_CONTROL_MIN_FRESH        | 0x00040000           | 262144                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |
| CACHE_CONTROL_MAX_STALE        | 0x00080000           | 524288                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |
| CACHE_CONTROL_CHUNKED          | 0x00100000           | 1048576                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |                      |               |                         |            |   |                        |            |   |                      |            |   |                       |            |   |                        |            |    |                            |            |    |                               |            |    |                                |            |     |                              |            |     |                    |            |     |                      |            |      |                        |            |      |                         |            |      |                      |            |      |                       |            |       |                        |            |        |                         |            |        |                         |            |        |                       |            |         |

**Table 4-46    cacheInfo Subfields**

| Name                | Data Type                | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| storeBypassCause    | Cache<br>Bypass<br>Cause | <p>Indicates the reason for bypassing the cache when attempting to store the data. The following values are defined:</p> <ul style="list-style-type: none"> <li>■ 1 = CACHE_BYPASS_INTERNAL_ERROR</li> <li>■ 2 = CACHE_BYPASS_UPDATING</li> <li>■ 3 = CACHE_BYPASS_METHOD_NOT_CACHEABLE</li> <li>■ 4 = CACHE_BYPASS_SERVICE_RULE_BYPASS</li> <li>■ 5 = CACHE_BYPASS_RESPONSE_CODE_NOT_CACHEABLE</li> <li>■ 6 = CACHE_BYPASS_CACHE_CONTROL</li> <li>■ 7 = CACHE_BYPASS_RESPONSE_CHUNKED</li> <li>■ 8 = CACHE_BYPASS_CACHE_DISABLED</li> <li>■ 9 = CACHE_BYPASS_SERVICE_RULE_DONT_CACHE_POLICY</li> <li>■ 10 = CACHE_BYPASS_MAX_AGE_ZERO</li> <li>■ 11 = CACHE_BYPASS_MULTIPART_REQUEST</li> <li>■ 12 = CACHE_BYPASS_BEV_NO_CACHE</li> <li>■ 13 = CACHE_BYPASS_FE_PROXY</li> <li>■ 14 = CACHE_BYPASS_CONTENT_OPTIMIZATION_MISMATCH</li> <li>■ 15 = CACHE_BYPASS_STALE_ENTRY</li> <li>■ 16 = CACHE_BYPASS_HEAD_INVALIDATE</li> <li>■ 17 = CACHE_BYPASS_RESOURCE_UNAVAILABLE</li> </ul> |
| retrieveBypassCause | Cache<br>Bypass<br>Cause | <p>Indicates the reason for bypassing the cache when retrieving data. The following values are defined:</p> <ul style="list-style-type: none"> <li>■ 1 = CACHE_BYPASS_INTERNAL_ERROR</li> <li>■ 2 = CACHE_BYPASS_UPDATING</li> <li>■ 3 = CACHE_BYPASS_METHOD_NOT_CACHEABLE</li> <li>■ 4 = CACHE_BYPASS_SERVICE_RULE_BYPASS</li> <li>■ 5 = CACHE_BYPASS_RESPONSE_CODE_NOT_CACHEABLE</li> <li>■ 6 = CACHE_BYPASS_CACHE_CONTROL</li> <li>■ 7 = CACHE_BYPASS_RESPONSE_CHUNKED</li> <li>■ 8 = CACHE_BYPASS_CACHE_DISABLED</li> <li>■ 9 = CACHE_BYPASS_SERVICE_RULE_DONT_CACHE_POLICY</li> <li>■ 10 = CACHE_BYPASS_MAX_AGE_ZERO</li> <li>■ 11 = CACHE_BYPASS_MULTIPART_REQUEST</li> <li>■ 12 = CACHE_BYPASS_BEV_NO_CACHE</li> <li>■ 13 = CACHE_BYPASS_FE_PROXY</li> <li>■ 14 = CACHE_BYPASS_CONTENT_OPTIMIZATION_MISMATCH</li> <li>■ 15 = CACHE_BYPASS_STALE_ENTRY</li> <li>■ 16 = CACHE_BYPASS_HEAD_INVALIDATE</li> <li>■ 17 = CACHE_BYPASS_RESOURCE_UNAVAILABLE</li> </ul>              |

**Table 4-46** cacheInfo Subfields

| Name            | Data Type | Description                                                                                                                                                                                                                                                                                                                                                    |
|-----------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cacheEntryStale | bool      | <p>Indicates whether the HTTP response in the cache is stale based on the Expires header field in the HTTP response. If this field is true, the Cache Bypass Cause When Retrieving subfield is set to CACHE_BYPASS_STALE_ENTRY.</p> <p>The following values are defined:</p> <ul style="list-style-type: none"> <li>■ 0 = False</li> <li>■ 1 = True</li> </ul> |

## manifestInfo

The following table lists the name, data type, and description for each subfield contained in the manifestInfo data record.

**Table 4-47** ManifestInfo Fields

| Name            | Data Type | Description                                                                                                                                                                                        |
|-----------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| manifestMatched | bool      | <p>Indicates whether the Content Provider (CP) Manifest is matched.</p> <p>The following values are defined:</p> <ul style="list-style-type: none"> <li>■ 0 = False</li> <li>■ 1 = True</li> </ul> |
| manifestMaxRate | uint64    | Max-Rate received from Manifest (kbps)                                                                                                                                                             |

## tcpClientConnectionInfo

The following table lists the name, data type, and description for each subfield contained in the tcpClientConnectionInfo data record used for congestion detection.

**Table 4-48 tcpClientConnectionInfo Fields**

| Name                                   | Data Type                              | Description                                                                                                                                                                                                                                                                 |
|----------------------------------------|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| minRTT                                 | uint64                                 | Minimum Round Trip Time of a connection, in microseconds                                                                                                                                                                                                                    |
| avgRTT                                 | uint64                                 | Average Round Trip Time of a connection, in microseconds                                                                                                                                                                                                                    |
| maxRTT                                 | uint64                                 | Maximum Round Trip Time of a connection, in microseconds                                                                                                                                                                                                                    |
| minDownstreamMSS                       | uint64                                 | Minimum Downstream Maximum Segment Size (MSS) of a connection, in bytes                                                                                                                                                                                                     |
| avgDownstreamMSS                       | uint64                                 | Average Downstream Maximum Segment Size (MSS) of a connection, in bytes                                                                                                                                                                                                     |
| maxDownstreamMSS                       | uint64                                 | Maximum Downstream Maximum Segment Size (MSS) of a connection, in bytes                                                                                                                                                                                                     |
| minDownstreamCWND                      | uint64                                 | Minimum Downstream Congestion Window of a connection                                                                                                                                                                                                                        |
| avgDownstreamCWND                      | uint64                                 | Average Downstream Congestion Window of a connection                                                                                                                                                                                                                        |
| maxDownstreamCWND                      | uint64                                 | Maximum Downstream Congestion Window of a connection                                                                                                                                                                                                                        |
| minDownstreamTPUT                      | uint64                                 | Minimum Downstream Throughput of a connection, in Kbps                                                                                                                                                                                                                      |
| avgDownstreamTPUT                      | uint64                                 | Average Downstream Throughput of a connection, in Kbps                                                                                                                                                                                                                      |
| maxDownstreamTPUT                      | uint64                                 | Maximum Downstream Throughput of a connection, in Kbps                                                                                                                                                                                                                      |
| numDownstreamSeg Retrans               | uint64                                 | Downstream Segments Retransmitted                                                                                                                                                                                                                                           |
| numDownstreamSegEst                    | uint64                                 | Estimated Downstream Segments                                                                                                                                                                                                                                               |
| tcpClientConn MeasurementBypass Reason | TcpClientConn MeasurementBypass Reason | TCP Client Connection Measurement Bypass Reason<br>The following values are defined: <ul style="list-style-type: none"> <li>■ 0 = Bypass_ConnTooShort</li> <li>■ 1 = Bypass_BrandNewConn</li> <li>■ 2 = Bypass_MissedSampInterval</li> <li>■ 3 = Bypass_BEFCConn</li> </ul> |
| congestionAlgorithm                    | string                                 | TCP congestion algorithm                                                                                                                                                                                                                                                    |
| congestionLevelNoneTime                | uint64                                 | Length of time (ms) that the client connection remains with no congestion level                                                                                                                                                                                             |
| congestionLevelMildTime                | uint64                                 | Length of time (ms) that the client connection remains in mild congestion level                                                                                                                                                                                             |

**Table 4-48 tcpClientConnectionInfo Fields**

| Name                            | Data Type | Description                                                                         |
|---------------------------------|-----------|-------------------------------------------------------------------------------------|
| congestionLevelModerate<br>Time | uint64    | Length of time (ms) that the client connection remains in moderate congestion level |
| congestionLevelSevere<br>Time   | uint64    | Length of time (ms) that the client connection remains in severe congestion level   |

## filterInfo

The following table lists the name, data type, and description for each subfield contained in the filterInfo data record.

**Table 4-49 filterInfo Fields**

| Name           | Data Type    | Description                                                                                                                                                                                                                                                                                                                      |
|----------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| filterName     | string       | Filter name                                                                                                                                                                                                                                                                                                                      |
| filterType     | CFFilterType | Filter type<br>The following values are defined:<br>0 = Global whitelist<br>1 = Global blacklist<br>2 = Global service-rule<br>3 = Instance phishing<br>4 = Instance IWF<br>5 = Instance service-rule<br>6 = Instance category<br>7 = Instance online category<br>8 = Instance malware category<br>9 = Instance filter-hash-list |
| actionApplied  | CFFilterType | Action to take for a filter match:<br>0 = BLOCK<br>1 = REDIRECT<br>2 = TERMINATE<br>3 = ALLOW<br>4 = NONE                                                                                                                                                                                                                        |
| filterInstName | string       | Filter instance name                                                                                                                                                                                                                                                                                                             |
| onlineCategory | CFOLCatType  | Online Category<br>1 = url<br>2 = malware<br>3 = phishing<br>4 = iwf                                                                                                                                                                                                                                                             |

## tcpServerConnectionInfo

The following table lists the name, data type, and description for each subfield contained in the tcpServerConnectionInfo data record. The data record is used to record Round Trip Time (RTT), Segments Retransmitted, and Segments Out of Order for the server connections (connections initiated by Proxy to the Origin Server).

**Table 4-50 tcpServerConnectionInfo Fields**

| Name                | Data Type | Description                                                                                                                     |
|---------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------|
| RTT                 | uint64    | Network Round Trip Time between proxy and Internet server, in microseconds                                                      |
| numSegRetransIn     | uint64    | Estimated Downstream Segments Retransmitted; number of TCP data segments retransmitted by Internet server and received by proxy |
| numSegOooIn         | uint64    | Estimated Downstream Segments Out of Order; number of TCP data segments received out of order by proxy (misordered by network)  |
| congestionAlgorithm | string    | TCP congestion algorithm                                                                                                        |

## entitlementInfo

The following table lists the name, data type, and description for each subfield contained in the entitlementInfo data record.

**Table 4-51 entitlementInfo Fields**

| Name              | Data Type | Description                       |
|-------------------|-----------|-----------------------------------|
| entitlementValues | string    | Field listing entitlement values. |

## udpProxyInfo

The following table lists the name, data type, and description for each subfield contained in the udpProxyInfo data record.

**Table 4-52** udpProxyInfo Fields

| Name                      | Data Type                     | Description                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| upstreamBytesRecd         | uint64                        | Number of bytes received from client                                                                                                                                                                                                                                                                                                                                                                                          |
| upstreamBytesSent         | uint64                        | Number of bytes sent to server                                                                                                                                                                                                                                                                                                                                                                                                |
| downstreamBytesRecd       | uint64                        | Number of bytes received from server                                                                                                                                                                                                                                                                                                                                                                                          |
| downstreamBytesSent       | uint64                        | Number of bytes sent to client                                                                                                                                                                                                                                                                                                                                                                                                |
| localIpAddr               | hexidecimal bytes             | Local IP address                                                                                                                                                                                                                                                                                                                                                                                                              |
| localPort                 | uint64                        | Local port number                                                                                                                                                                                                                                                                                                                                                                                                             |
| serverIpAddress           | hexidecimal bytes             | Server IP address                                                                                                                                                                                                                                                                                                                                                                                                             |
| serverPort                | uint64                        | Server port number                                                                                                                                                                                                                                                                                                                                                                                                            |
| proxyNetworkContext       | string                        | Network context of the proxy                                                                                                                                                                                                                                                                                                                                                                                                  |
| quicSni                   | string                        | Server Name Indication                                                                                                                                                                                                                                                                                                                                                                                                        |
| quicVersion               | string                        | QUIC Version                                                                                                                                                                                                                                                                                                                                                                                                                  |
| quicConnId                | uint64                        | QUIC connection ID                                                                                                                                                                                                                                                                                                                                                                                                            |
| quicRateLimitingApplied   | bool                          | Indicates whether rate limiting is applied: <ul style="list-style-type: none"> <li>■ 0 = False</li> <li>■ 1 = True</li> </ul>                                                                                                                                                                                                                                                                                                 |
| quicRateLimitingBitrate   | unit64                        | Rate limiting bit rate, in bits per second.                                                                                                                                                                                                                                                                                                                                                                                   |
| quicRLBypassReason        | QuicRateLimitingBypass Reason | Rate limiting bypass cancel reason: <ul style="list-style-type: none"> <li>■ 0 = Flow was not QUIC</li> <li>■ 1 = Rate-limiting disabled</li> <li>■ 2 = No Matching Policy</li> <li>■ 3 = No profile</li> <li>■ 4 = Protocol Error</li> <li>■ 5 = Internal Error</li> <li>■ 6 = Rate-limiting Opted Out by Subscriber</li> <li>■ 7 = Rate-limiting Subscriber Opt-in Req Not Met</li> <li>■ 8 = Savings Estimation</li> </ul> |
| quicRateLimitingBurstSize | unit64                        | Rate limiting burst size, in bytes                                                                                                                                                                                                                                                                                                                                                                                            |
| appliedUdpPolicyName      | percent encoded string        | Applied Policy Name                                                                                                                                                                                                                                                                                                                                                                                                           |



**Table 4-52    udpProxyInfo Fields**

| Name                             | Data Type             | Description                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| quicRateLimitingBytes Saved      | unit64                | UDP rate limiting bytes saved                                                                                                                                                                                                                                                                                                                  |
| udpRlSessionID                   | unit64                | Unique session ID for rate-limiting flow in a UDP rate-limiting session                                                                                                                                                                                                                                                                        |
| udpRlSessionDuration             | unit64                | UDP rate-limiting session duration, in milliseconds. Reported only when the last rate-limiting flow in that UDP rate-limited session has ended.                                                                                                                                                                                                |
| udpRlSessionBitrate              | unit64                | Bit rate of the UDP rate-limiting session, in bps. Reported only when the last rate-limiting flow in that UDP rate-limited session has ended.                                                                                                                                                                                                  |
| bytesDroppedDueToBuffer Limiting | uint64                | Number bytes dropped by buffer limiting                                                                                                                                                                                                                                                                                                        |
| failureReason                    | UdpProxyFailureReason | UDP failure reason<br>The following values are defined:<br>0 = UDPPROXY_FAILURE_NO_REASON<br>1 = UDPPROXY_FAILURE_REASON_SERVER_CONNECTION_INIT_FAILURE<br>2 = UDPPROXY_FAILURE_REASON_OUT_OF_SOCKET<br>3 = UDPPROXY_FAILURE_REASON_NO_CONNECTION_FOUND<br>4 = UDPPROXY_FAILURE_REASON_NO_PACKETS<br>5 = UDPPROXY_FAILURE_REASON_OTHER_REASONS |

## DpiInformation

The following table lists the name, data type, and description for each subfield contained in the DpiInformation data record.

**Table 4-53    DpiInformation Fields**

| Name        | Data Type | Description                             |
|-------------|-----------|-----------------------------------------|
| Application | bytes     | Application received from DPI detection |

## SHIPInformation

The following table lists the name, data type, and description for each subfield contained in the SHIPInformation data record.

**Table 4-54 SHIPInformation Fields**

| Name             | Data Type     | Description                                                                                                                                                                                         |
|------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Content Provider | string        | Content Provider                                                                                                                                                                                    |
| AppID            | uint64        | Not used                                                                                                                                                                                            |
| Consent Status   | ConsentStatus | Consent status<br>The following values are defined:<br>0 = Consent Not Available<br>1 = Consent Not Matched<br>2 = Consent Matched                                                                  |
| Ship Status      | ShipStatus    | SHIP status<br>The following values are defined:<br>0 = SHIP success<br>1 = Content Provider Not Found<br>2 = Info Not Available<br>3 = Consent Validation Failed<br>4 = SHIP Internal Server Error |

## HttpRecord

The following table lists the name, data type, and description for each subfield contained in the HttpRecord data record.

**Table 4-55 HttpRecord Fields**

| Name                  | Data Type                             | Description        |
|-----------------------|---------------------------------------|--------------------|
| keys                  | <a href="#">HttpRecordKeys</a>        | Contains subfields |
| httpTransactionRecord | <a href="#">HTTPTransactionRecord</a> | Contains subfields |

## HttpRecordKeys

The following table lists the name, data type, and description for each subfield contained in the HttpRecordKeys data record.

**Table 4-56 HttpRecordKeys Fields**

| Name            | Data Type | Description                   |
|-----------------|-----------|-------------------------------|
| sessionId       | uint64    | PDN session identifier        |
| bearerId        | uint64    | PDN bearer identifier         |
| flowId          | uint64    | Connection flow identifier    |
| transactionId   | uint64    | Transaction identifier        |
| cPlaneSessionId | uint64    | Session identifier on C-Plane |

## Real Time Troubleshooting (RTT) Event Data Record

This file contains the message definition for the RTT EDR message. The RTT EDR records provide visibility into the network and subscriber events to help diagnose networking and subscriber issues.

The following table lists the control plane procedures that result in the RTT EDRs being generated and sent to the EDR server at the end of the procedure. The data and attribute list available vary by procedure.

**Table 4-57 Procedures that Result in RTT EDR Generation**

| Procedure Number | Procedure Name                       | Supported Interfaces                     |
|------------------|--------------------------------------|------------------------------------------|
| 1                | EDR_PROC_GTP_CREATE_SESSION          | S4 / S11 / S5 / S8 / S2a / S2b / Gn / Gp |
| 2                | EDR_PROC_GTP_DELETE_SESSION          | S4 / S11 / S5 / S8 / S2a / S2b / Gn / Gp |
| 3                | EDR_PROC_GTP_CREATE_BEARER           | S4 / S11 / S5 / S8 / S2a / S2b / Gn / Gp |
| 4                | EDR_PROC_GTP_DELETE_BEARER           | S4 / S11 / S5 / S8 / S2a / S2b / Gn / Gp |
| 5                | EDR_PROC_GTP_MODIFY_BEARER           | S4 / S11 / S5 / S8 / S2a / S2b / Gn / Gp |
| 6                | EDR_PROC_GTP_UPDATE_BEARER           | S4 / S11 / S5 / S8 / S2a / S2b / Gn / Gp |
| 7                | EDR_PROC_GTP_REL_ACC_BEARER          | S4 / S11                                 |
| 8                | EDR_PROC_GTP_DL_DATA_NOTIFICATION    | S4 / S11                                 |
| 9                | EDR_PROC_DIAM_CCR_INITIAL            | Gx / Gy                                  |
| 10               | EDR_PROC_DIAM_CCR_UPDATE             | Gx / Gy                                  |
| 11               | EDR_PROC_DIAM_CCR_TERMINATE          | Gx / Gy                                  |
| 12               | EDR_PROC_DIAM_CCR_EVENT              | Gx / Gy                                  |
| 13               | EDR_PROC_DIAM_REAUTH                 | Gx / Gy / S6b / SWm                      |
| 14               | EDR_PROC_DIAM_ABORT_SESSION          | Gy / S6b / SWm                           |
| 15               | EDR_PROC_DIAM_SESSION_TERMINATION    | S6b / SWm                                |
| 16               | EDR_PROC_DIAM_AUTHORIZE_AUTHENTICATE | S6b / SWm                                |
| 17               | EDR_PROC_DIAM_EAP                    | SWm                                      |
| 18               | EDR_PROC_IKE_INIT                    | SWu                                      |
| 19               | EDR_PROC_IKE_AUTH                    | SWu                                      |
| 20               | EDR_PROC_IKE_INFO                    | SWu                                      |
| 21               | EDR_PROC_IKE_CHILD_SA                | SWu                                      |

**Table 4-57 Procedures that Result in RTT EDR Generation**

|    |                     |     |
|----|---------------------|-----|
| 22 | EDR_PROC_ROUTER_ADV | SWu |
|----|---------------------|-----|

## RTT Common Attribute List

The following table lists the attribute name, data type, description, and relevant procedures for GTP, Diameter, IKE, and all RTT messages. These attributes are in RttCommonInfo message.

**Table 4-58 RTT Common Attribute List**

| Name              | Data Type   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Relevant Procedures |
|-------------------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| timestamp         | EdrTimeInfo | Timestamp expressed by two fields: <ul style="list-style-type: none"> <li>1 = Epoch timestamp (uint32)</li> <li>2 = Fraction of the current second in microseconds (uint32)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                     | All                 |
| sourceId          | uint32      | EDR Interface name and task name that generated the record                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | All                 |
| sourceSequenceNum | uint32      | Incremental Record sequence number per sourceId                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | All                 |
| recordType        | EdrType     | Type of record: <ul style="list-style-type: none"> <li>7 = RTT Record Type</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | All                 |
| sessionId         | uint64      | PDN session identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | All                 |
| rtt_record_type   | EdrSubType  | RTT record subtype. The following values are defined: <ul style="list-style-type: none"> <li>8 = EDR_RTT_GTP_S5_RECORD</li> <li>9 = EDR_RTT_GTP_S8_RECORD</li> <li>10 = EDR_RTT_GTP_S11_RECORD</li> <li>11 = EDR_RTT_GTP_S4_RECORD</li> <li>12 = EDR_RTT_GTP_S2A_RECORD</li> <li>13 = EDR_RTT_GTP_S2B_RECORD</li> <li>14 = EDR_RTT_GTP_GN_RECORD</li> <li>15 = EDR_RTT_GTP_GP_RECORD</li> <li>16 = EDR_RTT_DIAMETER_GX_RECORD</li> <li>17 = EDR_RTT_DIAMETER_GY_RECORD</li> <li>18 = EDR_RTT_DIAMETER_S6B_RECORD</li> <li>19 = EDR_RTT_DIAMETER_SWM_RECORD</li> <li>20 = EDR_RTT_IKE_SWU_RECORD</li> </ul> | All                 |
| rtt_proc_num      |             | RTT procedure number (see <a href="#">Procedures that Result in RTT EDR Generation</a> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | All                 |

**Table 4-58 RTT Common Attribute List**

| Name            | Data Type                              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Relevant Procedures |
|-----------------|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| intf_type       |                                        | Interface Type. The following values are defined: <ul style="list-style-type: none"> <li>■ 1 = EDR_INTF_GTP_S4</li> <li>■ 2 = EDR_INTF_GTP_S5</li> <li>■ 3 = EDR_INTF_GTP_S8</li> <li>■ 4 = EDR_INTF_GTP_S11</li> <li>■ 5 = EDR_INTF_GTP_S2A</li> <li>■ 6 = EDR_INTF_GTP_S2B</li> <li>■ 7 = EDR_INTF_GTP_GN</li> <li>■ 8 = EDR_INTF_GTP_GP</li> <li>■ 9 = EDR_INTF_DIAM_GX</li> <li>■ 10 = EDR_INTF_DIAM_GY</li> <li>■ 11 = EDR_INTF_DIAM_S6B</li> <li>■ 12 = EDR_INTF_DIAM_SWM</li> <li>■ 13 = EDR_INTF_DIAM_STA</li> <li>■ 14 = EDR_INTF_IKE_SWU</li> </ul> | All                 |
| subscriberInfo  | <a href="#">SessionSubscriber Info</a> | Subscriber information. Contains subfields.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | All                 |
| gw_type         | EdrGatewayType                         | Gateway type. The following values are defined: <ul style="list-style-type: none"> <li>■ 1 = EDR_GW_TYPE_PGW</li> <li>■ 2 = EDR_GW_TYPE_SGW</li> <li>■ 3 = EDR_GW_TYPE_GGSN</li> <li>■ 4 = EDR_GW_TYPE_MCC</li> <li>■ 5 = EDR_GW_TYPE_WAG</li> <li>■ 6 = EDR_GW_TYPE_EPDG</li> </ul>                                                                                                                                                                                                                                                                          | All                 |
| gateway_info    | <a href="#">GatewayInfo</a>            | Gateway information. Contains subfields.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | All                 |
| proc_start_time | uint64                                 | Procedure start time                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | All                 |
| proc_stop_time  | uint64                                 | Procedure stop time                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | All                 |

## RTT Attributes for GTP Interface Procedures

The following table lists the attribute name, data type, description, and relevant procedures for GTP interface procedures. These attributes are in the RttGtpInfo message.

**Table 4-59 GTP Interface Attribute List**

| Name                           | Data Type                             | Description                                                                                                                                        | Relevant Procedures    |
|--------------------------------|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| serving_nw_info                | <a href="#">ServingNetworkInfo</a>    | Serving network information. Contains subfields.                                                                                                   | 1, 5                   |
| session_qos_info               | <a href="#">SessionQosInformation</a> | Quality of service information. Contains subfields.                                                                                                | 1, 5, 6                |
| pco_info                       | <a href="#">RttGtpPCOInfo</a>         | Protocol Configuration Options. Contains subfields.                                                                                                | 1, 2, 3, 4, 5, 6       |
| cause_codes                    | uint32                                | GTP cause codes information from 3GPP TS 29.274.                                                                                                   | 1, 2, 3, 4, 5, 6, 7, 8 |
| user_csg_info                  | <a href="#">SessionUserCSGInfo</a>    | Closed subscriber group information. Contains subfields.                                                                                           | OBSOLETE               |
| Indication_flags               | Grouped AVP:                          |                                                                                                                                                    | 1, 2, 5                |
|                                | dual_address_bearer_flag              | Boolean Dual Address Bearer Flag                                                                                                                   |                        |
|                                | handover_indication                   | Handover Indication. The following values are defined: <ul style="list-style-type: none"> <li>■ 0 = New Session</li> <li>■ 1 = Handover</li> </ul> |                        |
|                                | operation_indication                  | Boolean                                                                                                                                            |                        |
|                                | unauthenticated_imsi                  | Boolean                                                                                                                                            |                        |
|                                | piggybacking_supported                | Boolean                                                                                                                                            |                        |
|                                | csg_change_reporting_support          | Boolean                                                                                                                                            |                        |
|                                | cs_fallback_indication                | Boolean Circuit Switching Fall Back Indication                                                                                                     |                        |
|                                | cs_to_ps_srvcc_indication             | Boolean                                                                                                                                            |                        |
| piggybackRec Indicator         | bool                                  | Piggyback Record Indicator: <ul style="list-style-type: none"> <li>■ 0 = no</li> <li>■ 1 = yes</li> </ul>                                          | 3, 5                   |
|                                |                                       |                                                                                                                                                    |                        |
| charging_gateway_function_host | uint32                                | FQDN of Charging Gateway Function (CGF)                                                                                                            | OBSOLETE               |

**Table 4-59 GTP Interface Attribute List**

| Name                           | Data Type                     | Description                                                                                                                                                                                                                                             | Relevant Procedures    |
|--------------------------------|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| linked_bearer_id               | uint32                        | Default bearer ID                                                                                                                                                                                                                                       | 1, 2, 3, 4, 5          |
| chargingcharacteristics        | uint32                        | Charging Characteristics                                                                                                                                                                                                                                | 1                      |
| bearer_info_list_add           | <a href="#">RttBearerInfo</a> | Repeated RttBearerInfo. Contains subfields.                                                                                                                                                                                                             | 1, 3                   |
| bearer_info_list_delete        |                               | Repeated RttBearerInfo - Bearer Contexts to be removed                                                                                                                                                                                                  | 1, 4, 5                |
| bearer_info_list_modify        |                               | Repeated RttBearerInfo - Bearer Contexts to be modified                                                                                                                                                                                                 | 5, 6                   |
| bearer_info_list_failed        |                               | Repeated RttBearerInfo - Bearer Contexts failed to create                                                                                                                                                                                               | 4                      |
| selection_mode                 | uint32                        | Selection mode.                                                                                                                                                                                                                                         | 1                      |
| Maximum_apn_restriction        | uint32                        | APN Restriction in Request                                                                                                                                                                                                                              | 1                      |
| Apn_restriction                | uint32                        | APN Restriction in Response                                                                                                                                                                                                                             | 1, 5                   |
| recovery                       | uint32                        | Recovery (Restart Counter)                                                                                                                                                                                                                              | OBSOLETE               |
| ue_time_zone                   | uint32                        |                                                                                                                                                                                                                                                         |                        |
| user_csg_information           | string                        | User CSG Information                                                                                                                                                                                                                                    | 1, 5                   |
| signalling_priority_indication | bool                          | Signalling Priority Indication                                                                                                                                                                                                                          | OBSOLETE               |
| change_reporting_action        | string                        | Change Reporting Action IE                                                                                                                                                                                                                              | 1, 3, 4, 5             |
| charging_id                    | uint32                        | Charging Id IE                                                                                                                                                                                                                                          | 1, 2, 3, 4, 5, 6, 7, 8 |
| ue_requested_pdp_address       | string                        | Requested PDN address                                                                                                                                                                                                                                   |                        |
| ue_requested_pdp_type          | EdrIpType                     | Requested PDP address type. The following values are defined: <ul style="list-style-type: none"> <li>■ 0 = PDP_TYPE_INV</li> <li>■ 1 = PDP_TYPE_IPV4</li> <li>■ 2 = PDP_TYPE_IPV6</li> <li>■ 3 = PDP_TYPE_IPV4V6</li> <li>■ 4 = PDP_TYPE_PPP</li> </ul> |                        |
| ue_requested_pdp_accepted      | bool                          |                                                                                                                                                                                                                                                         |                        |
| request_cause_code             | uint32                        | Cause IE in request, from 3gpp TS 29.274.                                                                                                                                                                                                               | 8                      |



**Table 4-59 GTP Interface Attribute List**

| Name                     | Data Type       | Description                                                                                                                 | Relevant Procedures |
|--------------------------|-----------------|-----------------------------------------------------------------------------------------------------------------------------|---------------------|
| originating_node         | uint32          | Originating node                                                                                                            | 2                   |
| eps_bearer_id_list       |                 | Repeated Integer - list of EPS Bearer IDs, in create bearer request                                                         | 4, 7, 8             |
| msg_direction            | GtpMsgDirection | The following values are defined:<br><ul style="list-style-type: none"> <li>■ UeInitiated</li> <li>■ NwInitiated</li> </ul> | 2, 4, 6             |
| delay_dl_pkt_notify_req  | uint32          | Delay Downlink Packet Notification Request IE                                                                               | 5, 8                |
| procedure_transaction_id | uint32          | Procedure Transaction ID IE                                                                                                 | 3, 4, 6             |

## RttGtpPCOInfo

The following table lists the name, data type, and description for each subfield contained in the RttGtpPCOInfo data record.

**Table 4-60 RttGtpPCOInfo Subfields**

| Name             | Data Type | Description                      |
|------------------|-----------|----------------------------------|
| dns_v4_address   |           | Repeated String DNS IPv4 Address |
| dns_v6_address   |           | Repeated DNS IPv6 Address        |
| pcscf_v4_address |           | Repeated PCSCF IPv4 Address      |
| pcscf_v6_address |           | Repeated PCSCF IPv6 Address      |
| ipv4_mtu         | uint32    | IPv4 MTU size                    |

## RttBearerInfo

The following table lists the name, data type, and description for each subfield contained in the RttBearerInfo data record.

**Table 4-61 RttBearerInfo Subfields**

| Name              | Data Type                            | Description                                         |
|-------------------|--------------------------------------|-----------------------------------------------------|
| bearer_info       | <a href="#">BearerInfo</a>           | Bearer information. Contains subfields.             |
| bearer_qos_info   | <a href="#">BearerQosInformation</a> | Quality of service information. Contains subfields. |
| bearer_cause_code | uint32                               | GTP cause codes information from 3GPP TS 29.274.    |
| bearer_flags      | <a href="#">RttGtpBearerFlags</a>    | Bearer flags IE. Contains subfields.                |

## RttGtpBearerFlags

The following table lists the name, data type, and description for each subfield contained in the RttGtpBearerFlags data record.

**Table 4-62 RttGtpBearerFlags Subfields**

| Name         | Data Type | Description                   |
|--------------|-----------|-------------------------------|
| voice_bearer | bool      | Boolean Voice Bearer Flag     |
| vind         | bool      | Boolean vSRVCC indicator Flag |

## RTT Attributes for Diameter SWm/Gx/Gy/S6b Interface Procedures

The following table lists the attribute name, data type, description, and relevant procedures for Diameter SWm/Gx/Gy/S6b interface procedures. These attributes are in the RttDiameterInfo message.

**Table 4-63 Diameter SWm/Gx/Gy/S6b Interface Attribute List**

| Name                        | Data Type       | Description                                                                                                                                                                                                     | Relevant Procedures | Relevant Interfaces |
|-----------------------------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|---------------------|
| sessionid                   | string          | PDN session identifier                                                                                                                                                                                          | 9 - 16              | Gx / Gy / S6b       |
| originhost                  | string          |                                                                                                                                                                                                                 | 9 - 16              | Gx / Gy / S6b       |
| originrealm                 | string          |                                                                                                                                                                                                                 | 9 - 16              | Gx / Gy / S6b       |
| destinationhost             | string          |                                                                                                                                                                                                                 | 9 - 16              | Gx / Gy / S6b       |
| destinationrealm            | string          |                                                                                                                                                                                                                 | 9 - 16              | Gx / Gy / S6b       |
| authapplicationid           | uint32          |                                                                                                                                                                                                                 | 9 - 16              | Gx / Gy / S6b       |
| servicecontextid            | string          | Service Context ID                                                                                                                                                                                              | 9 - 16              | Gx / Gy / S6b       |
| requesttype                 | uint32          |                                                                                                                                                                                                                 | 9 - 16              | Gx / Gy / S6b       |
| requestnumber               | uint32          |                                                                                                                                                                                                                 | 9 - 16              | Gx / Gy / S6b       |
| originstateid               | uint32          |                                                                                                                                                                                                                 | 9 - 16              | Gx / Gy / S6b       |
| terminationcause            | enum            | Termination Cause: <ul style="list-style-type: none"> <li>■ EdrTerminationCause: Procedures 9, 10, 11, 12, 15</li> <li>■ EdrSessionReleaseCause: Procedure 13</li> <li>■ EdrAbortCause: Procedure 14</li> </ul> | 9 - 15              | Gx / Gy / S6b       |
| serviceselection            | string          |                                                                                                                                                                                                                 | 16                  | S6b                 |
| charginggatewayfunctionhost | string          |                                                                                                                                                                                                                 | 16                  | S6b                 |
| charginggroupid             | string          |                                                                                                                                                                                                                 | 16                  | S6b                 |
| authrequesttype             | uint64          |                                                                                                                                                                                                                 | 13, 16              | Gx / Gy / S6b       |
| reauthrequesttype           | uint64          |                                                                                                                                                                                                                 | 13                  | Gx / Gy / S6b       |
| visitednetworkidentifier    | string          | PLMNID to lookup PGW                                                                                                                                                                                            | 16                  | S6b                 |
| eventtrigger                | Repeated uint32 | Event trigger type                                                                                                                                                                                              | 9 - 13              | Gx / Gy / S6b       |
| ueipv4addr                  | uint32          | Framed IP address                                                                                                                                                                                               | 9 - 13, 16          | Gx / Gy / S6b       |

**Table 4-63 Diameter SWm/Gx/Gy/S6b Interface Attribute List**

| Name               | Data Type             | Description                                       | Relevant Procedures | Relevant Interfaces |
|--------------------|-----------------------|---------------------------------------------------|---------------------|---------------------|
| ueipv6prefix       | uint64                | Framed IPv6 prefix                                | 9 - 13, 16          | Gx / Gy / S6b       |
| servername         | string                | Contains the P-CSCF FQDN                          | 16                  | S6b                 |
| framedpool         | string                |                                                   | 16                  | S6b                 |
| framedipv6pool     | string                |                                                   | 16                  | S6b                 |
| ratinggroup        | uint32                |                                                   | 13                  | Gy                  |
| qosinfo            | Repeated group AVP:   |                                                   | 9 - 13              |                     |
|                    | qosClassId            | uint32<br>QoS class ID                            |                     |                     |
|                    | mbrUplink             | uint32<br>Uplink maximum bit-rate in kbps         |                     |                     |
|                    | mbrDownlink           | uint32<br>Downlink maximum bit-rate in kbps       |                     |                     |
|                    | gbrUplink             | uint32<br>Uplink guaranteed bit-rate in kbps      |                     |                     |
|                    | gbrDownlink           | uint32<br>Downlink guaranteed bit-rate in kbps    |                     |                     |
|                    | bearerid              | string                                            |                     |                     |
|                    | apnambUplink          | uint32<br>Uplink APN Aggregate maximum Bit-Rate   |                     |                     |
|                    | apnambDownlink        | uint32<br>Downlink APN Aggregate maximum Bit-Rate |                     |                     |
| ipcantype          |                       |                                                   | 9 - 13, 16          | Gx / Gy / S6b       |
| chargingruleremove | Repeated group AVP:   |                                                   | 9 - 13              |                     |
|                    | chargingrule name     | string                                            |                     |                     |
|                    | chargingrule basename | string                                            |                     |                     |

**Table 4-63 Diameter SWm/Gx/Gy/S6b Interface Attribute List**

| Name                  | Data Type                      | Description                                                                                                                                     | Relevant Procedures | Relevant Interfaces |
|-----------------------|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|---------------------|
| chargingrulerinstall  | Repeated group AVP:            |                                                                                                                                                 | 9 - 13              |                     |
|                       | chargingrule name              | string                                                                                                                                          |                     |                     |
|                       | chargingrule basename          | string                                                                                                                                          |                     |                     |
|                       | bearerid                       | string                                                                                                                                          |                     |                     |
| multiserviceindicator |                                |                                                                                                                                                 | 9 - 12              | Gy                  |
| mscreditcontrol_req   | Repeated group AVP in request: |                                                                                                                                                 | 9 - 12              | Gy                  |
|                       | requested serviceunit          | Repeated Grouped attribute for Total-Octets requested                                                                                           |                     |                     |
|                       | ratinggroup                    | uint32                                                                                                                                          |                     |                     |
|                       | grantedservice unit            | Grouped attribute for Total-Octets granted                                                                                                      |                     |                     |
|                       | usedservice unit               | Repeated Grouped attribute for Total-Octets used                                                                                                |                     |                     |
|                       | validitytime                   | uint32                                                                                                                                          |                     |                     |
|                       | volumequota threshold          | uint32                                                                                                                                          |                     |                     |
|                       | resultcode                     | uint32                                                                                                                                          |                     |                     |
|                       | finalunit indication           | Grouped AVP:                                                                                                                                    |                     |                     |
|                       |                                | enum EdrFinalUnitAction<br>finalunitaction                                                                                                      |                     |                     |
|                       |                                | string<br>restrictionfilterrule                                                                                                                 |                     |                     |
|                       |                                | string<br>filterid                                                                                                                              |                     |                     |
|                       |                                | redirectserver<br>Grouped AVP:                                                                                                                  |                     |                     |
|                       |                                | <ul style="list-style-type: none"> <li>■ redirectaddresstype (enum EdrRedirectAddressType)</li> <li>■ redirectserveraddress (string)</li> </ul> |                     |                     |

**Table 4-63 Diameter SWm/Gx/Gy/S6b Interface Attribute List**

| Name                | Data Type                     | Description                      | Relevant Procedures | Relevant Interfaces |
|---------------------|-------------------------------|----------------------------------|---------------------|---------------------|
| mscreditcontrol_rsp |                               | Repeated Grouped AVP in response | 9 - 12              | Gy                  |
| serviceinfo         | Grouped AVP:                  |                                  | 9 - 12              | Gy                  |
|                     | chargingid                    | uint32                           |                     |                     |
|                     | pdnconnection id              | uint32                           |                     |                     |
|                     | pdptype                       | uint32                           |                     |                     |
|                     | servedpdpip addr              | Repeated AVP string              |                     |                     |
|                     | qosinfo                       | Grouped AVP                      |                     |                     |
|                     | nsapid                        | string                           |                     |                     |
|                     | apnid                         | string                           |                     |                     |
|                     | chargingcharct                | string                           |                     |                     |
|                     | charging selectionmode        | uint32                           |                     |                     |
|                     | chargingrule basename         | string                           |                     |                     |
|                     | charging gateway functionhost | string                           |                     |                     |
|                     | charginggroup id              | string                           |                     |                     |
| apnconfiguration    | Repeated Grouped AVP          |                                  | 16                  | ePDG / S2b          |
| apnid               | string                        |                                  | 9 - 16              |                     |
| userequipmentinfo   | Grouped AVP:                  |                                  | 9 - 12              | Gx / Gy             |
|                     | userequipment type            | enum                             |                     |                     |
|                     | userequipment value           | string                           |                     |                     |

**Table 4-63 Diameter SWm/Gx/Gy/S6b Interface Attribute List**

| Name                     | Data Type               | Description                                        | Relevant Procedures | Relevant Interfaces |
|--------------------------|-------------------------|----------------------------------------------------|---------------------|---------------------|
| chargingrulereport       | Repeated Grouped AVP:   |                                                    | 9 - 13              | Gx / Gy / S6b       |
|                          | chargingrule name       | String repeated AVP                                |                     |                     |
|                          | chargingrule basename   | String repeated AVP                                |                     |                     |
|                          | bearerid                | String                                             |                     |                     |
|                          | pccrulestatus           | Enum                                               |                     |                     |
|                          | rulefailurecode         | Enum                                               |                     |                     |
|                          | finalunit indication    | Grouped AVP                                        |                     |                     |
| revalidationtime         | uint32                  |                                                    | 9 - 13              | Gx / Gy / S6b       |
| experimentalresult       | Grouped Attribute:      |                                                    | 9 - 16              | Gx / Gy / S6b       |
|                          | Vendorid                | uint32                                             |                     |                     |
|                          | experimental resultcode | uint32                                             |                     |                     |
| failedavp                |                         | Repeated list of AVPs                              | 9 - 16              | Gx / Gy / S6b       |
| errormessage             |                         | String                                             | 9 - 16              | Gx / Gy / S6b       |
| assumepositive treatment |                         | String                                             | 9 - 12              | Gx                  |
| originationtime stamp    |                         | uint64                                             | 9 - 12, 16          | Gx / Gy / S6b       |
| maxwaittime              |                         | uint32                                             | 9 - 12, 16          | Gx / Gy / S6b       |
| authorizationlifetime    |                         | uint32                                             | 16                  | S6b                 |
| authgraceperiod          |                         | uint32                                             | 16                  | S6b                 |
| sessiontimeout           |                         | uint32                                             | 16                  | S6b                 |
| idletimeout              |                         | uint32                                             | 16                  | S6b                 |
| selfactivationstatus     |                         | Activation Status                                  | 16                  | S6b                 |
| resultcode               |                         | Enum for the Diameter Result codes in response     | 9 - 16              | Gx / Gy / S6b       |
| procedure_result         |                         | Internal Procedure Result – Failed/Success/Timeout | 9 - 16              | Gx / Gy / S6b       |

**Table 4-63 Diameter SWm/Gx/Gy/S6b Interface Attribute List**

| Name                    | Data Type                                                    | Description                                                                                             | Relevant Procedures | Relevant Interfaces |
|-------------------------|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|---------------------|---------------------|
| usagemonitoringinfo_req | Repeated grouped attribute with data usage info in request:  |                                                                                                         | 9 - 13              | Gx / Gy / S6b       |
|                         | usage monitoringkey                                          | Usage Monitoring Key in bytes                                                                           |                     |                     |
|                         | grantedservice unit                                          | Grouped attribute for Total-Octets granted                                                              |                     |                     |
|                         | usedservice unit                                             | Grouped attribute for Total-Octets used                                                                 |                     |                     |
|                         | usage monitoring level                                       | Usage Monitoring Level: <ul style="list-style-type: none"> <li>■ session</li> <li>■ pcc_rule</li> </ul> |                     |                     |
| usagemonitoringinfo_rsp | Repeated grouped attribute with data usage info in response: |                                                                                                         | 9 - 13              | Gx / Gy / S6b       |
|                         | usage monitoringkey                                          | Usage Monitoring Key in bytes                                                                           |                     |                     |
|                         | grantedservice unit                                          | Grouped attribute for Total-Octets granted                                                              |                     |                     |
|                         | usedservice unit                                             | Grouped attribute for Total-Octets used                                                                 |                     |                     |
|                         | usage monitoring level                                       | Usage Monitoring Level: <ul style="list-style-type: none"> <li>■ session</li> <li>■ pcc_rule</li> </ul> |                     |                     |
| eapauthsessionstate     | string                                                       | EAP Session State                                                                                       | 16                  | SWm                 |
| eapmastersessionkey     | bytes                                                        | EAP Master Session Key                                                                                  | 16                  | SWm                 |
| eapstatuscode           | uint32                                                       | EAP Status Code: <ul style="list-style-type: none"> <li>■ success</li> <li>■ failure</li> </ul>         | 16                  | SWm                 |
| eapmessageid            | bytes                                                        | EAP Message Identifier                                                                                  | 16                  | SWm                 |
| eaptype                 | string                                                       | EAP Type (such as EAP-AKA)                                                                              | 16                  | SWm                 |
| wlanusername            | string                                                       | WLAN User Name                                                                                          | 16                  | SWm                 |



**Table 4-63 Diameter SWm/Gx/Gy/S6b Interface Attribute List**

| Name   | Data Type | Description       | Relevant Procedures | Relevant Interfaces |
|--------|-----------|-------------------|---------------------|---------------------|
| ratype | uint32    | Radio Access Type | 16                  | SWm                 |

## RTT Attributes for IKE Interface Procedures

The following table lists the attribute name, data type, description, and relevant procedures for IKE interface procedures. These attributes are in the RttIkeInfo message.

**Table 4-64 Diameter IKE Interface Attribute List**

| Name           | Data Type | Description                        | Relevant Procedures | Relevant Interfaces |
|----------------|-----------|------------------------------------|---------------------|---------------------|
| ueipaddress    | bytes     | UE IP Address                      | 18 - 22             | SWu                 |
| ueudpport      | uint32    | UE UDP Port                        | 18 - 22             | SWu                 |
| epdgidentity   | string    | ePDG Identity (IP address or FQDN) | 18 - 22             | SWu                 |
| epdgudpport    | uint32    | ePDG UDP Port                      | 18 - 22             | SWu                 |
| ispi           | bytes     | Initiator SPI                      | 18 - 19, 21         | SWu                 |
| rspi           | bytes     | Responder SPI                      | 18 - 19, 21         | SWu                 |
| transformtype  | string    | Transform and Type                 | 18 - 19, 21         | SWu                 |
| dhgroup        | uint32    | DH Group                           | 18                  | SWu                 |
| notifytype     | string    | Notify Message Type                | 18 - 21             | SWu                 |
| idi            | string    | IDi (identity – initiator)         | 19 - 20             | SWu                 |
| idr            | string    | IDr (identity – responder)         | 19 - 20             | SWu                 |
| tsi            | string    | TSi (traffic selector - initiator) | 19                  | SWu                 |
| tsr            | string    | TSr (traffic selector - responder) | 19                  | SWu                 |
| cfgauthmethod  | string    | Configured IKE Auth Method         | 18 - 20             | SWu                 |
| cfgipv4address | bytes     | Configured IKE IPv4 Address        | 18 - 20             | SWu                 |
| cfgipv4netmask | bytes     | Configured IKE IPv4 Netmask        | 18 - 20             | SWu                 |
| cfgipv4dns     | string    | Configured IKE IPv4 DNS            | 18 - 20             | SWu                 |
| saprotocol     | string    | Child SA Protocol                  | 21                  | SWu                 |

**Table 4-64 Diameter IKE Interface Attribute List**

| <b>Name</b>  | <b>Data Type</b> | <b>Description</b>        | <b>Relevant Procedures</b> | <b>Relevant Interfaces</b> |
|--------------|------------------|---------------------------|----------------------------|----------------------------|
| saspi        | bytes            | Child SA SPI              | 21                         | SWu                        |
| infotype     | string           | Information Exchange Type | 20                         | SWu                        |
| raipv6prefix | string           | IPv6 Advertised Prefix    | 22                         | SWu                        |

## EncryptedVideoRateLimiting

The following table lists the name, data type, and description for each top-level EncryptedVideoRateLimiting field. The top-level EncryptedVideoRateLimiting fields contain a set of subfields.

**Table 4-65 EncryptedVideoRateLimiting Fields**

| Name            | Description                                        |
|-----------------|----------------------------------------------------|
| timeInfo        | For details, see <a href="#">timeInfo</a> .        |
| subscriberInfo  | For details, see <a href="#">subscriberInfo</a> .  |
| connectionInfo  | For details, see <a href="#">connectionInfo</a> .  |
| TcpSplicingInfo | For details, see <a href="#">tcpSplicingInfo</a> . |

### timeInfo

The following table lists the name, data type, and description for each subfield contained in the timeInfo data record.

**Table 4-66 timeInfo Subfields**

| Name                      | Data Type | Description                                                                                                                                                                                                                                                              |
|---------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| transactionTimestamp      | int64     | The timestamp indicating when the HTTP transaction starts, in Epoch time (seconds).                                                                                                                                                                                      |
| requestLatency            | uint64    | Indicates the amount of time, in milliseconds, it takes the HTTP proxy to forward an HTTP request to the server. (Starts counting from the moment the HTTP proxy receives the first byte of data from the client to when it sends the last byte of data to the server.)  |
| responseLatency           | uint64    | Indicates the amount of time, in milliseconds, it takes the HTTP proxy to forward an HTTP response to the client. (Starts counting from the moment the HTTP proxy receives the first byte of data from the server to when it sends the last byte of data to the client.) |
| transactionDuration       | uint64    | Indicates the total amount of time of the transaction, in milliseconds.                                                                                                                                                                                                  |
| transactionTimeSubSeconds | int32     | The millisecond portion of the timestamp, indicating when the HTTP transaction starts.<br><br><b>Note:</b> Field 1 and Field 5 together provide the date and time of the transaction to the millisecond level.                                                           |

## subscriberInfo

The following table lists the name, data type, and description for each subfield contained in the subscriberInfo data record.

**Table 4-67 subscriberInfo Subfields**

| Name          | Data Type | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| imsi          | uint64    | Indicates the International Mobile Subscriber Identity (IMSI) of the User Equipment (UE) that originated the request.                                                                                                                                                                                                                                                                                                                                                                        |
| imei          | uint64    | Indicates the International Mobile Equipment Identity (IMEI) of the UE that originated the request.                                                                                                                                                                                                                                                                                                                                                                                          |
| msisdn        | uint64    | Indicates the primary Mobile Station (MS) ISDN number (MSISDN) of the subscriber.                                                                                                                                                                                                                                                                                                                                                                                                            |
| sessionId     | uint32    | Identifies the Mobile Content Cloud's session ID for the subscriber's PDN connection.                                                                                                                                                                                                                                                                                                                                                                                                        |
| servingMcc    | uint32    | Indicates the Mobile Country Code (MCC) of the serving network.                                                                                                                                                                                                                                                                                                                                                                                                                              |
| servingMnc    | uint32    | Indicates the Mobile Network Code (MNC) of the serving network.                                                                                                                                                                                                                                                                                                                                                                                                                              |
| plmnIdLength  | uint32    | Indicates the length of the home Public Land Mobile Network (PLMN) Identifier.                                                                                                                                                                                                                                                                                                                                                                                                               |
| ratType       | uint32    | Indicates the Radio Access Technology (RAT) type currently used by the Mobile Station, when available. The following values are defined: <ul style="list-style-type: none"> <li>■ 0 = Reserved</li> <li>■ 1 = UTRAN</li> <li>■ 2 = GERAN</li> <li>■ 3 = WLAN</li> <li>■ 4 = GAN</li> <li>■ 5 = HSPA-Evolution</li> <li>■ 6 = EUTRAN</li> <li>■ 101 = IEEE 802.16e</li> <li>■ 102 = 3GPP2-eHRPD</li> <li>■ 103 = 3GPP2-HRPD</li> <li>■ 104 = 3GPP2-1xRTT</li> <li>■ 105 = 3GPP-EPS</li> </ul> |
| roamingStatus | bool      | Indicates whether the subscriber is roaming. The following values are defined: <ul style="list-style-type: none"> <li>■ 0 = False</li> <li>■ 1 = True</li> </ul>                                                                                                                                                                                                                                                                                                                             |
| sgwIp         | bytes     | Indicates the IP address of the SGW used (can be an IPv4 or IPv6 address).                                                                                                                                                                                                                                                                                                                                                                                                                   |
| realApn       | string    | Indicates the logical name of the mapped access point to the external packet data network.                                                                                                                                                                                                                                                                                                                                                                                                   |

**Table 4-67 subscriberInfo Subfields**

| Name        | Data Type | Description                                                                                   |
|-------------|-----------|-----------------------------------------------------------------------------------------------|
| originalApn | string    | Indicates the logical name of the connected access point to the external packet data network. |

## connectionInfo

The following table lists the name, data type, and description for each subfield contained in the connectionInfo data record.

**Table 4-68 connectionInfo Subfields**

| Name            | Data Type | Description                                                                                                                                                                                                                                                                                                                                               |
|-----------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| giNc            | string    | Indicates the network context of the Gi interface.                                                                                                                                                                                                                                                                                                        |
| sourceIp        | byte      | Indicates the source IP address of the IP packets issued by the UE (can be an IPv4 or IPv6 address).                                                                                                                                                                                                                                                      |
| sourcePort      | uint32    | Indicates the source port of the IP packets received from the UE.                                                                                                                                                                                                                                                                                         |
| destinationIp   | byte      | Indicates the destination IP address of the IP packets issued by the UE (can be an IPv4 or IPv6 address).                                                                                                                                                                                                                                                 |
| destinationPort | uint32    | Indicates the destination port of the IP packets received from the UE.                                                                                                                                                                                                                                                                                    |
| ipTransparency  | bool      | Indicates whether IP-transparency is enabled or disabled. With IP-transparency, HTTP requests forwarded by the HTTP proxy to origin servers contain the UE's original IP address as the source TCP/IP address in the IP packet. The following values are defined: <ul style="list-style-type: none"> <li>■ 0 = Disabled</li> <li>■ 1 = Enabled</li> </ul> |

## tcpSplicingInfo

The following table lists the name, data type, and description for each subfield contained in the tcpSplicingInfo data record.

**Table 4-69 tcpSplicingInfo Subfields**

| Name                      | Data Type         | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Upstream bytes received   | uint64            | Number of bytes received from client                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Upstream bytes sent       | uint64            | Number of bytes sent to server                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Downstream bytes received | uint64            | Number of bytes received from server                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Downstream bytes sent     | uint64            | Number of bytes sent to client                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| status                    | enum              | Status of the Request: <ul style="list-style-type: none"> <li>■ 0 = Init</li> <li>■ 1 = IdleTimeout</li> <li>■ 2 = ClientClose</li> <li>■ 3 = ServerClose</li> <li>■ 4 = ClientErrEvt</li> <li>■ 5 = ServerErrEvt</li> <li>■ 6 = ClientRXError</li> <li>■ 7 = ClientTXError</li> <li>■ 8 = ServerRXError</li> <li>■ 9 = ServerTXError</li> <li>■ 10 = IdleForce</li> <li>■ 11 = NoBuffer</li> <li>■ 12 = NoData</li> <li>■ 13 = ConnectFail</li> <li>■ 14 = Handoff</li> <li>■ 15 = ClientError</li> <li>■ 16 = ServerError</li> <li>■ 17 = DnsReqFail</li> <li>■ 18 = DnsRspFail</li> <li>■ 19 = Reject</li> <li>■ 20 = TFDReject</li> </ul> |
| localIp                   | hexidecimal bytes | Local IP address                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| localPort                 | uint32            | Local port number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| serverIp                  | hexidecimal bytes | Server IP address                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| serverPort                | uint32            | Server port number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

**Table 4-69 tcpSplicingInfo Subfields**

| Name                           | Data Type | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proxyNC                        | string    | Network context of the proxy                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| protocol                       | enum      | Protocol type: <ul style="list-style-type: none"> <li>■ 0 = Invalid</li> <li>■ 1 = Http1xTls</li> <li>■ 2 = Http20</li> <li>■ 3 = Http20Tls</li> <li>■ 4 = WebSock</li> <li>■ 5 = Http</li> <li>■ 6 = Others</li> <li>■ 7 = WebRTC</li> <li>■ 8 = FaceBookZero</li> </ul>                                                                                                                                                                                                                             |
| TlsSNI                         | string    | Server Name Indication                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| TlsServerCommonName            | string    | Server common name                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| RateLimitingApplied            | bool      | Indicates whether ABR video rate limiting is applied: <ul style="list-style-type: none"> <li>■ 0 = False</li> <li>■ 1 = True</li> </ul>                                                                                                                                                                                                                                                                                                                                                               |
| RateLimitingBitrate            | unit64    | Rate limiting bit rate, in bits per second.                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| RateLimitingBypassCancelReason | enum      | Rate limiting bypass cancel reason: <ul style="list-style-type: none"> <li>■ 0 = Cancel_TooShort: Bytes to send is less than the min rate limiting bitrate</li> <li>■ 1 = Bypass_NoAbr: Subscriber Opt-out</li> <li>■ 2 = Bypass_Disable: Rate limiting admin-state is disabled</li> <li>■ 3 = Bypass_opt_in_NoAbr: Opt-in requirements are not met</li> <li>■ 4 = Bypass_No_MaxRate_in_CP_Manifest</li> <li>■ 5 = Bypass_Invalid_MaxRate_in_CP_Manifest</li> <li>■ 6 = Savings Estimation</li> </ul> |
| RateLimitingBurstSize          | unit64    | Rate limiting burst size, in bytes                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| negotiatedALPN                 | string    | Negotiated ALPN (Application Layer Protocol Negotiation)                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| tcpRISessionID                 | unit64    | Unique session ID for rate-limited connections in a TCP rate-limited session. Reported for all rate-limited connections in a TCP rate-limited session.                                                                                                                                                                                                                                                                                                                                                |
| tcpRISessionDuration           | unit64    | TCP rate-limited session duration, in seconds. Reported only when the last rate-limited connection in that TCP rate-limited session has ended.                                                                                                                                                                                                                                                                                                                                                        |
| tcpRISessionBitrate            | unit64    | Bit rate of the TCP rate-limited session, in bps. Reported only when the last rate-limited connection in that TCP rate-limited session has ended.                                                                                                                                                                                                                                                                                                                                                     |

**Table 4-69 tcpSplicingInfo Subfields**

| Name              | Data Type              | Description                                                                                                                                                                                                                                                                            |
|-------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| appliedPolicyName | percent encoded string | Name of the optimization policy that was matched and applied to the request. For example, if the request is video, MCC first evaluates about its eligibility for optimizations. If MCC finds it to be eligible, optimizations such as video pacing or rate limiting are applied to it. |
| tcpRlBytesSaved   | uint64                 | TCP rate-limited bytes saved                                                                                                                                                                                                                                                           |